

Malawi University of Science and Technology

Malawi Institute of Technology

PHYSICS 1

(MECHANICS AND THERMAL PROPERTIES OF MATTER)

Course Code

PHY 111

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Module Overview

This module is intended..... The goal of this module is to help you acquire a deep understanding of the principles of mechanics, thermal properties of matter and the gas laws.

Provide a general introduction to the whole module covering aims, objectives, outcomes, major activities and requirements for the module. Include pointers to references, tests and assignments, text books, etc.....

PART 1

MECHANICS

Outline

- 1.0 Introduction
- 1.1 Objectives
- 1.2 Key terms
- 1.3 Units and Measurements
 - 1.3.1 SI Units
- 1.4 Concepts of Motion and vectors
- 1.5 Forces and Motion
 - 1.5.1 Circular motion
 - 1.5.2 Rotation motion
- 1.6 Kinematics and Dynamics
- 1.7 Work and Energy
- 1.8 Impulse and Momentum

1.0 Introduction

Mechanics is “*an area of science concerned with the behaviour of physical bodies when subjected to forces or displacement and the subsequent effects of the bodies on their environment.*”^(wikipedia) the study of motion and forces. “

1.1 Objectives of the unit

By the end of this unit you should be able to:

Unit 1

Units and Measurement

Introduction

Physics can be described as the branch of science dealing with the study of properties of materials. To understand the properties of materials, measurement of physical quantities such as length, mass, time are involved. The uniqueness of physics lies in the measurement of these physical quantities. In this unit, we define what a physical quantity is, classifying physical quantities as base or derived. We will also classify these physical quantities as scalar or vector. Finally, we will carry out dimensional analysis which will help us in checking the consistency of equation and the derivation of an equation.

1.1 Objectives

By the end of this unit you should be able to:

- Define a physical quantity correctly
- State units of a given quantities
- Differentiate base quantities from derived quantities
- Applying dimensional analysis in checking the consistency of an equation
- Applying dimensional analysis in equation derivation

1.2 Key Terms

Physical quantity

SI Unit

Base quantities

Derived quantities

1.3 Measurement and unit

1.3.1 Physical quantity

A physical quantity is any quantity that can be measured e.g length, time, speed, density, electric charge, temperature

Units: A physical quantity is expressed as a number with units e.g length: $L=2\text{m}$, time; $t=1\text{s}$. Meter represented by m is a unit of length and second represented by s is unit of time.

A unit is a standard quantity (unit comes from latin for ‘one’) e.g length: 1 meter, time: 1 second, mass 1 kilogram.

SI Units

SI stands for *systeme internationale* in French, meaning “international system”. Therefore, SI units are agreed international standard units. Base quantities are measured using SI units.

Basic physical quantities are quantities which are not expressed in terms of other quantities. They are also called fundamental physical quantities.

The SI units of base quantities are called basic units or fundamental units.

There are seven base quantities as shown in Table 1 below:

Table 1: Base Physical Quantities

Quantity	SI Unit Name	Unit Symbol
Mass	kilogram	kg
Length	meter	m
Time	second	s
Electric current	Ampere	A
Temperature	Kelvin	K
Amount of substance	mole	mole
Luminous intensity	candela	cd

Derived quantities are expressed in terms of basic quantities by multiplication or division. Table 2 shows some examples of derived quantities.

Table 2: Examples of Derived Physical Quantities

Derived Quantity	Unit Name	Unit Symbol
Force	Newton	$\text{N} = \text{kgms}^{-2}$
Acceleration	Meter per second square	ms^{-2}
Pressure	Pascal	$\text{Pa} = \text{Nm}^{-2}$
Work	Joule	$\text{J} = \text{kgm}^2$
Power	Watt	$\text{W} = \text{kgm}^2\text{s}^{-3}$
Density	Mass per volume	kgm^{-3}

Unit Prefixes

For larger and smaller units, we write them in a short hand by multiply by powers of 10 as in Figure 3.

Figure 3: Unit Prefixes

Prefix	Symbol	Factor	Prefix	Symbol	Factor
Peta	P	10^{15}	deci	d	10^{-1}
Tera	T	10^{12}	centi	c	10^{-2}
Giga	G	10^9	milli	m	10^{-3}
Mega	M	10^6	micro	μ	10^{-6}
Kilo	k	10^3	nano	n	10^{-9}
Hector	h	10^2	pico	p	10^{-12}
Deca	da	10	fento	f	10^{-15}

Examples: $\text{cm} = 0.01\text{m} = 10^{-2} \text{ m}$, $\text{mm} = 0.001 \text{ m} = 10^{-3} \text{ m}$, $\text{km} = 1000 \text{ m} = 10^3 \text{ m}$

Changing units

Replace the unit you are given with the equivalent quantity in the unit you want

Example : Write 72km/hr in m/s

Solution

We know that 1km = 1000m (we are given km, we want m)

We also know that 1hr = 3600s (we are given hr, we want s)

$$\text{Then: } 72 \frac{\text{km}}{\text{hr}} = 72 \times \frac{1000 \text{ m}}{3600 \text{ s}} = 20 \text{ m/s}$$

SI standards

Length

Length is defined as the distance between two points. The SI unit of length is metre.

One standard metre is equal to 1 650 763.73 wavelengths of the orange – red light emitted by the individual atoms of krypton – 86 in a krypton discharge lamp.

Mass

Mass is the quantity of matter contained in an object? It is independent of temperature and pressure. It does not vary from place to place. The SI unit of mass is kilogram.

One kilogram is equal to the mass of the international prototype of the kilogram (a platinum-iridium alloy cylinder) kept at the International Bureau of Weights and Measures at Sevres, near Paris, France.

Time

Until 1960 the standard of time was based on the mean solar day, the time interval between successive passages of the sun at its highest point across the meridian. It is averaged over an year. In 1967, an atomic standard was adopted for second, the SI unit of time.

One standard second is defined as the time taken for 9 192 631 770 periods of the radiation corresponding to unperturbed transition between hyperfine levels of the ground state of cesium – 133 atom. Atomic clocks are based on this. In atomic clocks, an error of one second occurs only in 5000 years.

Activity 1.1

1. Define physical quantity
2. Explain the difference between basic quantities and derived quantities
3. Classify each of the following quantities as a basic quantity or a derived quantity:
 - a. $30\mu\text{s}$
 - b. 20ms^{-2}
 - c. 1000 kJ
 - d. 5W

1.3.2 Dimensional analysis

It deals with the physical quantity in question. It describes any quantity in terms of fundamental quantities.

Dimensions of a physical quantity are the powers to which the fundamental quantities must be raised. Table 4 shows some quantities with their dimensions.

Table 4: Some Physical quantities with their dimensions

Quantity	Dimension
Mass	M
Length	L
Time	T
Acceleration	LT^{-2}

We write [X] meaning “dimension of quantity X”

Activity 1.2

1. What are the dimensions of the following physical quantities:
 - a. Area of a triangle
 - b. Stress
 - c. Frequency
 - d. Impulse

Application of Dimensional analysis

Checking if an equation is consistent/valid

An equation is said to be dimensionally consistent if the SI unit(s) on the left side are the same as the SI units on the right of the equation

Example

Let us consider the following equation of motion with constant acceleration:

$$s = ut + \frac{1}{2}at^2$$

where:

s : displacement with dimension of L and SI unit of m

u : velocity with dimensions of LT^{-1} and SI unit of ms^{-1}

a : acceleration with dimensions of LT^{-2} and SI units of ms^{-2}

t : time with dimension of T and SI unit of s

Then, we rewrite the equation, all quantities replaced by their dimensions, we get:

$$s = ut + \frac{1}{2}at^2 \rightarrow L = LT^{-1} \times T + LT^{-2} \times T^2$$

$$L = L + L$$

$$L = L$$

We finally get the same dimension of length both sides of the equation. Therefore, the equation is dimensionally correct.

We also use dimensional analysis to derive an equation

If we know that a quantity is related in same way to other quantities, then it is possible to derive an equation expressing the relationship.

Example: Suppose we are told that the acceleration a of a particle moving with uniform speed v in a circle of radius r is proportional to some power of r , say r^n , and some power of v , say v^m . Determine the values of n and m and write the simplest form of an equation for the acceleration.

We write the relation as

$$a \propto r^n v^m$$

$$a = k r^n v^m$$

where k is the proportionality constant

Writing the equation dimensionally, we get

$$LT^{-2} = k[L]^n[LT^{-1}]^m$$

k is dimensionless, so we can take it out of the equation

$$LT^{-2} = [L]^n[LT^{-1}]^m$$

$$LT^{-2} = L^n L^m T^{-m}$$

$$LT^{-2} = L^{n+m} T^{-m}$$

equating the powers, we get

$$1 = n + m \text{ and } -2 = -m$$

from which, we get $m = 2$ and $n = -1$

Therefore, the derived equation is

$$a = k \frac{v^2}{r}$$

Activity 1.3

1. Given that, the time period T of a *simple pendulum* depends upon mass m of the bob, length l of the pendulum and acceleration due to gravity g at the place where the pendulum is suspended. Find an expression for the time period T in terms of m , l and g .

1.3.3 Decimal point, scientific notation, order of magnitude, precision, accuracy and order of magnitude

Decimal point: the number 7.32 has two decimal places and 1500 has no decimal point.

Scientific notation: this is writing numbers in powers of 10. In general any number in the form of $x \times 10^y$ is in scientific notation where x is any number such that $1 \leq x \leq 10$ and y is an exponent. E.g 7.41×10^3 is in scientific notation.

Example

Convert 6342147.14 to scientific notation

Solution

$$6.3421714 \times 10^6$$

When counting decimal places from right to left, the exponent is positive

Example

Write 0.0042 in scientific notation

Solution

$$4.2 \times 10^{-3}$$

When counting decimal places from left to right, the exponent is negative

Order of magnitude: it is a rough approximation of a quantity to the closest power of 10 e.g 8~10 this means 8 is of the order of 10

Examples

123~100, and 0.07~0.1

To find order of magnitude, round up or down to nearest power of 10 e.g $7.62 \times 10^4 \sim 10^5$, $3.2 \times 10^2 \sim 10^2$, $8.417 \times 10^{-6} \sim 10^{-5}$, $1.5 \times 10^{-8} \sim 10^{-8}$

Significant figures (s.f) are the number of digits in scientific notation e.g 4.71 has 3 s.f.

Uncertainty in measurement: in any measurement there is always an error, and this is called uncertainty.

Accuracy refers to closeness of a measured value to a standard or known value. For example, if in a lab you obtain a weight measurement of 3.2 kg for a given substance, but the actual or the known weight is 10 kg, then your measurement is not accurate. In this case your measurement is not close to the known value.

Precision refers to the closeness of two or more measurements to each other. For example, if you weigh a given substance five times and get 3.2 kg each time, then your measurement is very precise.

1.3.4 Summary

In this unit we have discussed units and measurements as fundamental concepts of physical quantities.

Physical quantity is any quantity that can be measured

Physical quantity has unit. Unit is a standard quantity from which other quantities are compared.

Basic quantities are not expressed in terms of other quantities

Derived quantities are expressed in terms of basic quantities by multiplication or division.

We use unit prefixes as shorthand in presenting large or small number of units

Dimensional analysis describes any quantity in terms of fundamental quantities. It is used for checking the validity of an equation. It also helps in derivation of an equation.

Decimal points, significant figures, scientific notation, order of magnitude, precision and accuracy are worth to be considered in describing a physical quantity.

1.3.5 End of unit 1 exercise

1. It is advantageous that fundamental standards, such as those for length and time, be accessible (easy to compare to), invariable (do not change), indestructible and reproducible. Discuss why these are advantageous and whether any of these criteria can be incompatible with others.
2. What are the merits and drawbacks of using a person's foot as a standard? Consider both (a) a particular person's foot (b) any person's foot
3. You measure the radius of a wheel to be 4.16 cm. If you multiply by 2 to get the diameter, should you write the result as 8 cm or as 8.32 cm? Justify your answer.
4. If an equation is dimensionally correct, does this mean that the equation must be true? If an equation is not dimensionally correct, does this mean that the equation is cannot be true?
5. The age of the universe is believed to be somewhere around 14 billion year. Write this in powers of ten in (a) years (b) months

6. Write the following numbers in full with a decimal number of zeros:
(a) 8.69×10^4 (b) 7.1×10^3 (c) 6.6×10^{-1}
7. How many significant figures do each of the following numbers have: (a) 214
(b) 81.60 (c) 7.03 (d) 0.03 (e) 0.0086 (f) 3236 (g) 8700
8. Express the following using the prefixes: (a) 10^6 volts, (b) 10^{-1} meters,
(c) 5×10^3 days (d) 8×10^2 bucks (e) 8×10^{-9} pieces
9. Write the following as full (decimal) numbers with standard units: (a) 86.6
mm (b) $35 \mu V$ (c) 860 mg (d) 600 picoseconds (e) 12.6 femtometers (f)
250 gigavolts
10. The volume of an object is 1000 m^3 . Express this volume in (a) cm^3 (b) km^3
11. Write dimensional formula for each of the following: (a) impulse (b)
frequency (c) power (d) heat capacity
12. Check whether the equation $\lambda = \frac{h}{mv}$ is dimensionally correct. λ is wavelength,
 h is plank's constant which has dimensions ML^2T^{-1} , m is mass and v is the
velocity.

Unit 2

Scalar and Vector quantities

2.0 Introduction

In unit 1, we discussed physical quantities in terms of base and derived quantities which are used to describe the physical quantities. In this unit we classify these physical quantities into scalar and vector quantities. We will particularly pay more attention to mathematical techniques of vector addition.

2.1 Objectives

By the end of this unit you should be to:

- Define scalar quantity and vector quantity
- Classify given quantities into scalar and vector quantities
- Represent vectors geometrically
- Resolve a given vector into its components

2.2 Key Terms

- Scalar quantity
- Vector quantity
- Resultant vector
- Vector resolution

2.3 Scalar and Vector Quantities

Scalar quantity is a physical quantity that is completely described by magnitude only, e.g mass, temperature, distance and speed. A scalar quantity tells you how much a given quantity is. For example, 20 Kg of meat is a scalar quantity. It is simply telling you how much meat is there. No direction is required for it to make a complete sense.

Vector quantity is a physical quantity that requires the specification of magnitude and direction e.g velocity, displacement and acceleration. A vector tells you how much is the quantity and which way. For example if you drive a bicycle East along the road at 5ms^{-1} . What you have here is a vector-velocity. The bicycle is moving at 5ms^{-1} (this is the magnitude) and you know where you are going –East (this is the direction).

Activity 2.1

1. Explain the difference between vector and scalar quantities
2. Give examples of physical quantities that are both base and scalar
3. Give examples of physical quantities that are both derived and vector

2.3.1 Vector Notation

Mathematical representation: vectors are represented by a symbol with an arrow pointing to the right above it e.g. $\vec{a}$, $\vec{v}$ and $\vec{F}$ represents vectors acceleration, velocity and force respectively.

Two vectors $\vec{OA}$ and $\vec{BC}$ are equal if they have the same magnitude and the same direction. This property allows us to translate a vector parallel to itself in a diagram without affecting the vector. See Figure 1.3.1, vector $\vec{OA}$ is translated to vector $\vec{BC}$.

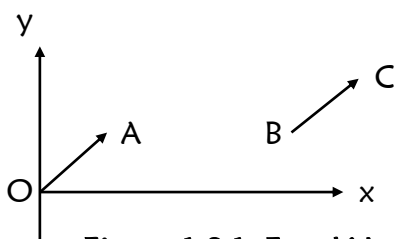


Figure 1.3.1: Equal Vectors

Graphical Representation: Vectors are drawn as arrows. An arrow has both a magnitude (how long it is) and a direction (the direction in which it points). The starting point of a vector is known as the tail and the end point is known as the head as shown in the Figure below:

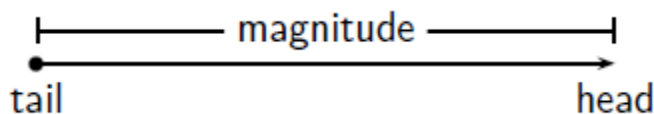
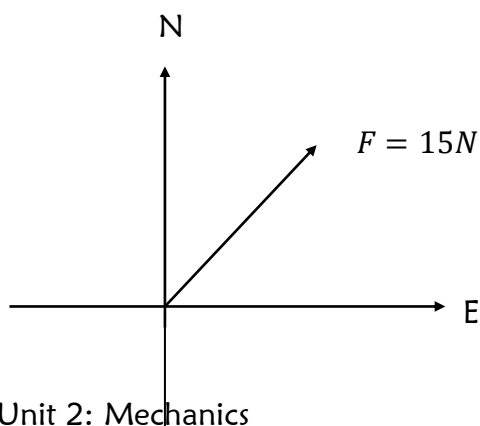


Figure 1.3.2: Parts of a vector

If you want to present a force of 15 N North East, you present it as follows:



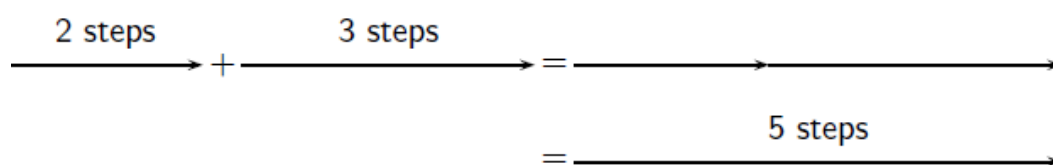
Activity 2.2

1. Represent the following vectors geometrically on a plane:

- i. Velocity, $v = 10\text{ms}^{-1}$ North
- ii. Velocity, $v = 10\text{ms}^{-1}$ North West
- iii. Acceleration $a = 5\text{ms}^{-2}$ at an angle of 300°

2.3.2 Adding vectors

When vectors are added, we need to add both a magnitude and a direction. For example, take 2 steps in the forward direction, stop and then take another 3 steps in the forward direction. The first 2 steps is a displacement vector and the second 3 steps is also a displacement vector. If we did not stop after the first 2 steps, we would have taken 5 steps in the forward direction in total. Therefore, if we add the displacement vectors for 2 steps and 3 steps, we should get a total of 5 steps in the forward direction. Graphically, this can be seen by first following the first vector two steps forward and then following the second one three steps forward:



The final answer when adding vectors is called the resultant. The resultant displacement in this case will be 5 steps forward.

Definition: Resultant of Vectors

The resultant of a number of vectors is the single vector whose effect is the same as the Individual vectors acting together

2.3.3 Techniques of Vector Addition

1. Graphical Techniques

Graphical techniques involve drawing accurate scale diagrams to denote individual vectors and their resultants.

We next discuss the two primary graphical techniques, the head-to-tail technique and the parallelogram method.

The Head-to-Tail Method

In describing the mathematical properties of vectors we used displacements and the head-to-tail graphical method of vector addition as an illustration. The head-to-tail method of graphically adding vectors is a standard method that must be understood.

Steps to follow: Head-to-Tail Method of Vector Addition

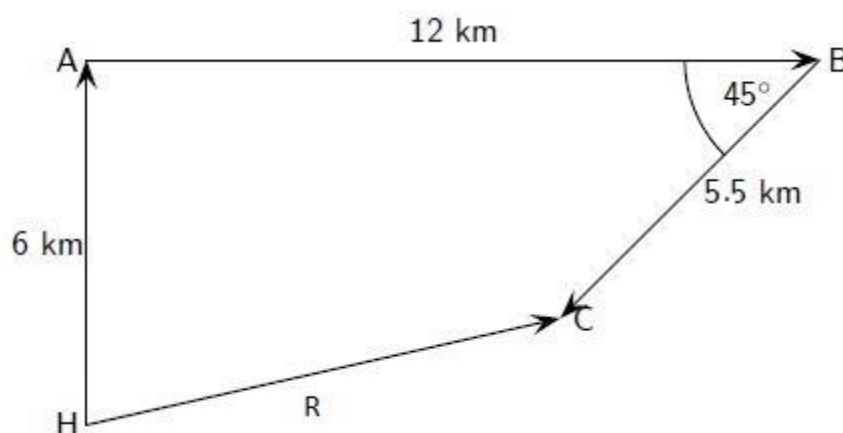
1. Choose a scale and include a reference direction.
2. Choose any of the vectors and draw it as an arrow in the correct direction and of the correct length – remember to put an arrowhead on the end to denote its direction.
3. Take the next vector and draw it as an arrow starting from the arrowhead of the first vector in the correct direction and of the correct length.
4. Continue until you have drawn each vector – each time starting from the head of the previous vector. In this way, the vectors to be added are drawn one after the other head-to-tail.
5. The resultant is then the vector drawn from the tail of the first vector to the head of the last. Its magnitude can be determined from the length of its arrow using the scale. Its direction too can be determined from the scale diagram.

Example 2.1

A ship leaves harbor H and sails 6 km north to port A. From here the ship travels 12 km east to port B, before sailing 5.5 km south-west to port C. Determine the ship's resultant displacement using the head-to-tail technique of vector addition.

Solution

Step1: Draw a rough sketch of the situation



Step 2 : Choose a scale and include a reference direction.

The choice of scale depends on the actual question – you should choose a scale such that your vector diagram fits the page.

It is clear from the rough sketch that choosing a scale where 1 cm represents 2 km (scale: 1 cm = 2 km) would be a good choice in this problem. The diagram will then take up a good fraction of an A4 page. We now start the accurate construction.

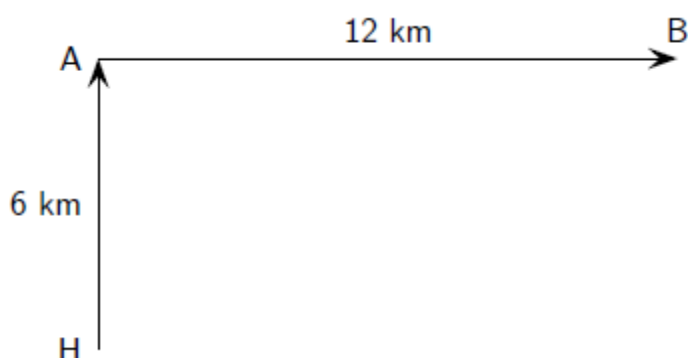
Step 3 : Choose any of the vectors to be summed and draw it as an arrow in the correct direction and of the correct length – remember to put an arrowhead on the end to denote its direction.

Starting at the harbour H we draw the first vector 3 cm long in the direction north.



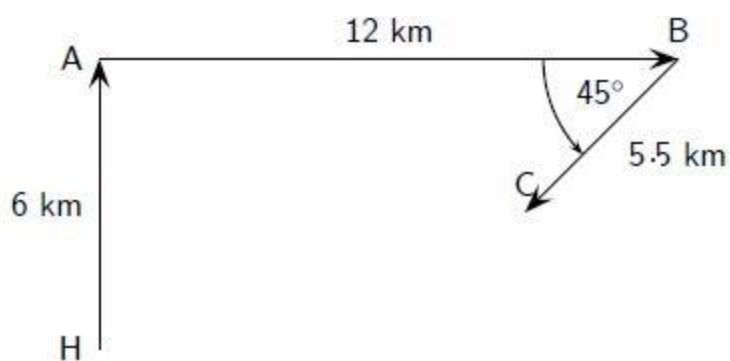
Step 4 : Take the next vector and draw it as an arrow starting from the head of the first vector in the correct direction and of the correct length.

Since the ship is now at port A we draw the second vector 6 cm long starting from point A in the direction east.



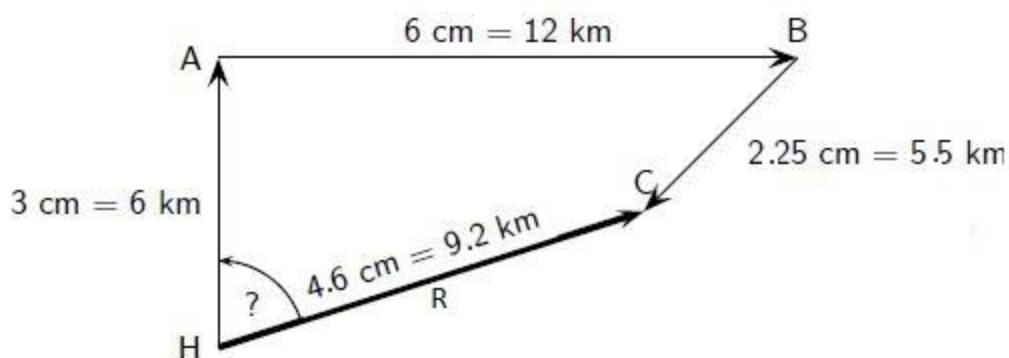
Step 5 : Take the next vector and draw it as an arrow starting from the head of the second vector in the correct direction and of the correct length.

Since the ship is now at port B we draw the third vector 2,25 cm long starting from this point in the direction south-west. A protractor is required to measure the angle of 45° .



Step 6 : The resultant is then the vector drawn from the tail of the first vector to the head of the last. Its magnitude can be determined from the length of its arrow using the scale. Its direction too can be determined from the scale diagram.

As a final step we draw the resultant displacement from the starting point (the Harbor H) to the end point (port C). We use a ruler to measure the length of this arrow and a protractor to determine its direction.



Step 7 : Apply the scale conversion.

We now use the scale to convert the length of the resultant in the scale diagram to the actual displacement in the problem. Since we have chosen a scale of 1 cm = 2 km in this problem the resultant has a magnitude of 9.2 km. The direction can be specified in terms of the angle measured either as 72.3° east of north or on a bearing of 072.3°.

Step 8 : Quote the final answer

The resultant displacement of the ship is 9.2 km on a bearing of 072.3°.

The Parallelogram Method

The parallelogram method is another graphical technique of finding the resultant of two vectors.

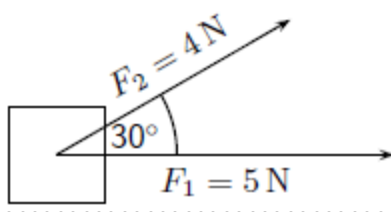
2. The Parallelogram Method

1. Choose a scale and a reference direction.
2. Choose either of the vectors to be added and draw it as an arrow of the correct length in the correct direction.
3. Draw the second vector as an arrow of the correct length in the correct direction from the tail of the first vector.
4. Complete the parallelogram formed by these two vectors.
5. The resultant is then the diagonal of the parallelogram. The magnitude can be determined from the length of its arrow using the scale. The direction too can be determined from the scale diagram.

Example 2.2 Parallelogram Method of Vector Addition

A force of $F_1 = 5\text{ N}$ is applied to a block in a horizontal direction.

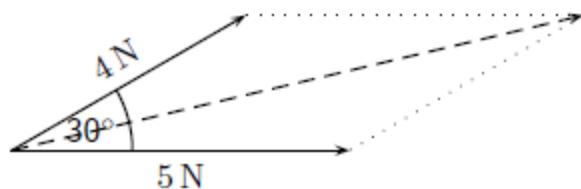
A second force $F_2 = 4\text{ N}$ is applied to the object at an angle of 30° above the horizontal.



Determine the resultant force acting on the block using the parallelogram method of accurate construction.

Solution

Step 1 : Firstly make a rough sketch of the vector diagram

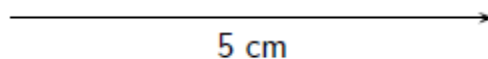


Step 2 : Choose a suitable scale

In this problem a scale of $1\text{ cm} = 1\text{ N}$ would be appropriate, since then the vector diagram would take up a reasonable fraction of the page. We can now begin the accurate scale diagram.

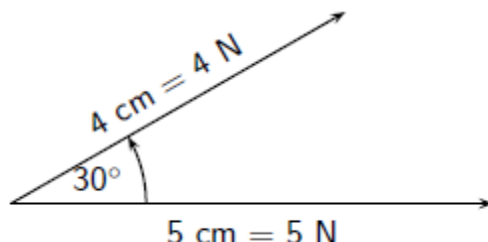
Step 3 : Draw the first scaled vector

Let us draw F_1 first. According to the scale it has length 5 cm.



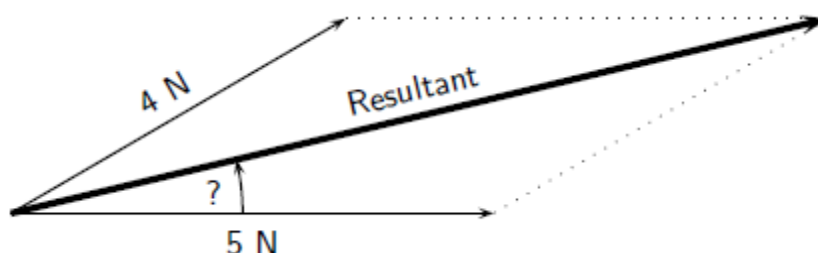
Step 4 : Draw the second scaled vector

Next we draw F_2 . According to the scale it has length 4 cm. We make use of a protractor to draw this vector at 30° to the horizontal.



Step 5 : Determine the resultant vector

Next we complete the parallelogram and draw the diagonal.



The resultant has a measured length of 8.7 cm.

Step 6 : Find the direction

We use a protractor to measure the angle between the horizontal and the resultant.

We get 13.3° .

Step 7 : Apply the scale conversion

Finally we use the scale to convert the measured length into the actual magnitude.

Since $1 \text{ cm} = 1 \text{ N}$, 8.7 cm represents 8.7 N. Therefore the resultant force is 8.7 N at 13.3° above the horizontal.

The parallelogram method is restricted to the addition of just two vectors. However, it is arguably the most intuitive way of adding two forces acting at a point.

3. Algebraic Addition and Subtraction of Vectors

Vectors in a Straight Line

Whenever you are faced with adding vectors acting in a straight line (i.e. some directed left and some right, or some acting up and others down) you can use a very simple algebraic technique:

Steps in Addition/Subtraction of Vectors in a Straight Line

1. Choose a positive direction. As an example, for situations involving displacements in the directions west and east, you might choose west as your positive direction. In that case, displacements east are negative.
2. Next simply add (or subtract) the vectors using the appropriate signs.
3. As a final step the direction of the resultant should be included in words (positive results are in the positive direction, while negative resultants are in the negative direction).

When vectors act at an angle, Simple geometric and trigonometric techniques can be used to find resultant vectors.

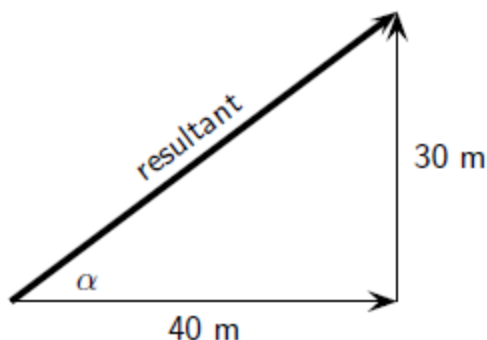
Example 2.3

A man walks 40 m East, then 30 m North. Calculate the man's resultant displacement.

Solution

Step 1: Draw a rough sketch

As before, the rough sketch looks as follows:



Step 2 : Determine the length of the resultant

Note that the triangle formed by his separate displacement vectors and his resultant displacement vector is a right-angle triangle. We can thus use the Theorem of Pythagoras to determine the length of the resultant. Let R represent the length of the resultant vector. Then:

$$R^2 = (40\text{m})^2 + (30\text{m})^2$$

$$R = \sqrt{(40\text{m})^2 + (30\text{m})^2}$$

$$\therefore R = 50\text{m}$$

Step 3 : Determine the direction of the resultant

Now we have the length of the resultant displacement vector but not yet its direction.

To determine its direction we calculate the angle α between the resultant displacement vector and East, by using simple trigonometry:

$$\tan\alpha = \frac{30\text{m}}{40\text{m}}$$

$$\alpha = \tan^{-1}(30\text{m}/40\text{m})$$

$$\alpha = 36.9^\circ$$

Step 4 : Quote the resultant

The resultant displacement is then 50 m at 36.9° North of East.

This is exactly the same answer we arrived at after drawing a scale diagram!

Activity 2.3

1. A man walks 40 m East, then 30 m North. Using Head-to-Tail Graphical Addition:
 - i. What was the total distance he walked?
 - ii. What is his resultant displacement?
2. A man walks from point A to point B which is 12 km away on a bearing of 45° . From point B the man walks a further 8 km east to point C. Calculate the resultant displacement using geometric and trigonometric techniques to find resultant vectors.

2.3.4 Vector Resolution

A vector can be resolved into x-component and y-component. The sum of these components is equal to the original resultant vector.

Vector resolution is the process of breaking a vector into its components. The components of a vector are also vectors and are perpendicular to each other. Vector resolution can be done anywhere in the Cartesian plane as long as the components of the vector are perpendicular to each other.

Consider vector **OA** shown in Figure 2.3 below which is acting at an angle of $\angle AOX$:

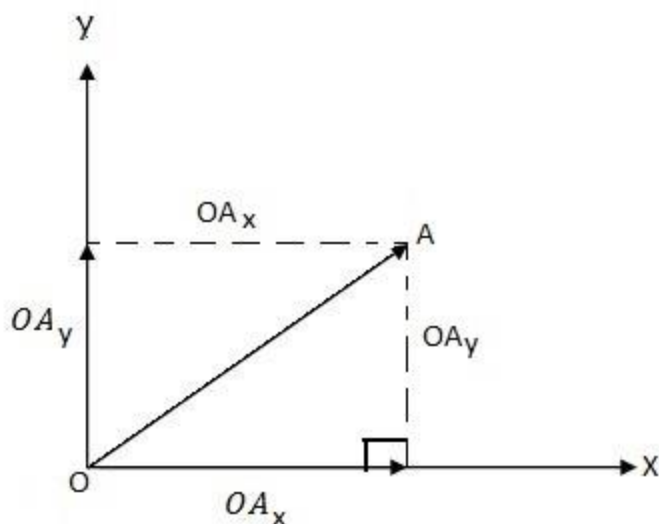


Figure 2.3 Components of Vector OA

In Figure 2.3 OA_x and OA_y are x-component and y-component of vector OA respectively.

Using trigonometry, you can get the components of the vector shown in Figure 2.3 by:

$$OA_x = OA \cos(\angle AOX)$$

and

$$OA_y = OA \sin(\angle AOX)$$

In general for any given vector Q at any angle of θ from X-axis, the x-component and y-components can be found as,

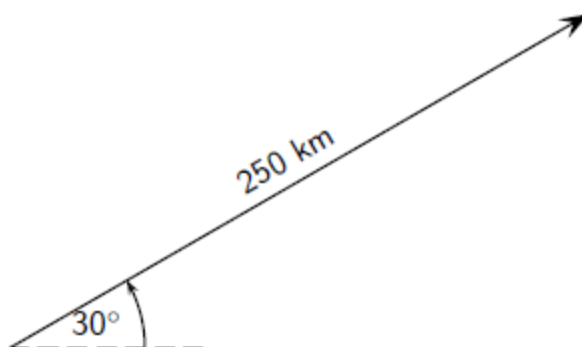
$$Q_x = Q \cos \theta \text{ and } Q_y = Q \sin \theta \text{ respectively}$$

Example 2.4

A motorist undergoes a displacement of 250 km in a direction 30° North of East.

Resolve this displacement into components in the directions north and east.

Solution

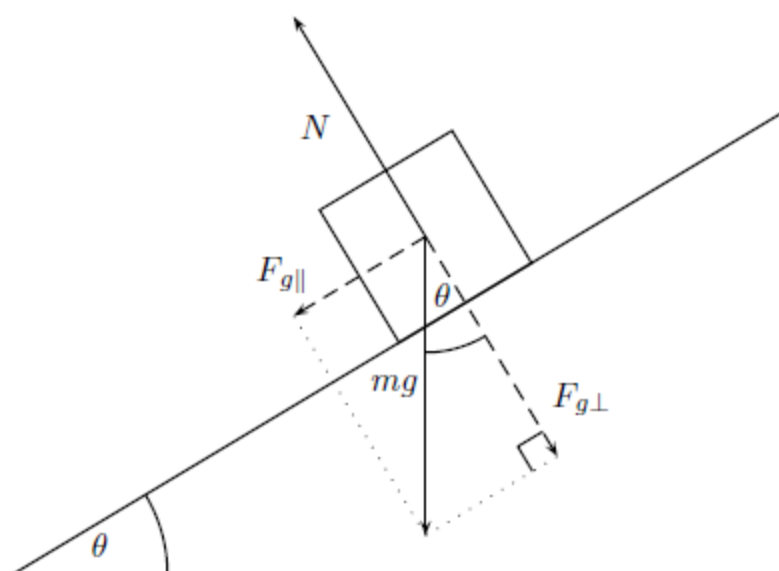


Then Eastern component is the same as x-component = $250 \cos 30^\circ = 125 \text{ km}$ and
 North component is the same as y-component = $250 \sin 30^\circ = 125 \text{ km}$

Example 2.5

As a further example of components let us consider a block of mass m placed on a frictionless surface inclined at some angle θ to the horizontal. The block will obviously slide down the incline, but what causes this motion?

The forces acting on the block are its weight, mg , and the normal force N exerted by the surface on the object and they are equal. These two forces are shown in the diagram below.



Resolving weight into its components, we get:

$$F_{g//} = mg \sin \theta$$

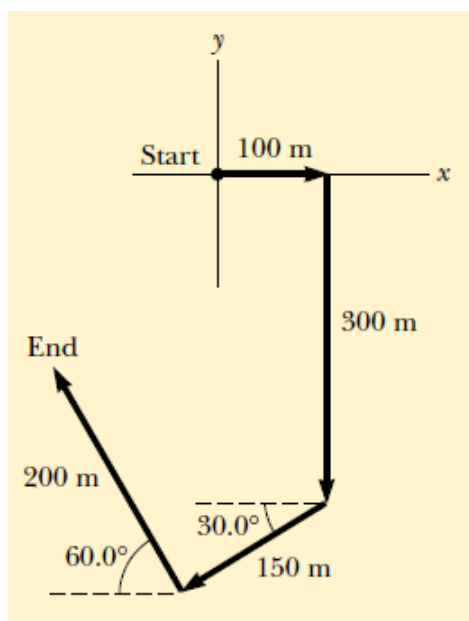
and

$$F_{g\perp} = mg \cos \theta$$

The component of the weight perpendicular to the slope $F_{g\perp}$ exactly balances the normal force N exerted by the surface. The parallel component, however, $F_{g//}$ is unbalanced and causes the block to slide down the slope.

Example 2.6

A person going for a walk follows the path shown in Figure below. The total trip consists of four straight-line paths. At the end of the walk, what is the person's resultant displacement measured from the starting point?

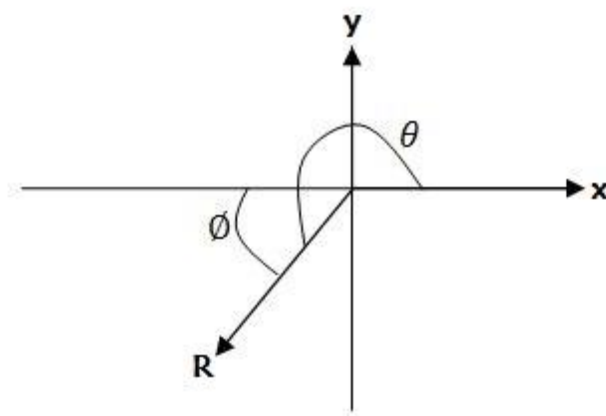


Solution

$$R_x = 100 - 150\cos 30^\circ - 200\cos 60^\circ = -129.90 \text{ m}$$

$$R_y = -300 - 150\sin 30^\circ + 200\sin 60^\circ = -201.79 \text{ m}$$

Then, the resultant displacement can be represented in the Figure below:



$$R = \sqrt{(-129.90)^2 + (-201.79)^2} = 239.99 \text{ m}$$

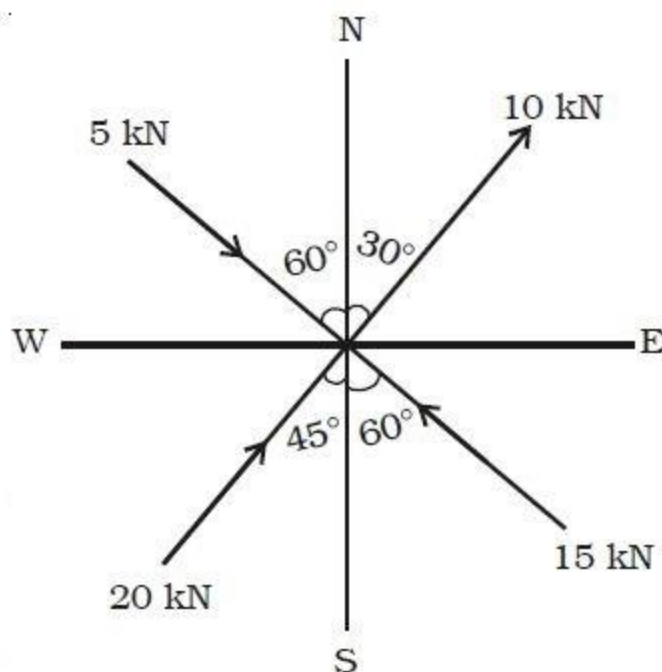
$$\phi = \tan^{-1}\left(\frac{-201.79}{-129.90}\right) = 57.23^\circ$$

$$\theta = 180^\circ + 57.23^\circ = 237.23^\circ$$

The resultant displacement is 239.99 m at an angle of 57.24° South West or at 237.23° .

Activity 2.4

Determine the magnitude and direction of the resultant of the following four forces acting at a point by vector resolution method.

**2.3.5 Summary**

A quantity that has magnitude only is called a scalar quantity. A quantity that has both magnitude and direction is called a vector quantity.

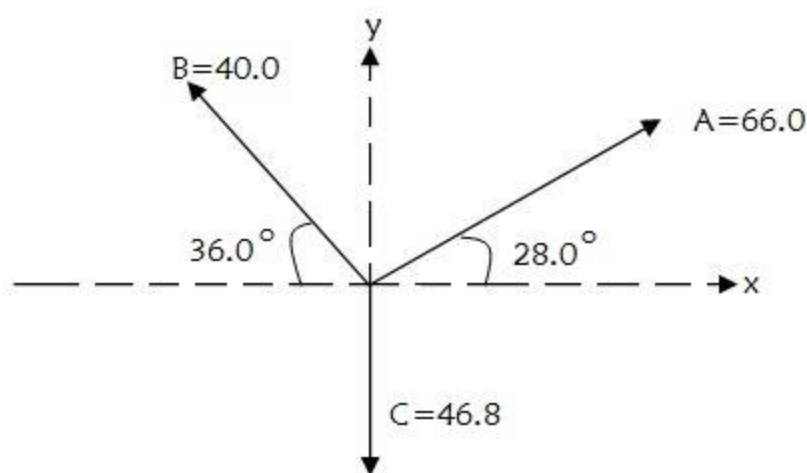
Addition of vectors can be done graphically by placing the tail of each successive arrow at the tip of the previous one. The resultant vector, is the arrow drawn from the tail of the first vector to the tip of the last vector. Two vectors can also be added using parallelogram method.

Vectors can be added more accurately by adding their components along chosen axis with the aid of trigonometric functions.

2.3.6 End of unit 2 Exercise

1. Does the odometer of a car measure a scalar or a vector quantity? What about the speedometer?
2. Two vectors are added together, and their vector sum is zero. What can you say about the magnitude of direction of the two vectors?

3. Can the displacement vector for an object moving in two dimensions ever be longer than the length of the path traveled by the object over the same time interval? Can it ever be less? Discuss.
4. Two vectors have magnitudes $v_1 = 3.5 \text{ km}$ and $v_2 = 4.0 \text{ km}$. What are the maximum and the minimum magnitudes of their vector sum?
5. An airplane is traveling 785 km/h in a direction 38.5° West of North.
 - a. Find the components of the velocity vector in the Northerly and Westerly directions.
 - b. How far North and how far West has the plane traveled after 3.00 h ?
6. A skier is accelerating down a 30.0° hill at 3.80 m/s^2 . What is the vertical component of her acceleration? How long will it take her to reach the bottom of the hill, assuming she starts from rest and accelerates uniformly if the elevation change is 335 m ?
7. Three vectors are shown in Figure below. Their magnitudes are given in arbitrary units. Determine the sum of the three vectors in terms of:
 - a. Components
 - b. Magnitude and angle with respect to x-axis



Unit 3

Motion in one Dimension

3.0 Introduction

This chapter is about how things move in a straight line (moving in one dimension) without the cause of the motion. This unit will help you to understand how any object moves in a straight line, for example a car on the freeway, a soccer ball being kicked towards the goal or your dog chasing the neighbour's cat. Three basic ideas about what it means when something is moving will be discussed. These three ideas describe different parts of exactly how an object moves and these are: (i) position or displacement which tells you exactly where the object is, (ii) speed or velocity which tells you exactly how fast the object's position is changing (how fast the object is moving) and (iii) acceleration which tells you exactly how fast the object's velocity is changing.

In this unit, you will also learn how to use position, displacement, speed, velocity and acceleration to describe the motion of simple objects. You will learn how to read and draw motion graphs. You will further learn about the equations that can be used to describe motion and how to apply these equations to objects moving in one dimension.

3.1 Objectives

By the end of this unit you should be able to:

- State the difference between distance and displacement, speed and velocity
- Explain the difference between average velocity and instantaneous velocity, average acceleration and instantaneous acceleration
- Draw distance-time graph, velocity-time graph
- Derive the equations of motion for uniform acceleration
- Solve problems of linear motion by applying the derived equations
- Analyse projectile motion

3.2 Key Terms

Frame of reference

Position

Displacement

Velocity

Speed

Acceleration

3.3 Kinematics in one dimension

Kinematics is the study of motion of an object. In this unit, you are going to learn motion of an object without considering its cause. The motion of an object can be described by displacement, velocity, acceleration and time. Recall that displacement, velocity and acceleration are vectors. They have both magnitudes and directions. This means that vector methods are employed when dealing with displacement, velocity and acceleration. To do so, you need a frame of reference.

3.3.1 Frame of Reference

A frame of reference is a reference point combined with a set of direction. For example, a boy is standing still inside a train as it pulls out of a station. You are standing on the platform watching the train move from left to right. To you it looks as if the boy is moving from left to right, because relative to where you are standing (the platform), he is moving. According to the boy, and his frame of reference (the train), he is not moving.

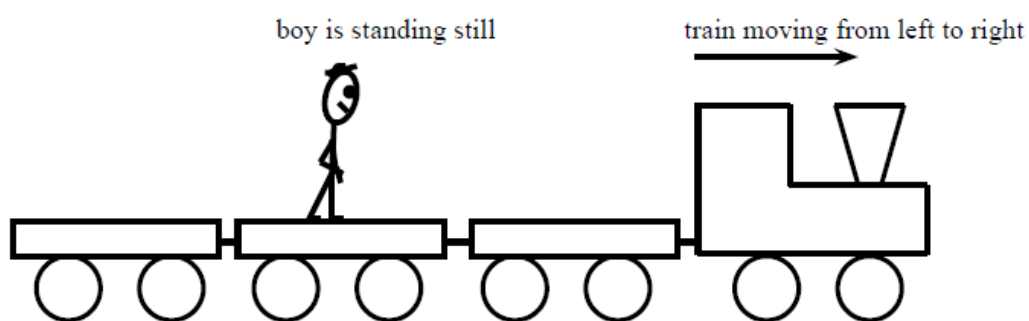


Figure 2.3.1 Frame of Reference

From your frame of reference the boy is moving from left to right.

Displacement is distance covered in a specified direction. It is measured in meters. E.g. 50 m West. Displacement is commonly denoted by letter, s . Figure 3.3.1 Illustrates displacement.

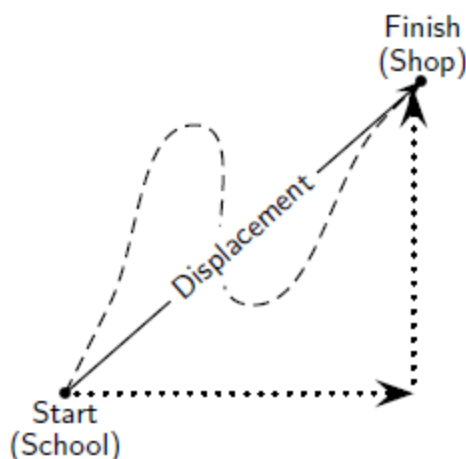


Figure 3.3.1 Illustrates displacement

From Figure 3.3.1, if you walk from school to the shop following the dotted path, then the displacement is the straight line joining the two points from school to the shop. The length of the dotted line gives you the distance.

Distance is the total length between two points. The magnitude of displacement gives you a distance. Distance is also measured in meters, m.

Velocity is the time rate of change of displacement (change in displacement divide by time). It is measured in meters per second (m/s). It is commonly denoted by u for initial velocity and v for final velocity.

Instantaneous velocity is the time rate of change of displacement at an instant time, t . (Change of displacement per infinitesimal time interval, limit of displacement as time approaches to zero i.e

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t}$$

Average velocity is change of displacement per unit time (total displacement divide by total time taken)

Acceleration a , is the time rate of change of velocity. It is measured in (ms^{-2}).

Average acceleration is change of velocity per unit time (total velocity divide by total time taken)

Instantaneous acceleration is the time rate of change of displacement at an instant time, t . (Change of velocity per infinitesimal time or limit of change of velocity as time approaches to zero i.e

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}$$

3.3.2 Motion graphs

Motion can be presented by a displacement-time graph and velocity-time graph.

3.3.2.1 Displacement-Time graph

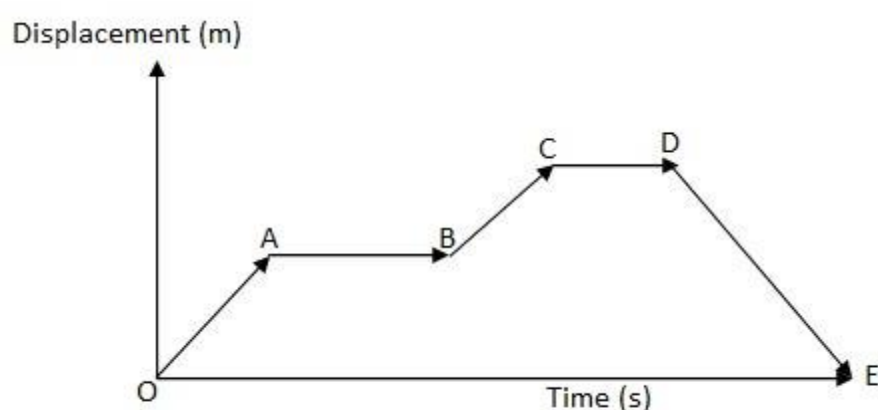


Figure 3.3.3. Displacement-Time graph

In a displacement time graph, the gradient (slope) of the graph gives you the velocity.

From figure 3.3.3, The object is starting from rest and moving with constant velocity between O and A and its velocity is the slope of line segment OA. Between A and B, the object is not moving and its velocity is zero. The motion of an object between B and C and C and D are the same as OA and AC respectively. Between D and E the slope is negative meaning that its velocity is negative. The object is moving back to the original position.

3.3.2.2 Velocity-Time graph

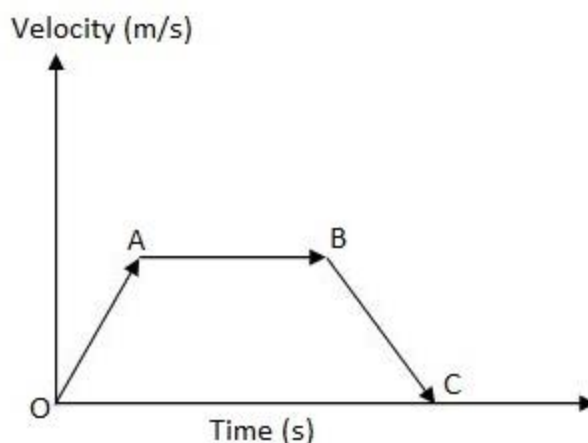


Figure 3.3.4

The slope of Velocity-Time graph gives you acceleration.

In Figure 3.3.4, the object moves with uniform acceleration between OA. This means that its velocity increases uniformly. Between A and B, the object moves with zero acceleration since its velocity is not changing (it is constant). Between B and C, the slope is negative. The object is moving with constant negative acceleration (Deceleration). The velocity between BC is decreasing uniformly.

3.3.3 Equations of Uniform motion (Motion with constant acceleration)

Consider an object which is initially moving at velocity u when you first see it (at $t = 0$). Then during time interval, t , the object increases its velocity uniformly to final value, v . During the same interval its displacement increases from zero to s .

The object's motion is represented by the graph in Figure 3.3.5 below:

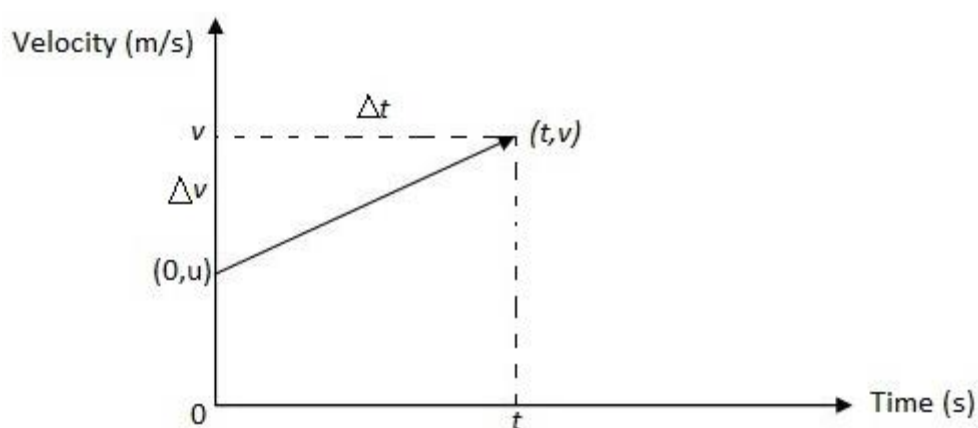


Figure 3.3.5

Recall that acceleration is equal to the gradient of velocity-time graph.

from Figure 3.3.5

$$\text{acceleration} = \frac{\text{Final velocity} - \text{Initial velocity}}{\text{time interval}}$$

$$a = \frac{\Delta v}{\Delta t}$$

$$a = \frac{v - u}{t}$$

$$\therefore v = u + at \quad (3.01)$$

This is the first equation of motion.

The area under velocity-time graph gives you displacement covered by an object during the time interval, t . From the graph in Figure 3.3.5, displacement (s) is equal to the area under the trapezium. Therefore,

$$s = 1/2(u + v)t$$

$$\therefore s = \frac{u+v}{2} t \quad (3.02)$$

This is the second equation of motion.

Substituting equation 3.3.1, into equation 2.3.2 for v , we have:

$$s = \frac{(u + at + u)t}{2}$$

This reduces to:

$$s = ut + \frac{1}{2}at^2 \quad (3.03)$$

This is the third equation of motion.

Making t the subject of the formula in equation 3.3.1 you get:

$$t = \frac{v-u}{a} \quad (3.04)$$

Substituting equation 3.04 into equation 3.02 for t , you get the last equation of motion:

$$s = \frac{v + u}{2} \times \frac{v - u}{a}$$

$$s = \frac{v^2 - u^2}{2a}$$

$$\therefore v^2 = u^2 + 2as \quad (3.05)$$

You will be using these equations to solve problems of motion with constant velocity. For free-fall motion i.e. motion under the influence of gravity, you substitute acceleration, a , with acceleration due to gravity, g .

Example 3.1

A race car starting from rest accelerates at a rate of 5.00 m/s^2 . What is the velocity of the car after it has travelled 100 ft? Given that $1 \text{ m} = 3.281 \text{ ft}$.

Solution

$$1\text{m}=3.281 \text{ ft} \rightarrow 1\text{ft}=0.305 \text{ m}$$

$$\text{Thus, } 100 \text{ ft} = 30.50 \text{ m}$$

We have been given

$$U=0 \text{ m/s, } a=+5 \text{ m/s}^2, x=30.5 \text{ m, } v=?$$

We apply the following equation:

$$v^2 = u^2 + 2a\Delta x$$

$$v^2 = 0^2 + 2(5\text{m/s}^2)(30.5 \text{ m})$$

$$v = \mp\sqrt{305 \text{ m}^2/\text{s}^2} = \mp 17.5 \text{ m/s}$$

The car is travelling to the right, we choose $+17.5 \text{ m/s}$ as the correct solution for final velocity v .

Example 3.2

A car travelling at a constant speed of 30.0 m/s passes a trooper hidden behind a billboard. One second after the speeding car passes the billboard, the trooper sets off in chase with a constant acceleration of 3.00 m/s^2 . How long does it take the trooper to overtake the speeding car?

Solution

The car is travelling at a constant speed, therefore, its acceleration is zero and

$$\Delta x = v_i t + \frac{1}{2}at^2$$

$$x_c = x_i + v_i t + \frac{1}{2}at^2$$

$$x_c = 30 + 30t$$

For the trooper who starts from rest i.e $t_i = 0$, $x_i = 0$, $v_i = 0$ and $a = 3\text{m/s}^2$, we have

$$x_T = x_i + v_i t + \frac{1}{2}at^2$$

$$x_T = \frac{1}{2}(3.00 \text{ m/s}^2)t^2$$

The trooper overtakes the car at the instant that $x_c = x_T$

$$30 + 30t = \frac{1}{2}(3.00)t^2$$

This reduces to:

$$t^2 - 20t - 20 = 0$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$t = \frac{-(-20) \pm \sqrt{(-20)^2 - 4 \times 1 \times (-20)}}{2 \times 1}$$

From which we get:

$$t = 21.0\text{s} \text{ or } t = -0.95\text{s}$$

Time is never negative. Therefore, the trooper overtakes the car at $t=21.0 \text{ s}$.

3.3.4 Freely falling object

A freely falling object is the one moving under the influence of gravity only regardless of its initial motion.

Objects thrown upward or downward and those released from rest are all falling freely once they are released.

Once they are in free fall, all objects have acceleration downward, which is free-fall acceleration g .

The equations describing free fall motion are the same as those of an object moving with constant acceleration along x-axis with the substitutions $x = y$ and $a = g$.

For a downward motion $a = g$, because the object moves in the direction of the gravity.

For upward motion $a = -g$, since the object is moves against the gravity.

For a freely falling object, $a = g$ and $v_i = 0$, since it starts from rest.

Example 3.3

A golf ball is released from rest at top of a very tall building. Neglecting air resistance, calculate the position and velocity of the ball after 1.00 s.

Solution

Taking the downward motion to be positive we have:

$$y_i = 0 \text{ at } t = 0, v_i = 0, a = g = 9.8 \text{ m/s}^2$$

$$\Delta y = v_i t + \frac{1}{2} a t^2$$

$$y = \frac{1}{2} (9.8) (1.00)^2$$

$$y = +4.90 \text{ m}$$

$$v_f = v_i + at$$

$$v_f = 0 + 9.8 \text{ m/s}^2 (1.00 \text{ s})$$

$$\therefore v_f = 9.8 \text{ m/s}$$

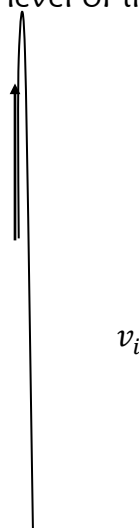
Example 3.4

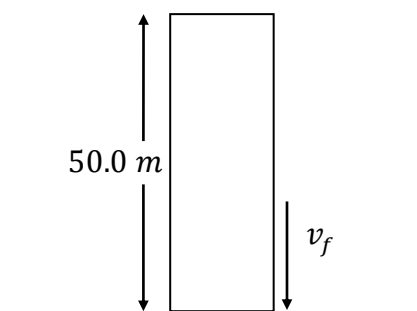
A stone is thrown from the top of a building with an initial velocity of 20 m/s straight upward. The building is 50.0 m high, and the stone just misses the edge of the roof on its way down. Determine:

- The time needed for the stone to reach its maximum height
- The maximum height
- The time needed for the stone to return to the level of the thrower
- The velocity of the stone at this instant
- The velocity of the stone at $t=5.00 \text{ s}$.

Solution

Sketch





- (a) Taking the upward direction to be negative and downward to be positive and noticing that at maximum height $v_f = 0$, we get:

$$v_f = v_i + at$$

$$-20.0 \text{ m/s} + 9.8 \text{ m/s}^2 t = 0$$

$$t = \frac{20.0}{9.8} \text{ s} = 2.04 \text{ s}$$

- (b) To find the maximum height we use:

$$y = v_i t + \frac{1}{2} a t^2$$

$$y_{\max} = \left(-20.0 \frac{\text{m}}{\text{s}}\right)(2.04 \text{ s}) + \frac{1}{2} \left(\frac{9.80 \text{ m}}{\text{s}^2}\right)((2.04 \text{ s})^2) = -20.4 \text{ m}$$

- (c) When the stone is back, $\Delta y = 0$, then,

$$\Delta y = -v_i t + \frac{1}{2} g t^2$$

$$0 = -20.0 t + \frac{1}{2} (9.8 \text{ m/s}^2) t^2$$

Solving for t , we get $t = 0$ or $t = 4.08 \text{ s}$

$t = 0$ Corresponds to the time at which the stone starts its motion. $t = 4.08$ is the time taken for the ball to return to the level of the thrower.

- (d) The velocity of the ball as it reaches back to the level of the thrower:

$$v_f = v_i + at$$

$$v_f = -20.0 \text{ m/s} + 9.8 \text{ m/s}^2 (4.08 \text{ s}) = +20.0 \text{ m/s}$$

Notice that the velocity of the stone when it arrives back at its original height is equal in magnitude to the stone's initial velocity but opposite in direction.

- (e) To get the velocity and position of the stone at $t=5.00 \text{ s}$

$$v_f = v_i + at$$

Taking the upward motion to be negative and downward motion to be positive, we have:

$$v_f = -20.0 \text{ m/s} + (9.8 \text{ m/s}^2)(5.00 \text{ s}) = 29.0 \text{ m/s}$$

We now find the position by

$$\Delta y = (-20.0 \text{ m/s})(5 \text{ s}) + \frac{1}{2}(9.80 \text{ m/s}^2)(5.00 \text{ s})^2 = 22.5 \text{ m}$$

Activity 3.1

1. In example 2.4 find the velocity of the stone just before it hits the ground.
2. An airplane accelerates down a runway at 3.20 m/s^2 for 32.8 s until it finally lifts off the ground. Determine the distance travelled before taking off.
3. A car starts from rest and accelerates uniformly over a time of 5.21 s for a distance of 100 m . Determine the acceleration of the car.
4. The driver of a car travelling at 72 km/h observes the light 300 m ahead of turning red. The traffic light is timed to remain red for 20 s before it turns green. If the motorist wishes to pass the light without stopping to wait for it to turn green, determine:
 - i. The required uniform acceleration of the car
 - ii. The speed with which the motorist crosses the traffic light
5. A stone is dropped from the top of the tower 50 m high. At the same time another stone is thrown up from the foot of the tower with a velocity of 25 ms^{-1} . At what distance from the top and after how much time the stones cross each other?

3.3.5 Unit 3 Summary

Kinematics deals with the description of object's motion. The description of the motion any object must always be given relative reference frame.

The displacement of an object is the change in position of the object

Average speed is the distance travelled divided by the elapsed time. An object's velocity over a particular time interval Δt is the displacement Δx divided by Δt .

Instantaneous velocity, whose magnitude is the average velocity taken over an infinitesimal short time interval

Acceleration is the change of velocity per unit time.

Instantaneous acceleration is the average acceleration taken over infinitesimal time interval.

If an object moves in a straight line with constant acceleration, the velocity v and position Δx are related to the acceleration a , the elapse time t and the initial velocity u by the following equations:

$$v_{av} = \frac{v+u}{2} \quad (1)$$

$$v = u + at \quad (2)$$

$$\Delta x = ut + \frac{1}{2}at^2 \quad (3)$$

$$v^2 = u^2 + 2a\Delta x \quad (4)$$

Objects that move vertically near the surface of the Earth, either falling or having been thrown up or down, move with constant downward acceleration due to gravity with magnitude $g = 9.80 \text{ m/s}^2$, if air resistance is ignored.

3.3.6 End of unit 3 Exercise

1. A race car accelerates uniformly from 18.5 m/s to 46.1 m/s in 2.47 seconds. Determine the acceleration of the car and the distance travelled.
2. A feather is dropped on the moon from a height of 1.40 meters. The acceleration of gravity on the moon is 1.67 m/s^2 . Determine the time for the feather to fall to the surface of the moon.
3. Rocket-powered sleds are used to test the human response to acceleration. If a rocket-powered sled is accelerated to a speed of 444 m/s in 1.83 seconds, then what is the acceleration and what is the distance that the sled travels?
4. A bike accelerates uniformly from rest to a speed of 7.10 m/s over a distance of 35.4 m. Determine the acceleration of the bike.
5. A kangaroo is capable of jumping to a height of 2.62 m. Determine the takeoff speed of the kangaroo.
6. If Michael Jordan has a vertical leap of 1.29 m, then what is his takeoff speed and his hang time (total time to move upwards to the peak and then return to the ground)?
7. An 18 year old runner complete a 10.0 km course with an average speed of 4.39 m/s. a 50 year old runner can cover the same distance with an average speed of 4.27 m/s. How much later (in seconds) the younger runner starts the race in order to finish the course at the same time as the older runner?

8. The space Shuttle travels at a speed of about $7.6 \times 10^3 \text{ m/s}$. The blink of an astronaut's eye lasts about 110 ms. How many football fields (length = 91.4 m) does the Shuttle cover in the blink of an eye?
9. A tourist in Nyika national park being chased by an angry elephant is running in a straight line toward his car at a speed of 4.0 m/s. The car is at a distance d away. The elephant is 26 m behind the tourist and running at 6.0 m/s. The tourist reaches the car safely. What is the maximum possible value for d ?
10. The left ventricle of the heart accelerates blood from rest to a velocity of +26 cm/s.
 - a. If the displacement of the blood is +2.0 cm, determine its acceleration (in cm/s^2)
 - b. How much time does the blood take to reach its final velocity?
11. A dog running at 10 m/s is 30 m behind a rabbit moving at 5 m/s. When will the dog catch up with the rabbit assuming both their velocities remain constant during the chase?
12. A small rocket is launched vertically from rest and reaches an altitude of h . the acceleration is constant at 11.5 m/s^2 until the fuel burns out after 5.8 s. Assuming that from the point of burn-out onwards the rocket is travelling freely vertically upwards with constant deceleration $g = 9.8 \text{ m/s}^2$, find the value of h .

Unit 4

Motion in two dimensions

4.0 Introduction

In our discussion of one dimensional motion in Unit 3, we used the concept of vectors only to a limited extent. Positive sign or negative sign, up or down, North or South, East or West can be used to indicate direction of an object moving in one dimension. However these methods of indicating direction cannot work with an object moving in two dimensions. In this unit we will employ vector manipulation techniques we learned in unit 2 such as vector resolution when dealing with motion in two dimensions. We will apply these mathematical tools to special two dimensional motions called projectile and to the understanding of relative motion. As it was in chapter 3 we discuss motion in two dimensions in this unit, without considering the cause of the motion.

4.1 Objectives

By the end of this unit you should be able to:

- Define projectile motion
-

4.2 Key words

Projectile

Trajectory

Angle of projection

Velocity of projection

Time of flight

Range

4.3.1 Displacement, Velocity and Acceleration in Two dimensions

Consider an object moving in space as shown in Figure 3.1.

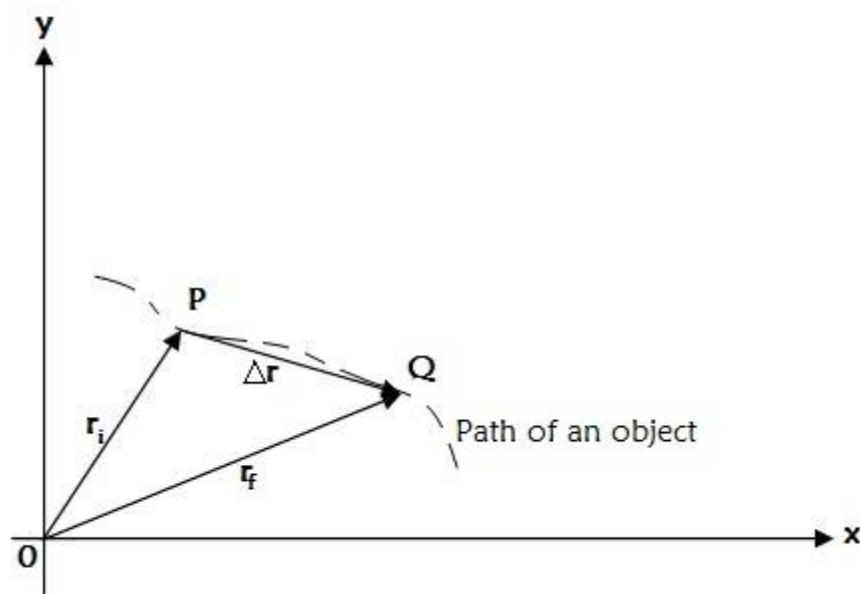


Figure 4.1 Displacement of an object in two dimensions

The object moves from initial point P at time t_i to point Q at time t_f then:

An object's displacement in two dimensions is defined as the change in its position vector given by:

$$\Delta \mathbf{r} = \mathbf{r}_f - \mathbf{r}_i \quad (4.01)$$

Its SI unit: meter (m)

We can now present the generalisations of the definitions of velocity and acceleration given in unit 3 as follows:

An object's average velocity during time interval Δt is its displacement divide by Δt :

$$\mathbf{v}_{av} = \frac{\Delta \mathbf{r}}{\Delta t} \quad (4.02)$$

Its SI unit: meter per second (m/s)

An object's instantaneous velocity is the limit of its average velocity as Δt goes to zero:

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{r}}{\Delta t} \quad (4.03)$$

Its SI unit: meter per second squared (m/s)

The direction of the instantaneous velocity is along the line that is tangent to the object's path and in the direction of motion.

An object's average acceleration is the change in its velocity divide by time interval:

$$\mathbf{a}_{av} = \frac{\Delta \mathbf{v}}{\Delta t} \quad (4.04)$$

SI unit: meter per second squared (m/s^2)

An object's instantaneous acceleration is the limit of its average acceleration as time interval goes to zero:

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{v}}{\Delta t} \quad (4.05)$$

SI unit: meter per second squared (m/s^2)

4.3.2 Projectile motion

Projectile motion is a good example of motion of an object moving in two dimensions. A projectile moves in both x-and y-directions simultaneously.

A body thrown with some initial velocity and then allowed to move under the action of gravity alone is known as a projectile.

The projectile moves in a path, which can be considered as a part of parabola. Such a motion is known as **projectile motion**.

Some examples of projectiles are:

- (i) a bomb thrown from an aero plane
- (ii) a javelin or a shot-put thrown by an athlete
- (iii) motion of a ball hit by a cricket bat

The different types of projectiles are shown in Fig. 4.01.

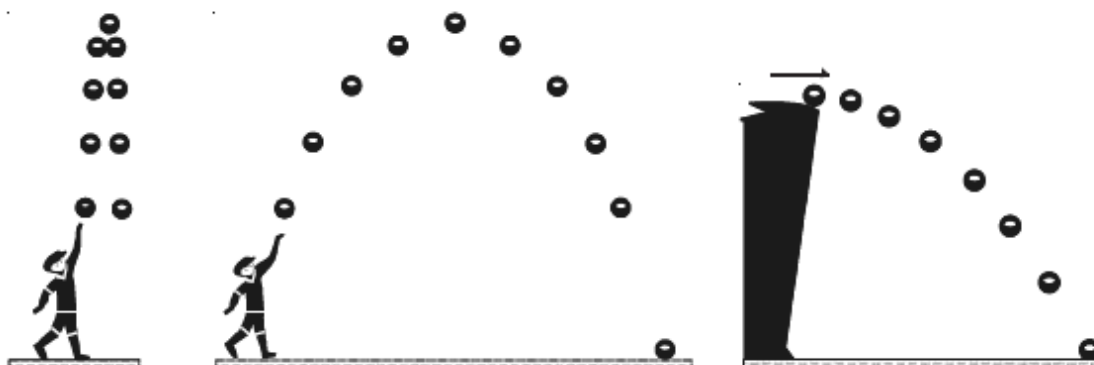


Figure 4.01 Types of projectile motion

A body can be projected in two ways:

- i. It can be projected horizontally from a certain height.
- ii. It can be thrown from the ground in a direction inclined to it.

The projectiles undergo a vertical motion as well as horizontal motion. The two components of the projectile motion are

- i. vertical component and
- ii. Horizontal component

These two components of projectile are perpendicular to each other and are independent of each other

4.3.2.1 Parts of a Projectile motion:

- i. **Angle of projection**

The angle between the initial direction of projection and the horizontal direction through the point of projection is called the angle of projection.

- ii. **Velocity of projection**

The velocity with which the body is projected is known as velocity of projection.

- iii. **Range**

Range of a projectile is the horizontal distance between the point of projection and the point where the projectile hits the ground.

- iv. **Trajectory**

The path described by the projectile is called the trajectory.

- v. **Time of flight**

Time of flight is the total time taken by the projectile from the instant of projection till it strikes the ground

4.3.2.2 Motion of a projectile projected at an angle with the horizontal (oblique projection)

Consider a body projected from a point O on the surface of the

Earth with an initial velocity u at an angle θ with the horizontal as shown in Fig. 4.02

The velocity u can be resolved into two components

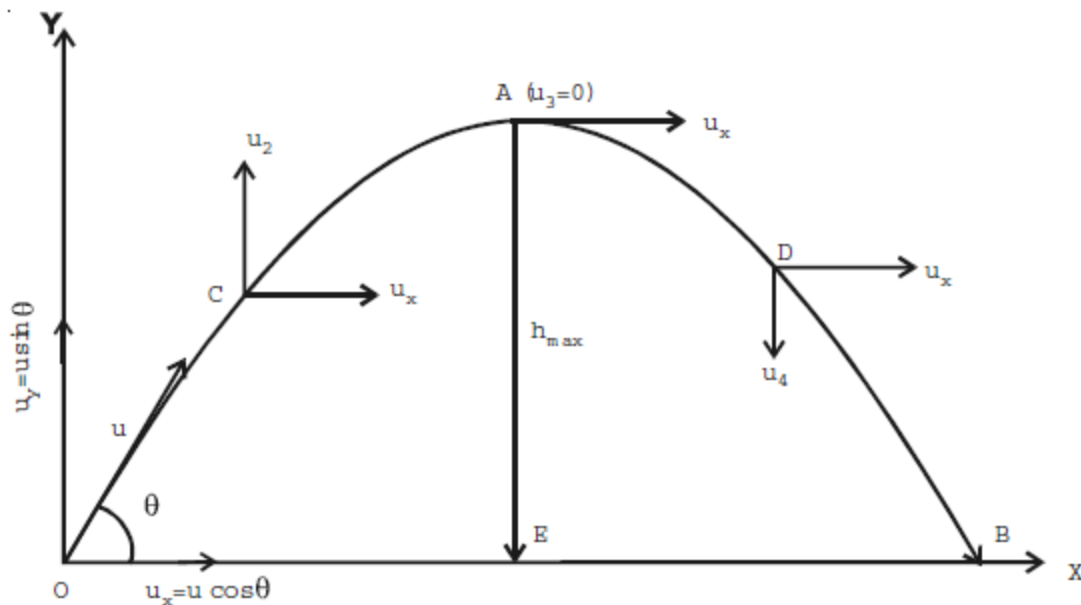


Figure 4.02

(i) $u_x = u \cos \theta$, along the horizontal direction OX and

(ii) $u_y = u \sin \theta$, along the vertical direction OY

The horizontal velocity u_x of the object shall remain constant as no acceleration is acting in the horizontal direction. But the vertical component u_y of the object continuously decreases due to the effect of the gravity and it becomes zero when the body is at the highest point of its path. After this, the vertical component u_y is directed downwards and increases with time till the body strikes the ground at B.

Path of the projectile:

Let t_1 be the time taken by the projectile to reach the point C from the instant of projection.

Horizontal distance travelled by the projectile in time t_1 is,

$$\begin{aligned} x &= u_x t \\ x &= (u \cos \theta) t \rightarrow t = \frac{x}{u \cos \theta} \end{aligned} \quad (4.06)$$

The vertical distance traveled by the projectile, y at time t, is given by

$$\begin{aligned} y &= u_y t - \frac{1}{2} g t^2 \\ y &= (u \sin \theta) t - \frac{1}{2} g t^2 \end{aligned} \quad (4.07)$$

Substituting equation 3.6 into equation 3.7 for time t, you have:

$$\begin{aligned} y &= (u \sin \theta) t - \frac{1}{2} g t^2 \\ y &= (u \sin \theta) \left(\frac{x}{u \cos \theta} \right) - \frac{1}{2} g \left(\frac{x}{u \cos \theta} \right)^2 \\ y &= (\tan \theta) x - \frac{g}{2(u \cos \theta)^2} x^2 \end{aligned} \quad (4.08)$$

Equation 4.3.8 describes the path taken by a projectile as parabolic (quadratic) in nature which can be rewritten as:

$y = ax^2 + bx + c$ where a , b and c are constants such that $a = -\frac{g}{2(ucos\theta)^2}$, $b = tan\theta$ and $c = 0$

Resultant velocity of the projectile at any instant t

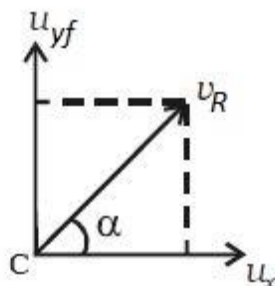


Figure 4.03

At any time, t , $u_x = ucos\theta$ and from the equation of motion $u_{yf} = usin\theta - gt$. The resultant is found by applying Pythagoras theorem:

$$u_R = \sqrt{u_x^2 + u_{yf}^2}$$

$$u_R = \sqrt{(ucos\alpha)^2 + (usin\alpha - gt)^2} \quad (4.09)$$

The direction of the resultant velocity can be found by:

$$tan\alpha = \frac{u_{yf}}{u_x}$$

$$\alpha = tan^{-1}((usin\alpha - gt) / (ucos\alpha)) \quad (4.10)$$

where α is the angle made by u_R with the horizontal line as shown in Figure 3.6

Maximum height reached by the projectile

The maximum vertical displacement produced by the projectile is known as the maximum height reached by the projectile. In Figure 4.02, EA is the maximum height attained by the projectile. It is represented as h_{max} .

At O, $u_{yi} = usin\theta$ and $u_{yf} = 0$

Using equation 1.3.5 which is $u_{yf}^2 = u_{yi}^2 - 2gs$ substituting appropriate variables, you get $0 = (usin\theta)^2 - 2gh_{max}$ and making h_{max} the subject of the formular, you finally get the maximum height of projectile as:

$$h_{max} = \frac{u^2 cos^2 \theta}{2g} \quad (4.11)$$

Time taken to attain maximum height (t_{max})

You are now familiar with the fact that at O, $u_{yi} = usin\theta$ and $u_{yf} = 0$

Applying the equation of motion ($u_{yf} = u_{yi} - gt$) with proper substitutions, you get:

$$0 = u \sin \theta - gt \rightarrow t_{\max} = \frac{u \sin \theta}{g}$$

$\therefore$ the time taken for projectile to reach maximum height is given by:

$$t_{\max} = \frac{u \sin \theta}{g} \quad (4.12)$$

You may also find $h_{\max}$ by substituting $t_{\max}$ into equation $y = u_{yi}t - \frac{1}{2}gt^2$. Do it as an activity.

Time taken by the projectile for the whole flight (t_{ft})

You want to find time taken for the projectile to move from O to B in Figure 4.7

At point B, the projectile has returned to where it started. Therefore, its displacement at point B is zero. Applying equation $y = u_{yi}t - \frac{1}{2}gt^2$ and making substitutions, you get:

$0 = (u \sin \theta)t_{ft} - \frac{1}{2}gt_{ft}^2$ and solving for t_{ft} , you get time for the whole flight as:

$$t_{ft} = \frac{2u \sin \theta}{g} \quad (4.13)$$

Note that, the time taken by the projectile for the whole flight is just twice the time for the projectile to reach maximum height.

Horizontal range (R)

From Figure 4.02, Range R is the distance between OB.

Along OB, the projectile moves with constant velocity i.e it has zero acceleration.

Therefore:

$R = u_x t = (u \cos \theta)t$, t is time for the whole flight.

$$\begin{aligned} R &= (u \cos \theta)t_{ft} = (u \cos \theta)\left(\frac{2u \sin \theta}{g}\right) \\ R &= \frac{2u^2 \cos \theta \sin \theta}{g} \\ R &= \frac{u^2 \sin 2\theta}{g} \end{aligned} \quad (4.14)$$

Maximum Range

From equation 4.3.13, The Range R is maximum only if $\sin 2\theta$ is maximum

$$i.e \sin 2\theta = 1$$

$$2\theta = \sin^{-1}(1)$$

$$2\theta = 90^\circ$$

$$\therefore \theta = 45^\circ$$

Then

$$R_{max} = \frac{u^2 \sin(2 \times 45^\circ)}{g} = \frac{u^2 \times 1}{g} = \frac{u^2}{g}$$

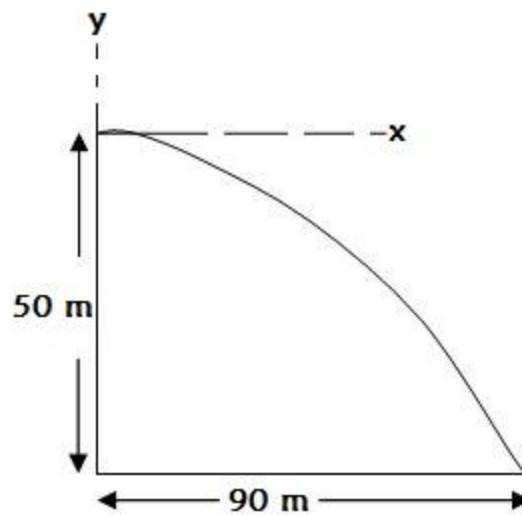
$$\therefore R_{max} = \frac{u^2}{g} \quad (4.15)$$

Example 4.1

A movie student driver on a motorcycle speeds horizontally off a 50 m high cliff. How fast must the motorcycle leave the cliff-top if it is to land on level ground below, 90.0 m from the base of the cliff where the cameras are?

Solution

Sketch diagram



Taking the downward motion to be positive, we have $v_0 = ?$, $y = 50 \text{ m}$, $g = 9.8 \text{ m/s}^2$, $x = 90 \text{ m}$, and $v_{0y} = 0$ since it speeds horizontally,

$$y = v_{0y}t + \frac{1}{2}gt^2$$

$$50 = \frac{1}{2}(9.8)t^2$$

Solving for t , we get $t = 3.19 \text{ s}$.

To get the initial velocity, v_{0x} , we apply:

$$x = v_{0x}t$$

$$90 = v_{0x}(3.19)$$

We solve for v_{0x} , and get:

$$v_{0x} = 28.2 \text{ m/s}$$

Example 4.2

A football is kicked at an angle $\theta = 37.0^\circ$ with velocity of 20 m/s . Calculate the maximum height reached by the ball.

Solution

At maximum height $v_{oy} = 0$, and this occurs at time

$$t = \frac{v_{oy}(\sin \theta)}{g}$$

$$t = \frac{20.0 \sin 37.0^\circ}{9.8} = 1.22 \text{ s}$$

We then find the maximum height:

$$y_{max} = v_{oy}t + \frac{1}{2}gt^2$$

And taking the upward to be negative and the downward to be positive, we have:

$$y_{max} = (-20 \sin 37^\circ)(1.22 \text{ s}) + \frac{1}{2}\left(9.8 \frac{\text{m}}{\text{s}^2}\right)(1.22 \text{ s})^2 = 7.35 \text{ m}$$

Activity 4.1

1. A tiger leaps horizontally from a 7.5 m high rock with a speed of 4.5 m/s . how far from the base of the rock will she land?
2. A diver running 1.6 m/s dives out horizontally from the edge of a vertical cliff and reaches the water below 3.0 s later. How high was the cliff and how far from its base did the diver hit the water?
3. In example 3.2 calculate:
 - (a) The time of travel before the football hits the ground
 - (b) How far away it hits the ground
 - (c) The velocity vector at the maximum height
 - (d) The acceleration vector at the maximum height

4.3.3 Relative motion

Relative motion is about relating the measurements of two different observers, one moving with respect to the other. Measurements of velocities depend on the reference frame of the observer. Recall that reference frames are just the coordinate systems from which motion measurements are based. Most of the time we use a stationary frame of reference relative to the Earth, but occasionally we use a moving frame of reference associated with a bus, car or a plane moving with constant velocity relative to the Earth.

In two dimensions, let E be an observer, assumed stationary with respect to the Earth.

Let two cars labeled A and B, and introduce the following notation see Figure 4.02.

r_{AE} = the position of Car A as measured by E (in a coordinate system fixed with respect to the Earth)

r_{BE} = the position of Car B as measured by E

r_{AB} = the position of Car A as measured by an observer in Car B

Then:

$$\mathbf{r}_{AB} = \mathbf{r}_{AE} - \mathbf{r}_{BE} \quad (4.16)$$

And the rate of change of these quantities with time gives us relative velocities as:

$$\mathbf{v}_{AB} = \mathbf{v}_{AE} - \mathbf{v}_{BE} \quad (4.17)$$

For any two objects or reference frames, A and B, the velocity of A relative to B has the same magnitude, but opposite direction, as the velocity of B relative to A:

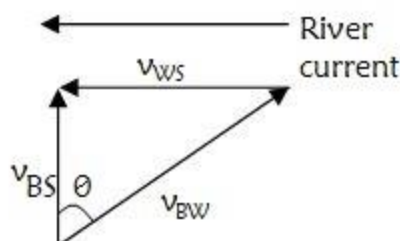
$$\mathbf{v}_{BA} = -\mathbf{v}_{AB} \quad (4.17)$$

Example 4.1

A boat's speed in still water is $v_{BW} = 2.00 \text{ m/s}$. If the boat is to travel directly across a river whose current has speed of $v_{WS} = 1.50 \text{ m/s}$, at what upstream angle must the boat head?

Solution

Letting v_{BS} be the velocity of boat as seen from the shore, the situation can be presented diagrammatically as follows:



$$\sin \theta = \frac{v_{WS}}{v_{BW}} = \frac{1.50 \text{ m/s}}{2.00 \text{ m/s}}$$

$$\theta = \sin^{-1} \left(\frac{1.50}{2.00} \right)$$

$$\therefore \theta = 48.59^\circ$$

So the boat must head upstream at 48.59° angle.

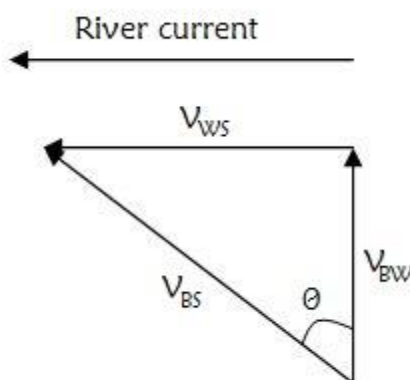
Example 4.2

The same boat in example 4.1, ($v_{BW} = 2.00 \text{ m/s}$) now heads directly across the stream whose current is still 1.5 m/s .

- What is the velocity of (magnitude and direction) of the boat relative to the shore?
- If the river is 110 m wide, how long will it take to cross and how far downstream will the boat be then?

Solution

(a)



$$\begin{aligned} \mathbf{v}_{BS} &= \mathbf{v}_{BW} + \mathbf{v}_{WS} \\ v_{BS} &= \sqrt{v_{BW}^2 + v_{WS}^2} = \sqrt{(2.00 \text{ m/s})^2 + (1.50 \text{ m/s})^2} \\ v_{BS} &= 2.5 \text{ m/s} \end{aligned}$$

- (b) Given the river's width $D = 110 \text{ m}$, and using the definition of velocity, we solve for

$$\begin{aligned} t &= \frac{D}{v_{BW}} \\ t &= \frac{110 \text{ m}}{2.00 \text{ m/s}} = 55 \text{ s} \end{aligned}$$

The boat will have been carried downstream in this time a distance of:

$$d = v_{WS}t = (1.50 \text{ m/s})(55 \text{ s}) = 82.5 \text{ m}$$

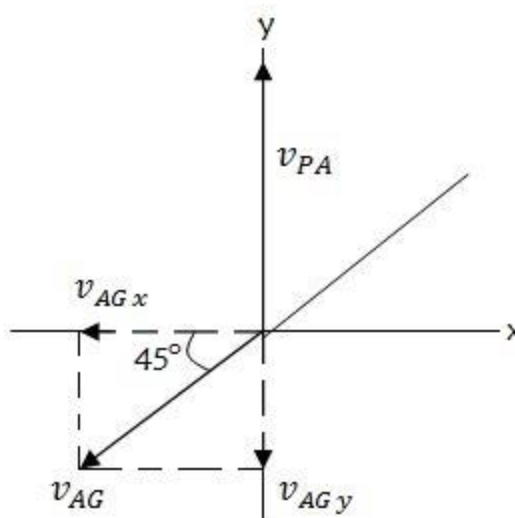
Activity 4.2

- A plane whose air speed is 200 km/h heads due North. But a 100 km/h northeast wind suddenly begins to blow. What is the resulting velocity of the plane with respect to the ground?

Solution

Let v_{AP} be the velocity of the plane relative to air, v_{AG} be the velocity of air relative to the ground and v_{PG} be the velocity of the plane relative to the ground:

$$\text{Then } v_{PG} = v_{PA} + v_{AG}$$



From the sketch diagram, we find:

$$v_{AGx} = -v_{AG} \cos 45^\circ = -(100 \text{ km/h})(\cos 45^\circ) = -70.7 \text{ km/h}$$

$$v_{AGy} = -v_{AG} \sin 45^\circ = -(100 \text{ km/h})(\sin 45^\circ) = -70.7 \text{ km/h}$$

$$v_{PAx} = 0 \text{ km/h}$$

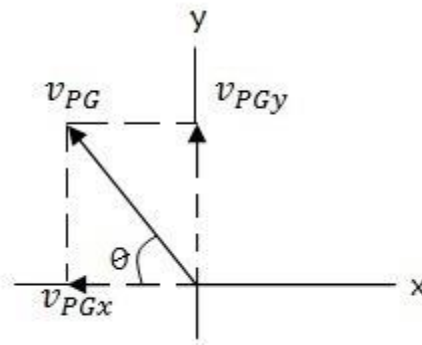
$$v_{PAy} = 200 \text{ km/h}$$

Then:

$$v_{PGx} = 0 \text{ km/h} + (-70.7 \text{ km/h}) = -70.7 \text{ km/h}$$

$$v_{PGy} = 200 \text{ km/h} + (-70.7 \text{ km/h}) = 129 \text{ km/h}$$

We now sketch the resultant v_{PG} as follows:



Applying Pythagoras theorem

$$v_{PG} = \sqrt{v_{PGx}^2 + v_{PGy}^2}$$

$$= \sqrt{(-70.7)^2 + (129)^2} = 147 \text{ km/h}$$

To find the angle θ that v_{PG} makes with the x axis, we use:

$$\tan \theta = \frac{v_{PGy}}{v_{PGx}} = \frac{129 \text{ km/h}}{-70.7 \text{ km/h}}$$

And

$$\theta = \tan^{-1} \left(\frac{129}{-70.7} \right) = -61.3^\circ \text{ with respect to negative x-axis.}$$

4.3.4 Summary

An object moving in both x-direction and y-direction simultaneously is said to undergo two dimensional motions. Graphical methods of vector addition and vector resolution are much involved in analysis of two dimensional motions.

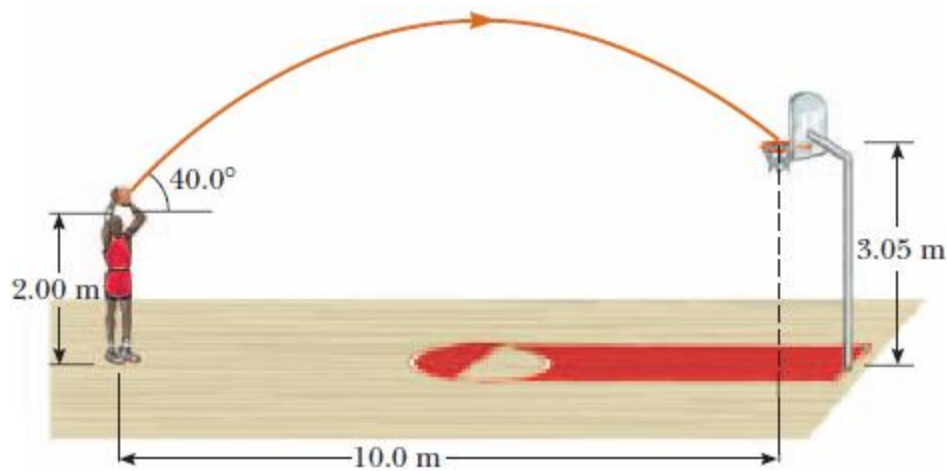
An object that moves under the influence of acceleration due to gravity is said to undergo projectile motion. The path it takes is called trajectory and it is parabolic in nature.

The velocity of an object relative to one frame of reference can be found by vector addition if its velocity relative to a second frame of reference and the relative velocity of the two reference frames are known.

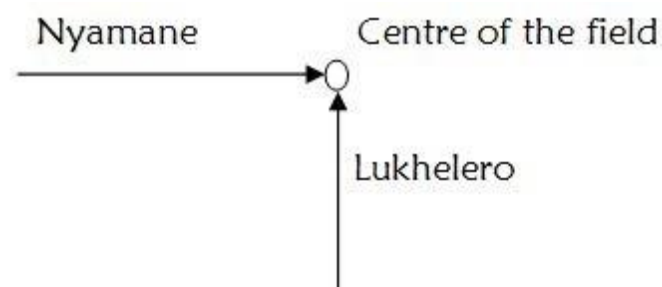
4.3.5 End of Unit 4 Exercise

A boy on a small hill aims his water-balloon slingshot horizontally, straight at a second boy hanging from a tree branch a distance d away. At the instant the water balloon is released, the second boy lets go and falls from the tree, hoping to avoid being hit. Show that he made the wrong move; he had not studied physics yet.

1. A ball is thrown horizontally from the roof of a building 56 m tall and lands 45 m from the base. What was the ball's initial speed?
2. A fire hose held near the ground shoots water at a speed of 6.5 m/s. at what angle(s) should the nozzle point in order that the water land 2.0 m away? Why are there two different angles?
3. Romeo is chucking pebbles gently up to Juliet's window, and he wants the pebbles to hit the window with only a horizontal component of velocity. He is standing at the edge of a rose garden 8.0 m below her window and 9.0 m from the base of the wall. How fast are the pebbles going when they hit her window?
4. A projectile is fired in such a way that its horizontal range is equal to three times its maximum height. What is the angle of projection?
5. A rock is thrown upward from the level ground in such a way that the maximum height of its flight is equal to its horizontal range d . (a) At what angle θ is the rock thrown? (b) **What If?** Would your answer to part (a) be different on a different planet? (c) What is the range $d_{\max}$ the rock can attain if it is launched at the same speed but at the optimal angle for maximum range?
6. An astronaut on a strange planet finds that she can jump a maximum horizontal distance of 15.0 m if her initial speed is 3.00 m/s. What is the free-fall acceleration on the planet?
7. A 2.00-m-tall basketball player is standing on the floor 10.0 m from the basket, as in Figure below. If he shoots the ball at a 40.0° angle with the horizontal, at what initial speed must he throw the basketball so that it goes through the hoop without striking the backboard? The height of the basket is 3.05 m.



8. If you are riding on a train that speeds past another train moving in the same direction on an adjacent track, it appears that the slower moving train is moving backwards. Why?
9. Two soccer players; Nyamane and Lukhelero, approach the centre of the field at right angle to each other. Nyamane runs with velocity of 30 km/h and Lukhelero runs with velocity of 50 km/h as shown in the diagram below:



What is the velocity of Nyamane as seen by Lukhelero? What is the velocity of Lukhelero as seen by Nyamane?

10. A passenger on a boat moving at 1.50 m/s on a still lake walks up a flight of stairs at a speed of 0.50 m/s. The stairs are angled at 45° pointing in the direction of motion of the boat. What is the velocity of the passenger relative to the water?
11. Two planes approach each other head-on. Each has a speed of 835 km/h, and they spot each other when they are initially 10.0 km apart. How much time do the pilots have to take evasive action? Answer: 21.60 s
12. A passenger on a boat moving at 1.50 m/s on a still lake walks up a flight of stairs at a speed of 0.50 m/s. The stairs are angled at 45° pointing in the direction of motion of the boat. What is the velocity of the passenger relative to the water?

Unit 5

Forces and Motion

5.0 Introduction

In Kinematics; units 3 and 4, we studied motion excluding what caused the motion. Motion was discussed in terms of displacement, acceleration and acceleration. Now we deal with the question of why objects move. What causes an object to accelerate or decelerate? We can answer in each case that a net force is needed to cause motion. We recall what a force is and learn about how force and motion are related. We will then be introduced to Newton's three laws of motion. We will finally deal with a retarding force called friction.

5.1 Objectives

By the end of this unit you should be able to:

- Define force
- Define friction force
- State Newton's laws of motion
- Apply Newton's laws of motion in problem solving
- Solve problems related to friction force

5.2 Key Terms

Force

Net force/ resultant force

Newton's first law of motion

Newton's second law of motion

Newton's third law of motion

5.3 Force and Motion

5.3.1 What is force?

A force is anything that can cause a change to an object. Forces can:

- Change the shape of an object
- Move or stop an object

- Change the direction of a moving object
- Produce heat in an object

A force can be contact or noncontact

A contact force is in contact with an object to cause a change e.g

- The force that is used to push or pull things, like on a door to open or close it
- The force that a sculptor uses to turn clay into a pot
- The force of the wind to turn the windmill

A non-contact force does not have to touch an object to cause a change. E.g

- Gravitational force, like the earth pulling the moon towards itself
- Electric force, like a proton and an electron attracting each other
- Magnetic force, like a magnet pulling a paper clip towards itself

The unit of force is Newton (N). 1 Newton is a force that produces an acceleration of 1ms^{-2} when applied on a mass of 1 kg. i.e $1\text{N} = 1\text{kgms}^{-2}$. Force is a vector physical quantity i.e it is completely described by both magnitude and direction.

Resultant force or net force is a single force representing the vector sum of the forces that act on the object.

Activity 5.1

1. Find the resultant force of the two forces $F_1 = 3\text{ N}$ and $F_2 = 4\text{ N}$ which are at an angle of 60° to each other.

5.3.2 Newton's laws of motion

5.3.2.1 Newton's first law of motion

It states that an object will remain at its state of rest or state of motion with constant velocity except when it is acted upon by a non-zero net force. This means that force is not needed to maintain its motion. According to this law, an object may either be in a static or dynamic equilibrium.

Newton's first law is also called the law of inertia. Inertia is the resistance (inability) of an object to start moving or to stop moving once it started.

This resistance is directly proportional to the mass of object.

There are three types of inertia:

(i) Inertia of rest

It is the inability of the body to change its state of rest by itself.

Examples

- (i) A person standing in a bus falls backward when the bus suddenly starts moving. This is because, the person who is initially at rest continues to be at rest even after the bus has started moving.
- (ii) A book lying on the table will remain at rest, until it is moved by some external agencies.
- (iii) When a carpet is beaten by a stick, the dust particles fall off vertically downwards once they are released and do not move along the carpet and fall off.

(ii) Inertia of motion

Inertia of motion is the inability of the body to change its state of motion by itself.

Examples

- (a) When a passenger gets down from a moving bus, he falls down in the direction of the motion of the bus.
- (b) A passenger sitting in a moving car falls forward, when the car stops suddenly.
- (c) An athlete running in a race will continue to run even after reaching the finishing point.

(iii) Inertia of direction

It is the inability of the body to change its direction of motion by itself.

Examples

When a bus moving along a straight line takes a turn to the right, the passengers are thrown towards left. This is due to inertia which makes the passengers travel along the same straight line, even though the bus has turned towards the right.

From the first law, you can infer that to change the state of rest or uniform motion, an external agency called force is required.

Therefore, force can also be defined as an agency which when acting on a body changes or tends to change the state of rest or of uniform motion of the body along a straight line.

Mass and Inertia

You recall that Inertia is the tendency of an object to resist being accelerated by a force and that inertia depends on mass.

Therefore mass, m , can be defined as a measure of an object's inertia

Mass can also be defined as a measure of the quantity of substance in a physical object

The SI unit of mass is kilogram (kg).

Mass is a scalar quantity

Mass is not weight; mass is a property of an object, whereas weight is the force exerted on the object by gravity.

5.3.2.2 Newton's Second Law of Motion and Weight

It can be stated as acceleration is equal to force divided by mass

Expressed by the equation:

$$\mathbf{F}_R = m\mathbf{a} \quad (5.04)$$

Equation 5.04 can also be written as:

$$\sum \mathbf{F} = m\mathbf{a} \quad (5.05)$$

Writing equation 5.05 in Vector components you get:

$$\sum F_x = m\mathbf{a}_x \quad (5.06)$$

$$\sum F_y = m\mathbf{a}_y \quad (5.07)$$

From Newton's Second Law of motion, the magnitude of gravitational force (weight) can be found by:

$$\mathbf{F}_g = m\mathbf{g} \quad (5.08)$$

Where F_g is gravitational force, m is the mass of an object and g is the object's acceleration due to gravity.

5.3.2.3 Newton's Third Law of Motion

This law is the action-reaction law. It states that

"To every action there is always an equal and opposite reaction"

OR

"If object A exerts a force on object B then object B exerts a force on object A that is equal in magnitude but opposite in direction"

Newton's third law of motion implies that if an object A exerts force on object B, then object B will exert an equal amount of force on object A but in opposite direction. These forces do not cancel each other because they act on different objects. The effect of third law of motion can be observed in many activities in our everyday life. Some examples are:

- (i) When a bullet is fired from a gun with a certain force (action), there is an equal and opposite force exerted on the gun in the backward direction (reaction).
- (ii) When a man jumps from a boat to the shore, the boat moves away from him. The force he exerts on the boat (action) is responsible for its motion and his motion to the shore is due to the force of reaction exerted by the boat on him.
- (iv) The swimmer pushes the water in the backward direction with a certain force (action) and the water pushes the swimmer in the forward direction with an equal and opposite force (reaction).

5.3.3 Application of Newton's Laws of motion

The following steps can be followed when solving problems related to Newton's laws of motion:

1. Read the problem carefully at least once
2. Draw a picture of the system, identify the object of primary interest, and indicate forces with arrows
3. Label each force in the picture in the way that will bring to mind what physical quantity the label stands for e.g T is for tension
4. Draw a free body diagram of the object of interest, based on the labeled picture. If additional objects are involved, draw separate free-body diagrams for them. Choose convenient coordinates for for each object
5. Apply Newton's second law. The x-and y-components of Newton's second law should be taken from the vector equation and written individually. This usually result in two equations and two unknowns
6. Solve for the desired unknown quantity, and substitute the numbers

5.3.3.1 Objects in translational Equilibrium

Objects that are at rest or are moving with constant velocity are said to be in equilibrium. In this situation acceleration $\mathbf{a}$ is zero and Newton's second law applied to an object in translational equilibrium gives:

$$\sum \mathbf{F} = 0 \quad (5.09)$$

This equations tells you that the vector sum of all forces (net force) acting on an object in equilibrium is equal to zero.

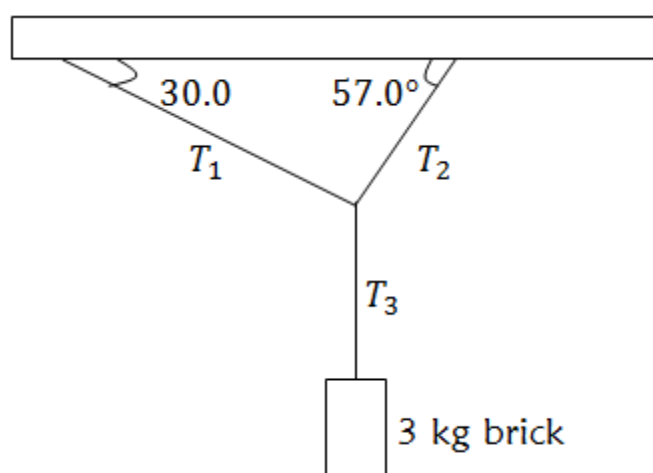
Equation 5.09 can be written in the following equivalent vector forms in two dimensions:

$$\sum F_x = 0 \text{ and } \sum F_y = 0 \quad (5.10)$$

Note that an object at rest and an object moving with constant velocity are both at translational equilibrium because in both cases acceleration is zero giving us a net force equal to zero.

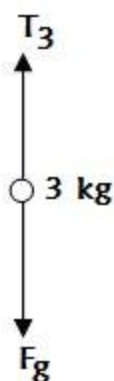
Example

A brick of mass 3 kg hangs from a vertical cable tied to two other cables that are fastened to a support as shown in the Figure below. Find the tension in each of the three cables.

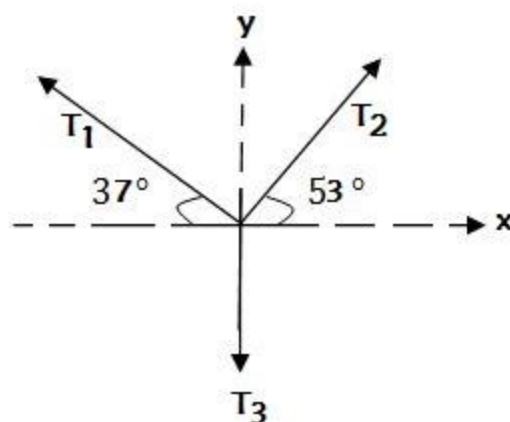


Solution

We draw the free-fall diagrams:



(a)



(b)

From the first free-fall diagram (a) we write the following equation:

$$\sum F_y = 0 \rightarrow T_3 - F_g = 0$$

$$T_3 = F_g = mg = 3kg(9.8 \text{ m/s}^2)$$

$$T_3 = 29.4 \text{ N}$$

We write the two equations from the second free-fall diagram (b):

$$\sum F_x = 0 \rightarrow T_2 \cos 53^\circ - T_1 \cos 37^\circ = 0 \quad (i)$$

$$\sum F_y = 0 \rightarrow T_2 \sin 53^\circ + T_1 \sin 37^\circ - T_3 = 0 \quad (ii)$$

From Equation i, $T_2 = T_1 \left(\frac{\cos 37^\circ}{\cos 53^\circ} \right) = 1.327 T_1$

Substitute this result into ii,

$$1.327 T_1 \sin 53^\circ + T_1 \sin 37^\circ - T_3 = 0$$

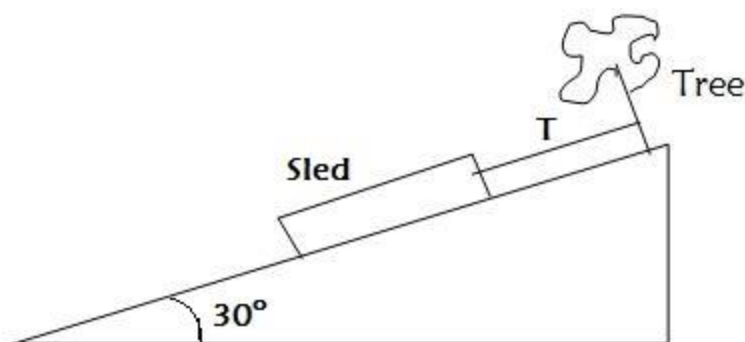
$$T_1 = \frac{T_3}{1.327 \sin 53^\circ + \sin 37^\circ} = \frac{29.4}{1.662} = 17.690 \text{ N}$$

$$T_2 \cos 53^\circ = T_1 \cos 37^\circ \rightarrow T_2 = \frac{T_1 \cos 37^\circ}{\cos 53^\circ}$$

$$= \frac{17.690 \cos 37^\circ}{\cos 53^\circ} = 23.475 \text{ N}$$

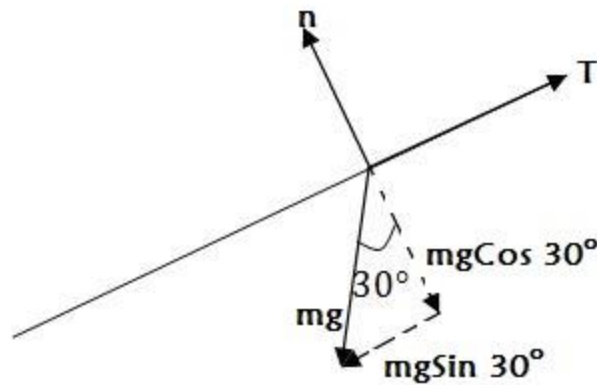
Example

A sled is tied to a tree on a frictionless, snow-covered hill as shown in the Figure below. If the sled weighs 77.0 N, find the magnitude of the tension T exerted by the rope on the sled and the normal force n exerted by the hill on the sled.



Solution

Draw a free-fall diagram



$$n - mg \cos 30^\circ = 0$$

$$n = mg \cos 30^\circ = 77.0 \cos 30^\circ = 66.68 \text{ N}$$

$$T + f - mg \sin 30^\circ$$

$$T + 0 - 77.0 \sin 30^\circ$$

$$T = 77.0 \sin 30^\circ = 38.5 \text{ N}$$

$\therefore$ The tension $T = 66.68 \text{ N}$ and normal force $n = 38.50 \text{ N}$

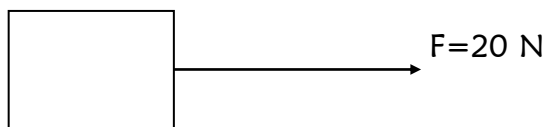
5.3.3.2 Accelerating objects and Newton's second law

When a net force acts on an object, the object accelerates and Newton's second Law is applied.

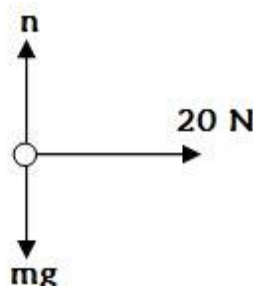
Example

The weight of a box containing books is 300 N. If a student pulls on the rope with a constant force of 20 N, what is the acceleration of the crate and how far will it move in 2.00 s? Assume the box starts from the rest and that there is no friction between the surface and the box.

Solution



And its free-fall diagram is



There is no motion in y direction and therefore no acceleration in y direction

We find mass from $mg = 300 \text{ N} \rightarrow m = \frac{300}{9.8} = 30.6 \text{ kg}$

$$ma_x = F$$

$$a_x = \frac{20 \text{ N}}{30.6 \text{ kg}} = 0.654 \text{ m/s}^2$$

We now apply the following equation of motion:

$$x_f - x_i = v_i t + \frac{1}{2}at^2$$

$$x_f - 0 = (0)(2.00 \text{ s}) + \frac{1}{2}(0.654)(2.00 \text{ s})^2 = 1.308 \text{ m}$$

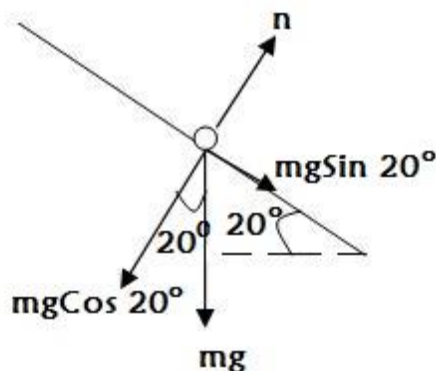
Example

A toy car of mass 0.5 kg is on a frictionless inclined plane at an angle of 20° .

- Determine its acceleration down the incline plane.
- If the length of the incline plane is 5.0 m and the car starts from rest at the top, how long does it take to travel to the bottom?
- What is the car's speed at the bottom?

Solution

(a) Free-fall diagram:



$$mg \sin 20^\circ = ma_x$$

$$a_x = \frac{mg \sin 20^\circ}{m} = g \sin 20^\circ = \left(\frac{9.8 \text{ m}}{\text{s}^2} \right) \sin 20^\circ = 3.352 \text{ m/s}^2$$

(b)

$$\Delta x = v_i t + \frac{1}{2}at^2$$

$$5.0 = (0)t + \frac{1}{2}(3.352)t^2$$

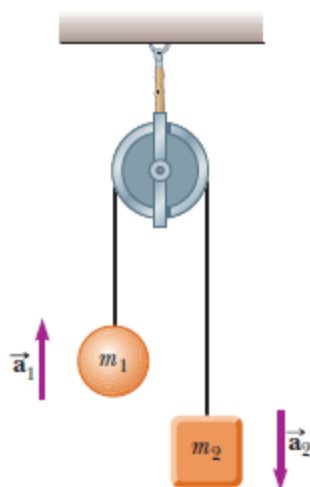
$$t = \sqrt{\frac{5.0 \times 2}{3.352}} = 1.727 \text{ s}$$

$$(c) \quad v_f = v_i + at$$

$$v_f = 0 + (3.352 \text{ m/s}^2)(1.727 \text{ s}) = 5.789 \text{ m/s}$$

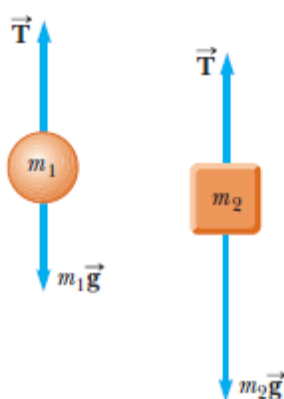
Example

Two objects of mass m_1 and m_2 , with $m_2 > m_1$, are connected by a light, inextensible cord and hung over a frictionless pulley, as in the Figure below. Both cord and pulley have negligible mass. Find the magnitude of the acceleration of the system and the tension in the cord.



Solution

the following are the free-fall diagrams for m_1 and m_2 :



For m_1 , we write the following Equation:

$$T - m_1g = m_1a_y \quad (i)$$

For m_2 , we have the following Equation:

$$m_2g - T = m_2a_y \quad (ii)$$

Adding the two equations, you get:

$$(m_2 - m_1)g = (m_2 + m_1)a_y$$

$$a_y = \frac{(m_2 - m_1)g}{(m_2 + m_1)}$$

Lets use Equation I to get T:

$T = m_1 a_y + m_1 g$ and Substituting a_y , we get:

$$T = m_1 \left(\frac{(m_2 - m_1)g}{(m_2 + m_1)} \right) + m_1 g$$

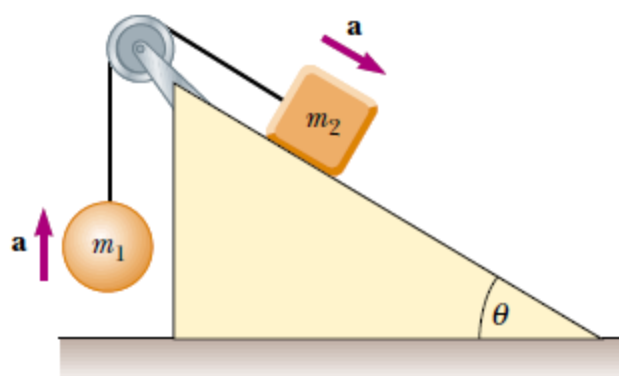
After further simplification, we get:

$$T = \frac{2m_1 m_2 g}{(m_2 + m_1)}$$

Note that the acceleration of the second block is the same as that of the first, but negative. When m_2 gets very large compared with m_1 , the acceleration of the system approaches g , as expected, because m_2 is falling nearly freely under the influence of gravity. Indeed, m_2 is only slightly restrained by the much lighter m_1 .

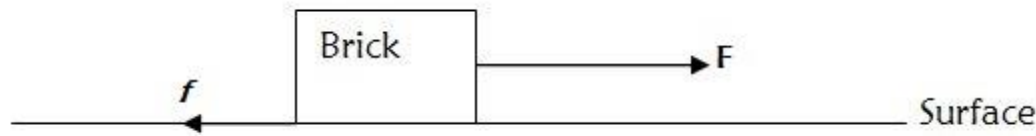
Activity 5.1

1. Why does a child in a wagon seem to fall backward when you give the wagon a sharp pull?
2. A ball of mass m_1 and a block of mass m_2 are attached by a lightweight cord that passes over a frictionless pulley of negligible mass, as in Figure below. The block lies on a frictionless incline of angle θ . Find the magnitude of the acceleration of the two objects and the tension in the cord.

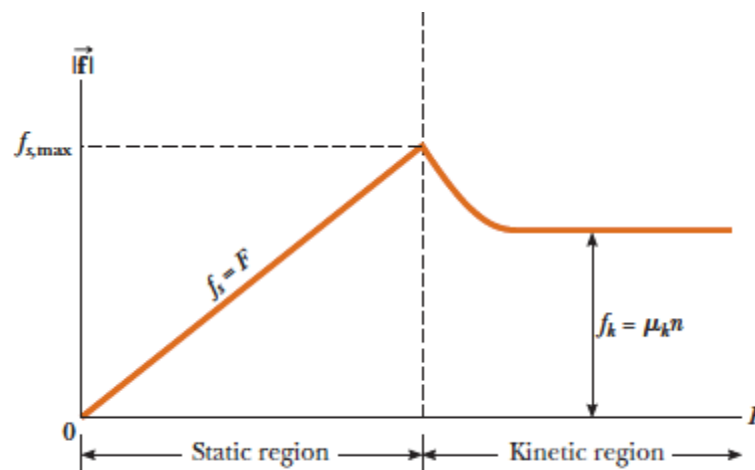


5.3.4 Friction Force

Consider a applied force F on a brick at rest on the surface as shown in the Figure below:



The applied force F is opposed by friction force, f . Before the brick starts moving F is equal to f and f_s is directly proportional to the applied force F until f_s reaches its maximum value $f_{s,max}$ as shown in the graph of Figure



f_s is maximum when the brick is on the verge of moving. At this point F exceeds $f_{s,max}$, the brick accelerates to the right. When the object is in motion the friction force is less than applied force. The friction force for an object in motion is called kinetic friction, f_k .

The magnitude of maximum static friction is given by:

$$f_{s,max} = \mu_s n$$

where μ_s is constant called coefficient of static friction. And n is the magnitude of the normal force exerted by one surface on the other.

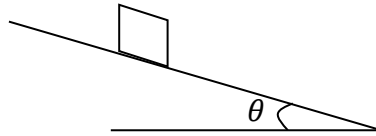
The magnitude of kinetic friction is found by:

$$f_k = \mu_k n$$

Where μ_k is constant called the coefficient of kinetic friction.

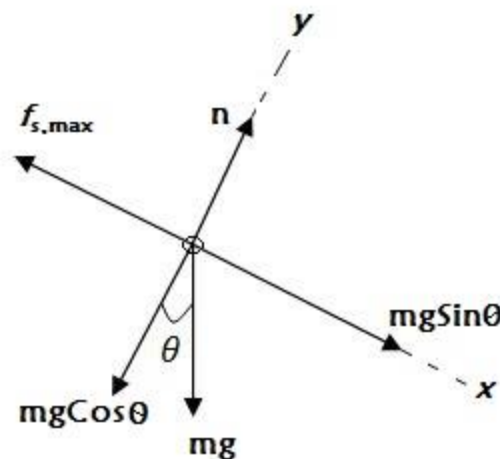
Example

Suppose a block with a mass of 2.50 kg is resting on a ramp. If the coefficient of static friction between the block and ramp is 0.350, what maximum angle can the ramp make with the horizontal before the block starts to slip down?



Solution

Free-fall diagram



$$n = mg \cos \theta$$

$$f_{s,max} = mg \sin \theta \quad (i)$$

And you know that

$$f_{s,max} = \mu_s n = \mu_s (mg \cos \theta) \quad (ii)$$

Equating the two Equations, you have:

$$\mu_s (mg \cos \theta) = mg \sin \theta$$

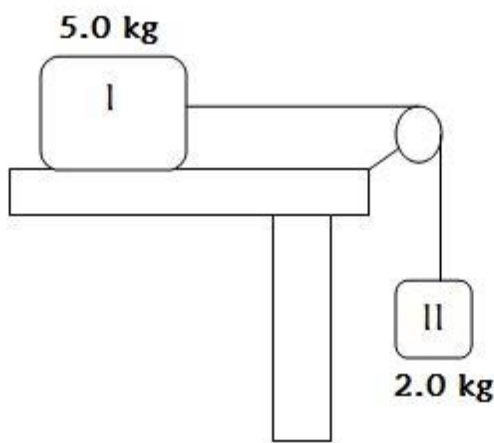
$$\tan \theta = \mu_s$$

$$\theta = \tan^{-1}(\mu_s) = \tan^{-1}(0.350) = 19.29^\circ$$

Activity 5.2

1. Your little sister wants a ride on her sled. If you are on flat ground, will you exert less force if you push her or pull her? Assume you push her at the same angle θ to the ground.

2. Two boxes are connected by a cord running over a pulley. The coefficient of kinetic friction between box I and the table is 0.20. We ignore the mass of the cord and the pulley and any friction in the pulley, which means we can assume that a force applied to one end of the cord will have the same magnitude at the other end. Find the acceleration of both boxes.



3. The skier has just begun descending the 30° slope. Assuming the coefficient of kinetic friction is 0.10, (a) draw the free body diagram (b) calculate her acceleration (c) calculate the speed she will reach after 4.0 s.

5.3.5 Summary

Newton's three laws of motion are the basic classical laws describing motion.

Newton's first law (the law of inertia) states that if the net force on an object is zero, an object originally at rest remains at rest and an object in motion remains in motion in a straight line with constant velocity.

Newton's second law states that the acceleration of an object is directly proportional to the net force acting on it, and inversely proportional to its mass:

$$\sum \mathbf{F} = m\mathbf{a}$$

Newton's third law states that whenever one object exerts a force on a second object, the second object always exerts a force on the first object which is equal in magnitude but opposite in direction:

$$\mathbf{F}_{AB} = -\mathbf{F}_{BA}$$

The tendency of an object to resist a change in its motion is called inertia. Mass is a measure of an object's inertia.

Weight refers to the force of gravity on an object and is equal to the product of mass, m , and the acceleration due to gravity g :

$$\mathbf{F}_g = m\mathbf{g}$$

Force can be considered as a push or pull. From Newton's second law of motion, force can be defined as an action capable of giving rise to acceleration. The net force on an object is the vector sum of all forces acting on it.

Friction force is one which opposes motion.

The maximum static friction is given by

$$f_{s,max} = \mu_s F_N$$

where μ_s is the constant called coefficient of static friction and F_N is the normal force.

Normal force is the one which an object exerts on the other object perpendicular to their contact surfaces.

Kinetic friction is given by:

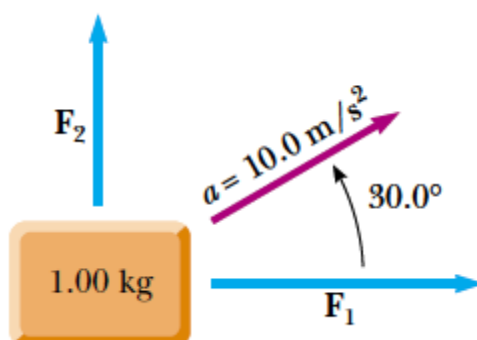
$$f_k = \mu_k F_N$$

where μ_k is the constant called coefficient of kinetic friction.

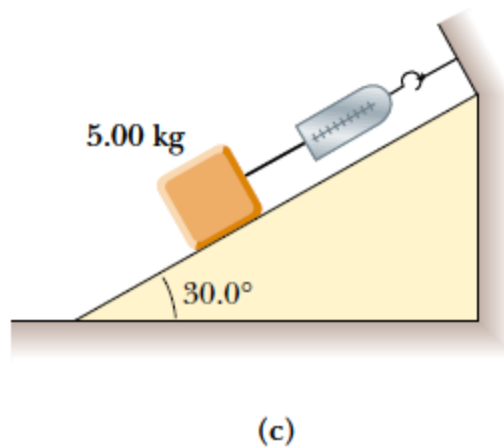
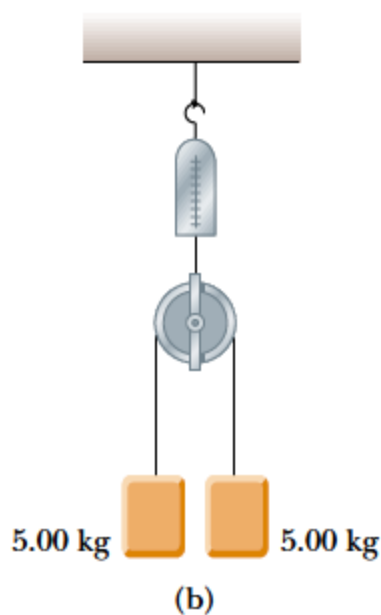
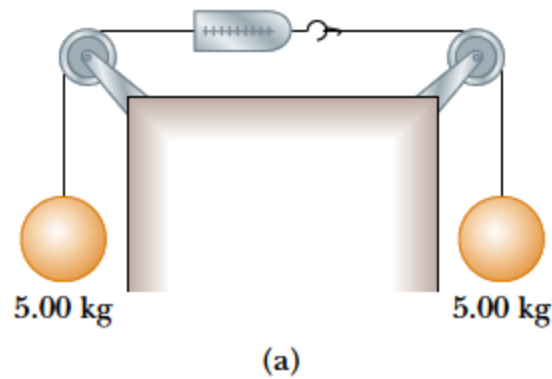
5.3.6 End of unit 5 exercise

1. A ball is held in a person's hand:
 - a. Identify all the external forces acting on the ball and the reaction to each
 - b. If the ball is dropped, what force is exerted on it while it is falling? Identify the reaction force in this case (neglect air resistance)
2. If a car is traveling Eastward with a constant speed of 20 m/s. what is the resultant force acting on it?
3. What is wrong with the statement "Because the car is at rest, there are no forces acting on it." How would you correct this sentence?
4. A rubber ball is dropped onto the floor. What force causes the ball to bounce?
5. While the football is in flight, what forces act on it? What are the action-reaction pair while the football is being kicked and while it is in flight?
6. What force causes automobile to move? A propeller-driven airplane? A rowboat?
7. Find the weight on Earth of an object whose mass is (a) 3.00 Kg (b) 200 g.
8. If a man weighs 900 N on the Earth, what would he weigh on Jupiter, where the acceleration due to gravity is 25.9 m/s²?

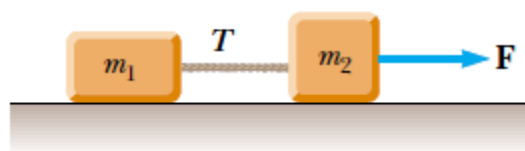
9. A 20.0 kg object that can move freely is subjected to a resultant force of 45.0 N in the x direction. Find the acceleration of the object
10. What force is needed to accelerate a child on a sled (total mass = 60.0 kg) at 1.15 m/s^2 ?
11. A net force of 255 N accelerates a bike and rider at 2.20 m/s^2 . What is the mass of the bike and rider?
12. An electron of mass $9.11 \times 10^{-31} \text{ kg}$ has an initial speed of $3.00 \times 10^5 \text{ m/s}$. It travels in a straight line, and its speed increases to $7.00 \times 10^5 \text{ m/s}$ in a distance of 5.00 cm. Assuming its acceleration is constant, (a) determine the force exerted on the electron and (b) compare this force with the weight of the electron, which we neglected.
13. A 1.00-kg object is observed to have an acceleration of 10.0 m/s^2 in a direction 30.0° north of east as shown in the figure below. The force F_2 acting on the object has a magnitude of 5.00 N and is directed north. Determine the magnitude and direction of the force F_1 acting on the object.



14. The systems shown in Figure below are in equilibrium. If the spring scales are calibrated in newtons, what do they read? (Neglect the masses of the pulleys and strings, and assume the incline in part (c) is frictionless.)

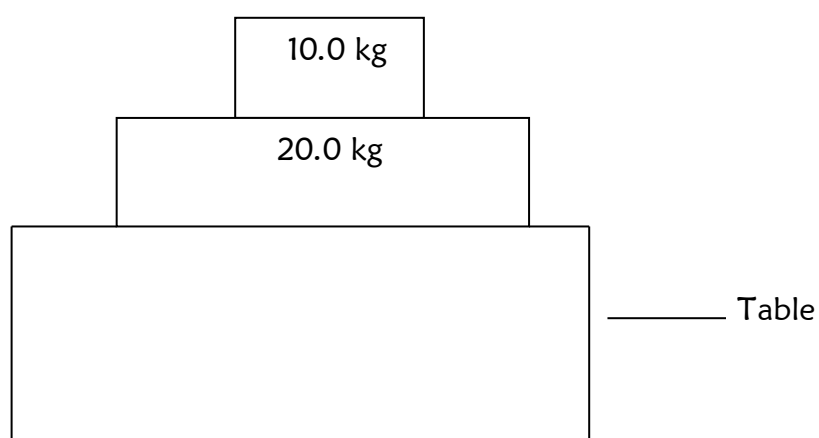


15. A car is traveling at 50.0 mi/h on a horizontal highway. (a) If the coefficient of static friction between road and tires on a rainy day is 0.100, what is the minimum distance in which the car will stop? (b) What is the stopping distance when the surface is dry and μ_s 0.600?
16. A 3.00-kg block starts from rest at the top of a 30.0° incline and slides a distance of 2.00 m down the incline in 1.50 s. Find (a) the magnitude of the acceleration of the block, (b) the coefficient of kinetic friction between block and plane, (c) the friction force acting on the block, and (d) the speed of the block after it has slid 2.00 m.
17. Two blocks connected by a rope of negligible mass are being dragged by a horizontal force $F = 18.0 \text{ kg}$, and the coefficient of kinetic friction between each block and the surface is 0.100. (a) Draw a free-body diagram for each block. (b) Determine the tension T and the magnitude of the acceleration of the system.



18. A 20.0 kg box rests on a table.

- What is the weight of the box and the normal force acting on it?
- A 10.0 kg box is placed on top of 20.0 kg as shown in the Figure below.
Determine the normal force that the table exerts on the 20.0 kg box and the normal force that the 20.0 kg box exerts on the 10.0 kg.



Unit 6

Work, Energy and Power

6.0 Introduction

Until now we have been studying the motion of an object in terms of Newton's three laws of motion. In that analysis, force played a central role as the quantity that causes an object's motion. In this unit and unit 7 we discuss the analysis of motion of an object in terms of energy and momentum. The importance of these quantities is that they are conserved. The conservation of energy and momentum gives us a deep understanding of the nature of the world and gives us alternative way to approach practical problems. This unit is devoted to energy and a very closely concept of work. Energy and work are scalar quantities as such they are easier to deal with than force and acceleration. The importance of energy is that it is conserved and is useful in the study of motion and in other areas of physics as well as in other sciences.

6.1 Objectives

By the end of this unit you should be able to:

- Define work
- Define energy
- Define power
- State work-energy theorem
- State different forms of energy
- Solve work and Energy related problems
- Derive equations of Kinetic and Potential energy
- State the law of conservation of energy
- Solve problems related to the conservation of Energy

6.2 Key words

- Work
- Energy
- Power
- Efficiency

6.3.1 Work done

Work is done when a force $\mathbf{F}$, exerted on an object causes the object to move in the direction of the force. Work is a scalar quantity.

If a constant force $\mathbf{F}$ acts on an object which moves a distance Δx in the direction of the force as shown in Figure 6.1, then work done on the object by this force is given by:

$$W = F\Delta x \quad (6.01)$$

Where W =work done in joules (J)

F =Constant applied force in newton (N)

Δx = displacement measured in meters (m)

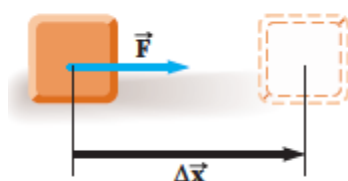


Figure 6.1 A constant force F in the same direction as the displacement Δx , does work $F\Delta x$

If a constant force F acts on an object at an angle which moves a distance Δx in the x -direction as shown in Figure 6.2, then work done on the object by this force is given by:

$$W = (F\cos\theta)\Delta x \quad (6.02)$$

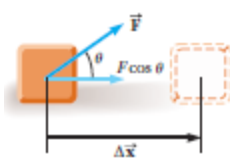


Figure 6.2 A constant force exerted at an angle θ with respect to the displacement, Δx , does work $(F\cos\theta)\Delta x$.

Where:

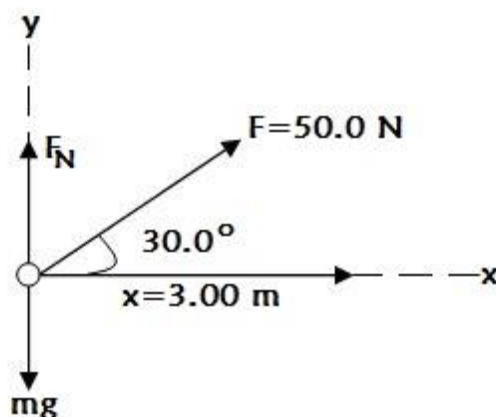
F = magnitude of the force, Δx is the magnitude of the displacement and θ is the angle between the directions of F and Δx .

Example 6.1

A man cleaning his apartment pulls the canister of a vacuum cleaner with a force of magnitude $F = 50.0 \text{ N}$ at an angle of 30.0° with the horizontal. He moves the vacuum cleaner a distance of 3.00 m . calculate the work done by the 50.0 N force.

Solution

We draw the free body diagram as follows:



From the free the free body diagram, we notice that F_N and mg do no work because there is no movement in y -direction. Therefore,

$$W = F_x s = (50.0 \cos 30.0^\circ)(3.00 \text{ m}) = 130 \text{ J}$$

Activity 6.1

1. Define work
2. A 2.0 kg object is moving at 3 m/s . A 4.0 N force is applied in the direction of motion and then removed after the object has traveled an additional 5.0 m . calculate the value of work done on the object
3. A man wishes to pull a crate 15 m across a rough floor by exerting a force of 100 N . the coefficient of kinetic friction is 0.25 . Find the angle between the force and the horizontal distance for the man to do the least work.

6.3.2 Energy

Energy is the ability to do work. The SI unit of energy is joule (J).

There are many forms of energy. Here the focus is on mechanical energy which consists of Kinetic and Potential energy.

6.3.2.1 Kinetic Energy and the Work-Energy Theorem

Kinetic energy is the energy possessed by an object by virtue of movement.

Suppose an object of mass, m , moves under the action of constant net force, F shown in Figure 6.3.

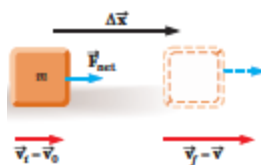


Figure 6.3

The force is constant and therefore the object moves with constant acceleration. If the object is displaced by Δx , then the work done by F is:

$$W_{net} = F_{net} \Delta x = ma \Delta x \quad (6.03)$$

You know that when the object moves with constant acceleration, then the following equation holds:

$$\begin{aligned} v^2 &= u^2 + 2a\Delta x \\ \rightarrow a\Delta x &= \frac{m(v^2 - u^2)}{2} \end{aligned} \quad (6.04)$$

Substituting Equation 6.4 into equation 6.3 for $a\Delta x$, you get:

$$\begin{aligned} W_{net} &= m \left(\frac{v^2 - u^2}{2} \right) \\ W_{net} &= \frac{1}{2}mv^2 - \frac{1}{2}mu^2 \end{aligned} \quad (6.05)$$

The Kinetic Energy (KE) of an object of mass m moving with speed v is

$$KE = \frac{1}{2}mv^2 \quad (6.06)$$

Equation 6.5 is Known as Work-Energy Theorem.

Work-Energy theorem states that the net work done on an object is equal to the change of object's Kinetic energy:

$$W_{net} = KE_f - KE_i = \Delta KE \quad (6.07)$$

The change in kinetic energy is due to the object's change in speed.

Example 6.2

A 1400 kg car has a net forward force of 4500 N applied to it. The car starts from rest and travels down a horizontal high-way. What are its kinetic energy and speed after it

has traveled 100 m? (Ignore losses in kinetic energy because of friction and air resistance.)

Solution

$v_i = 0$, and work is given by:

$$\begin{aligned} W_{net} &= KE_f - KE_i = \Delta KE \\ &= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 \\ &= \frac{1}{2}(1400)v_f^2 - \frac{1}{2}(1400)(0)^2 = 700v_f^2 \end{aligned} \quad (i)$$

And the work done by the net force on the car is:

$$W_{net} = Fs = 4500 \text{ N} \times 100 \text{ m} = 4.50 \times 10^5 \text{ J} \quad (ii)$$

To get the final speed, we equate the two equations:

$$\begin{aligned} 700v_f^2 &= 4.50 \times 10^5 \text{ J} \\ v_f &= \sqrt{\frac{4.50 \times 10^5}{700}} = 25.4 \text{ m/s} \end{aligned}$$

Activity 6.2

1. State the kinetic energy theorem
2. A 8000 N car is traveling at 12 m/s along a horizontal road when the brakes are applied. The car skids to a stop in 4.0 s. how much kinetic energy does the car lose in this time?
3. A 5.0 kg cart is moving horizontally at 6.0 m/s. Find the work required to be done in order to change its speed to 10.0 m/s.
4. A 4.0 kg cart starts up an inclined with a speed of 3 m/s and comes to rest 2 m up the inclined. Find the total work done on the car.

6.3.3 Conservative and non-conservative forces

A force is said to be conservative if the work it does in moving an object between two points is the same no matter what path is taken, otherwise it is non-conservative. Force of gravity is the best example of conservative force.

Non-conservative force disperses object's energy which is in the form of heat or sound energy. Examples of non-conservative forces are kinetic friction and air drag.

Suppose W_{nc} be work done by non – conservative force and W_c be work done by conservative force, the net work done is given by the sum of W_{nc} and W_c i.e

$$W_{net} = W_{nc} + W_c$$

From equation 5.7 $W_{net} = \Delta KE$, you get:

$$W_{nc} + W_c = \Delta KE \quad (6.08)$$

6.3.4 Potential Energy

Potential energy is the energy possessed by an object by virtue of its position.

6.3.4.1 Gravitational Work and Potential Energy

Gravity is a conservative force and for every conservative force potential energy can be found.

Consider a book of mass m falls from a height h_i to a height h_f and assume that the only force acting on the book is gravitation. The magnitude of gravitational force F_g is mg and the displacement Δh is $h_i - h_f$. The angle between F_g and Δh is 0° since both pointing down as shown in Figure 6.5

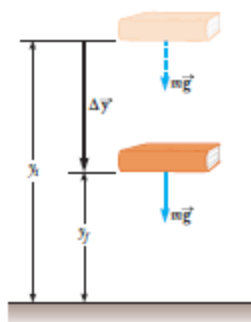


Figure 4.5

Applying the definition of work:

$$W_g = (F_g \cos \theta) \Delta h = mg(h_i - h_f) \cos 0^\circ = -mg(h_f - h_i) \quad (6.09)$$

Suppose that the system involves only conservative and non-conservative forces and you know that W_g is work done by conservative force, then:

$$W_{nc} + W_g = \Delta KE$$

and

$$W_{nc} - mg(h_f - h_i) = \Delta KE \quad (6.10)$$

Making W_{nc} the subject of the formula, you get:

$$\begin{aligned} W_{nc} &= \Delta KE + mg(h_f - h_i) \\ W_{nc} &= \Delta KE + mgh_f - mgh_i \end{aligned} \quad (6.11)$$

The gravitational energy of a system is:

$$PE = mgh \quad (6.12)$$

Where g is the acceleration due to gravity and h is the vertical position of the mass m relative to the surface of the earth.

From equation 6.3.9:

$$\begin{aligned} W_g &= -mg(h_f - h_i) \\ \therefore W_g &= -(mgh_f - mgh_i) \\ W_g &= -(PE_f - PE_i) \end{aligned} \quad (6.13)$$

Equation 6.13 tells you that the work done by gravity is the same as negative change in gravitational potential energy.

Using Equation 6.13 in Equation 6.11 then work-energy theorem is given by:

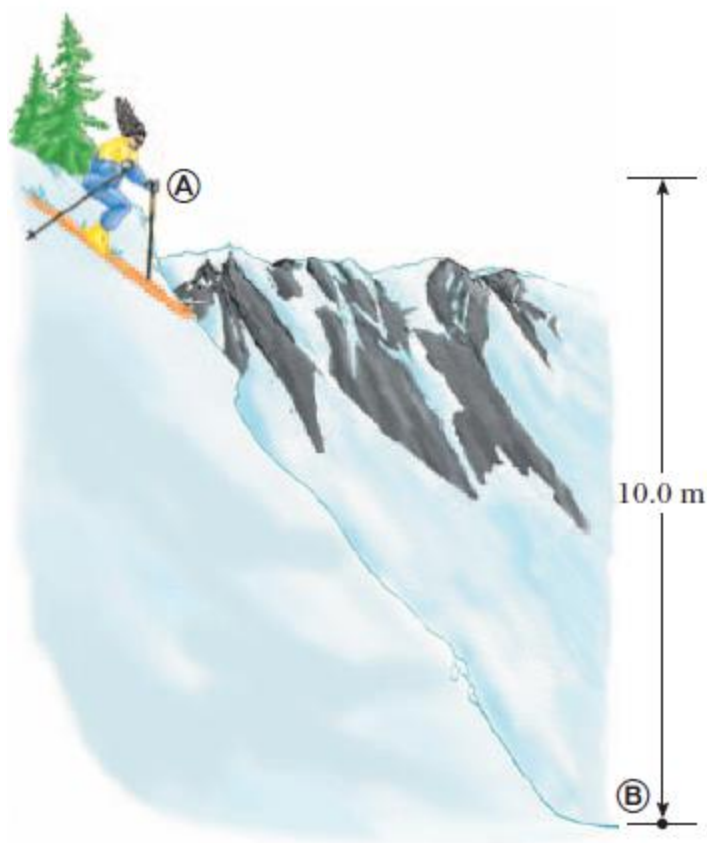
$$W_{nc} = (KE_f - KE_i) + (PE_f - PE_i) \quad (6.14)$$

which states that the work done by non-conservative forces is equal to the change of kinetic energy plus the change of gravitational potential energy.

Example 6.3

A 60.0 kg skier is at the top of a slope. At the initial point A the skier is 10.0 m vertically above point B.

- Setting the zero level for gravitational potential energy at B, find the gravitational potential energy at A and B.
- Repeat this problem with zero level at point A



Solution

a.

$$PE_i = mgy_i = 60 \text{ kg} \times 9.80 \times 10.0 \text{ m} = 5880 \text{ J}$$

Because $PE_f = 0$, the difference in potential energy is

$$PE_i - PE_f = 5880 \text{ J} - 0 \text{ J} = 5880 \text{ J}$$

b. $PE_f = mgy_f = 60 \text{ kg} \times 9.80 \times (-10.0 \text{ m}) = -5880 \text{ J}$

In this case the distance $y_f = 0$.

$$PE_i - PE_f = 0 \text{ J} - (-5880 \text{ J}) = 5880 \text{ J}$$

The potential energy at the top of the slope is greater than the potential energy at the bottom by 5880 J regardless of the zero level selected.

Activity 6.3

1. A projectile of mass 0.50 kg is fired with an initial speed of 10 m/s at an angle of 60° above the horizontal. Find the potential energy of the projectile.

6.3.5 Gravity and the Conservation of Mechanical Energy

In the absence of non-conservative forces on the system, $W_{nc} = 0$ and Equation 6.14 becomes:

$$0 = (KE_f - KE_i) + (PE_f - PE_i)$$

Putting initial terms and final terms together, you get:

$$KE_i + PE_i = KE_f + PE_f \quad (6.15)$$

You see from equation 6.15 that the sum of kinetic energy and potential energy remains constant all times i.e. mechanical energy is conserved for an isolated system.

Equation 6.15 can be rewritten as:

$$E_{mec} = KE + PE \quad (6.16)$$

where E_{mec} is the total mechanical energy.

6.3.6 The law of conservation of Mechanical energy

It states that in an isolated system in which only conservative force is involved the total mechanical energy of the system remains constant i.e. the total initial mechanical energy is equal to total final mechanical energy.

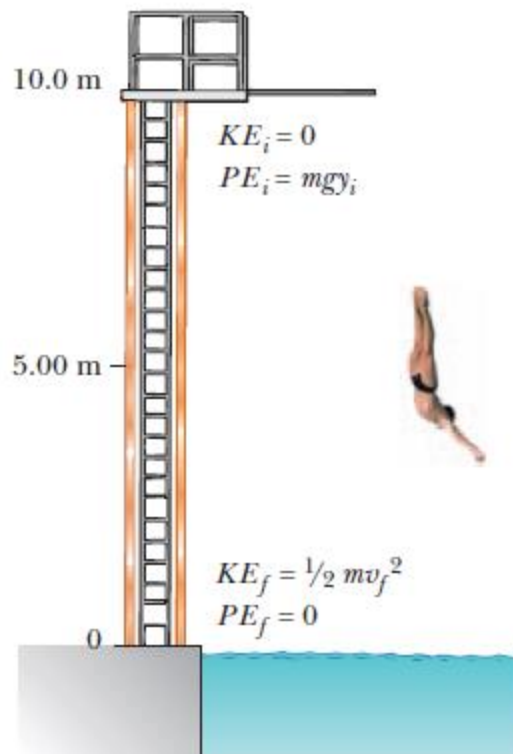
You can write equation 6.3.15 in the form:

$$\frac{1}{2}mu^2 + mgh_i = \frac{1}{2}mv^2 + mgh_f \quad (6.17)$$

if the only force doing work on an isolated system is conservative.

Example 6.4

A diver of mass m drops from a board 10.0 m above the water's surface, as in Figure 6.16. Neglect air resistance. **(a)** Use conservation of mechanical energy to find his speed 5.00 m above the water's surface. **(b)** Find his speed as he hits the water.



Solution

$$KE_i + PE_i = KE_f + PE_f$$

$$\frac{1}{2}mv_i^2 + mgy_i = \frac{1}{2}mv_f^2 + mgy_f$$

$$0 + gy_i = \frac{1}{2}v_f^2 + gy_f$$

$$v_f = \sqrt{2g(y_i - y_f)} = \sqrt{2(9.8)(10.0 - 5.0)} = 9.90 \text{ m/s}$$

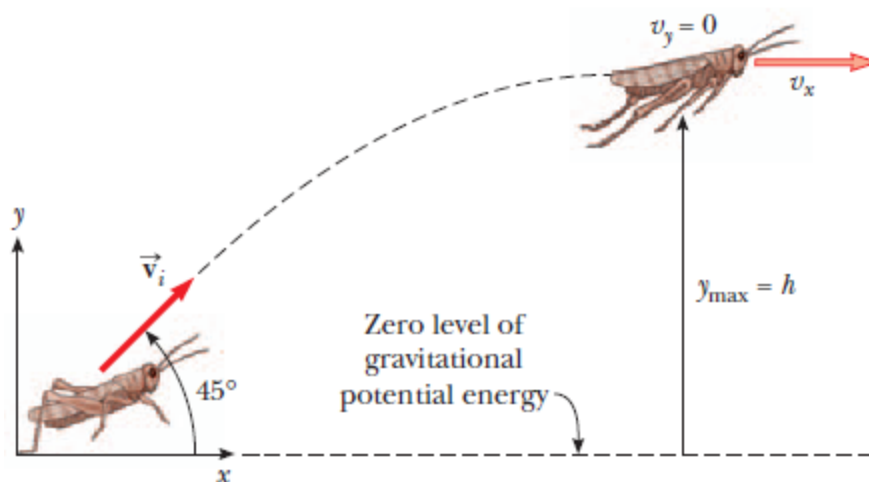
Find the diver's speed at the water's surface, $y_f = 0$:

$$0 = mgy_i = \frac{1}{2}mv_f^2 + 0$$

$$v_f = \sqrt{2gy_i} = \sqrt{2(9.8)(10.0)} = 14.0 \text{ m/s}$$

Activity 6.4

1. State the law of conservation of mechanical energy
2. A powerful grasshopper launches itself at an angle of 45° above the horizontal and rises to a maximum height of 1.00 m during the leap. (See Figure 5.17.) With what speed v_i did it leave the ground? Neglect air resistance.



6.3.7 Power

Power can be defined as:

- i. The rate at which work is done
- ii. The rate at which energy is transformed from one form to another
- iii. The rate at which energy is transferred from one point to another in a medium

Average amount of power can be found by:

$$P = \frac{\text{Work done}}{\text{Time}} = \frac{\text{Energy transformed}}{\text{time}} \quad (6.18)$$

Example 6.5

A 70.0 kg jogger runs up a long flight of stairs in 4.0 s. the vertical height of the stairs is 4.5 m. (a) estimate the jogger's power output in watts (b) how much energy did this require?

Solution

$$P = \frac{W}{t} = \frac{mgy}{t} = \frac{(70.0 \text{ kg})(9.8 \text{ m/s}^2)(4.5 \text{ m})}{4.0 \text{ s}} = 770 \text{ W}$$

The energy required is:

$$E = Pt = 770 \text{ J/s}(4.0\text{s}) = 3100 \text{ J}$$

6.3.8 Summary

Work is done on an object by a net force when the object moves a distance, d . If the direction of a constant force $\mathbf{F}$ makes an angle θ with the direction of motion, the work done by this force is:

$$W = Fd \cos \theta$$

Energy can be defined as the ability to do work. In SI units, work and energy are measured in **joules** ($1\text{J}=1\text{N}\cdot\text{m}$).

Kinetic Energy (KE) is energy of an object's motion. An object of mass m and speed v has translational kinetic energy

$$KE = \frac{1}{2}mv^2$$

Potential Energy (PE) is energy associated with forces that depend on the position or configuration of the body. Gravitational potential energy is given by

$$PE_g = mgh$$

where h is the height of the object of mass m above an arbitrary reference point. Elastic potential energy is equal to

$$PE_{ela} = \frac{1}{2}kx^2$$

for a compressed or stretched spring, where x is the displacement from the outstretched position and k is proportionality constant called elasticity constant.

Other potential energies include chemical, electrical and nuclear energy. The change in potential energy when an object changes position is equal to the external work needed to take the object from one position to the other.

The Work-Energy principle states that the net work done on an object by the net force equals the change in Kinetic energy of that object

$$W_{net} = \Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

The general law of conservation of energy states that the total energy is neither increased nor decreased in any process. Energy can be transformed from one form to another, and transferred from one body to another, but the total amount remains constant.

In an isolated system, mechanical energy is conserved:

$$KE + PE = \text{constant}$$

When non-conservative forces act, then:

$$W_{nc} = \Delta KE + \Delta PE$$

Power is defined as the rate at which work is being done or the work at which energy is transformed or is transferred. The SI unit of power is watt ($1\text{W}=1\text{J/s}$).

6.3.9 End of Unit 6 Exercise

1. Can a centripetal force ever do work on an object?
2. Why is it tiring to push hard against a solid wall even though no work is done?
3. Can kinetic energy ever be negative? Explain.
4. A 75.0 kg firefighter climbs a flight of stairs 10.0 m high. How much work is required?
5. A 900 N crate rests on the floor. How much work is required to move it at constant speed (a) 6.0 m along the floor against a friction force of 180 N (b) and 6.0 m vertically
6. How much work did the movers do horizontally pushing a 150 kg crate 12.3 m across a rough floor without acceleration, if the effective coefficient of friction was 0.70.
7. How much work must be done to stop a 100 kg car traveling at 110 km/h?
8. How long will it take a 1750 W motor to lift a 285 kg piano to a six-story window 16.0 m above?

Unit 7

Linear Momentum

7.0 Introduction

The law of conservation of energy we have just discussed in unit 6 is one of conservation laws in physics including conservation of linear momentum. We will discuss linear momentum and its conservation in this unit. We will then make use of the conservation of linear momentum and of Energy to analyze collisions. The law of conservation of linear momentum is particularly useful when dealing with two or more objects that interact with each other, as in collisions.

7.1 Objectives

By the end of this unit you should be able to:

- Define linear momentum
- Explain types of collisions
- State the law of conservation of linear momentum
- State impulse-momentum theorem

7.2 Key words

Momentum

Impulse

7.3. Linear Momentum

Linear momentum is mass multiply by velocity given by:

$$\mathbf{P} = m\mathbf{v} \quad (7.01)$$

Where $\mathbf{P}$ is linear momentum, m is mass and v the object's velocity.

The unit of momentum is $kgms^{-1}$ and it is a vector quantity.

Example 7.1

An object of mass 500g accelerates uniformly from rest at $4ms^{-2}$. Calculate its momentum at time $t = 6s$

Solution:

You need first of all find the final velocity of an object at $t = 6s$:

Applying equation; $v = u + at$ and noting that initial velocity is $u = 0$ (it accelerates from rest), then you get:

$$v = 0 + 4ms^{-2}(6s)$$

$$v = 24ms^{-2}$$

From the definition of momentum:

$$P = mv$$

$$P = 0.5kg(24ms^{-1})$$

$$P = 12ms^{-1}$$

7.3.1 Linear Momentum and Newton's second law of motion

Newton stated the second law of motion as the resultant force acting on an object is directly proportional to the time rate of change of linear momentum.

We now know that Linear momentum is mass multiply by velocity:

$$\mathbf{P} = m\mathbf{v} \quad (7.01)$$

Where $\mathbf{P}$ is linear momentum, m is mass and v the object's velocity.

Suppose an object of mass m , moving with initial velocity u and changes its velocity to v in time t , then the change of momentum is given by:

$$\Delta\mathbf{P} = m(\mathbf{v} - \mathbf{u}) \quad (7.02)$$

and the time rate of change of momentum becomes:

$$\frac{\Delta\mathbf{P}}{t} = \frac{m(\mathbf{v} - \mathbf{u})}{t}$$

From the definition of Newton's second law in terms of momentum:

$$\mathbf{F}_R \propto m\left(\frac{v-u}{t}\right) \quad (7.03)$$

Relationship 7.03 in equation form becomes:

$$\mathbf{F}_R = \frac{km(v-u)}{t} \quad (7.04)$$

where k is the proportionality constant.

Recall that the first equation of motion is $v = u + at$. Substituting this equation into equation 6.04 for v and taking $k = 1$, you get:

$$\mathbf{F}_R = \frac{km(\mathbf{u} + \mathbf{at} - \mathbf{u})}{t}$$

which reduces to:

$$\mathbf{F}_R = m\mathbf{a} \quad (7.05)$$

which is the same as equation 4.4 in unit 4.

You notice that the two statements of Newton's second law of motion are the same:

- i. Acceleration is equal to force divided by mass of an object.
- ii. The resultant force acting on an object is directly proportional to the time rate of change of linear momentum in the direction of the force.

7.3.2 The law of Conservation of Linear Momentum

It states that in a closed system, the total momentum of the system remains constant.

To show that linear momentum is conserved, let's consider Figure 7.1 in which two objects are involved in collision:

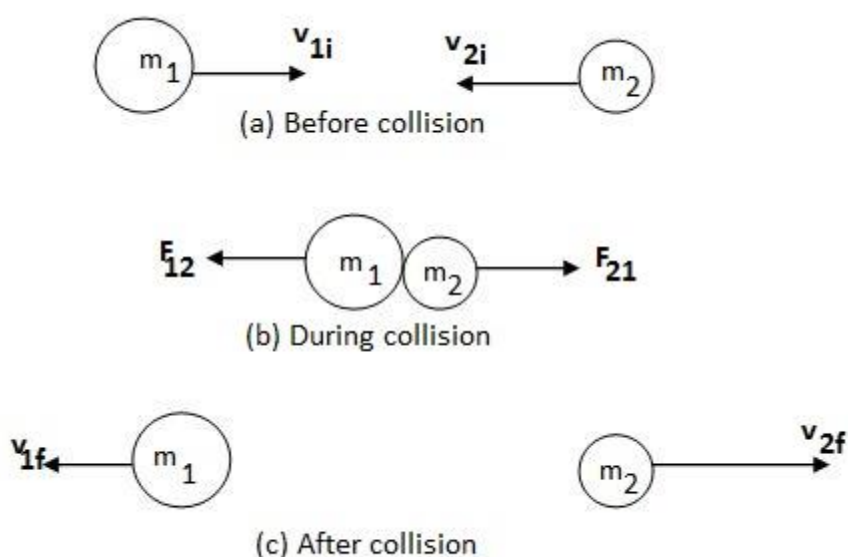


Figure 7.1 Conservation of Momentum

The force experienced by m_1 due to m_2 is F_{12} and the force experienced by m_2 due to m_1 is F_{21} .

Newton's third law of motion tells us that:

$$F_{12} = -F_{21} \quad (7.06)$$

Suppose that m_1 changes its velocity from initial velocity v_{1i} to final velocity v_{1f} during the time interval of Δt and object m_2 changes its velocity from initial value v_{2i} to final value v_{2f} during the same time interval of Δt , then according to Newton's second law:

$$F_{12} = \frac{m_1(v_{1i} - v_{1f})}{\Delta t} \quad (7.07)$$

$$F_{21} = \frac{m_2(v_{2i} - v_{2f})}{\Delta t} \quad (7.08)$$

Substituting 7.06 and 6.08 into 7.06 we get:

$$\frac{m_1(v_{1i}-v_{1f})}{\Delta t} = \frac{m_2(v_{2i}-v_{2f})}{\Delta t}$$

$$m_1v_{1i} - m_1v_{1f} = -m_2v_{2i} + m_2v_{2f}$$

and putting initial values and final values to the left and to the right hand sides respectively, we get:

$$m_1v_{1i} + m_2v_{2i} = m_1v_{1f} + m_2v_{2f} \quad (7.09)$$

Equation 7.09 is the **Law of Conservation of linear momentum**.

The **law of conservation of linear momentum** states that for an isolated system of objects, the total linear momentum before collision is equal to the total linear momentum after collision.

7.4 Impulse

Impulse is the product of the net force and the time interval for which the force acts.

Impulse is defined as:

$$\text{Impulse} = \mathbf{F}_{net} \cdot \Delta t \quad (7.10)$$

From Newton's second law of motion:

$$\mathbf{F}_{net} = \frac{\Delta \mathbf{P}}{\Delta t}$$

$$\mathbf{F}_{net} \cdot \Delta t = \Delta \mathbf{P}$$

Therefore,

$$\text{Impulse} = \Delta \mathbf{P} \quad (7.11)$$

Equation 7.11 is the **impulse-momentum theorem**.

Impulse-momentum theorem states that impulse is equal to the change of an object's linear momentum.

Example 7.2

A 150 N resultant force acts on 300 kg trailer. Calculate how long it takes this force to change the trailer's velocity from 2ms^{-1} to 6ms^{-1} in the same direction.

Solution:

$$m = 300\text{kg}, F_n = 150\text{ N}, u = 2\text{ms}^{-1} \text{ and } v = 6\text{ms}^{-1}$$

$$F_n \cdot \Delta t = \Delta P$$

$$F_n \cdot \Delta t = m(v - u)$$

$$\Delta t = \frac{m(v-u)}{F_n}$$

$$\Delta t = \frac{300\text{kg} (6\text{ms}^{-1} - 2\text{ms}^{-1})}{150\text{ N}} = 8\text{s}$$

7.4.1 Collision in two dimensions

So far you have looked at collisions in one dimension. You should be aware that there are also collisions in two dimensions which objects move off at some angle after collision. This is also called **glancing collision**.

Consider the glancing collision between two billiard balls shown in figure 8.14. Ball 1 is moving to the right at the velocity v_{1i} and ball 2 is at rest ($v_{2i} = 0$). After the collision, ball 1 is found to be moving at an angle 45.0° above the horizontal and ball 2 is moving at an angle $\alpha = 45.0^\circ$ below the horizontal. Let us find the velocities of both balls after the collision.

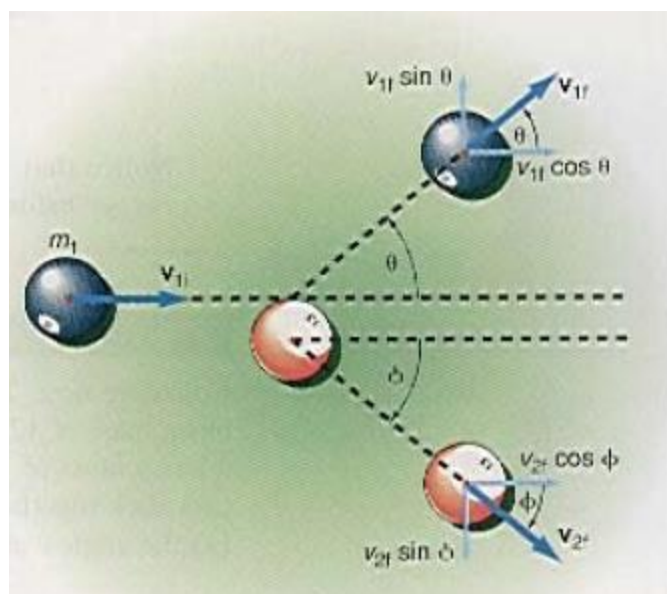


Figure 3.3.4 Glancing collision

As in all collisions, the law of conservation of momentum holds, that is,

$$\mathbf{P}_f = \mathbf{P}_i$$

$$\sum P_x = m_1 v_{1f} \cos \theta + m_2 v_{2f} \cos \vartheta = m_1 v_{1i} \quad (7.12)$$

$$\sum P_y = m_1 v_{1f} \sin \theta - m_2 v_{2f} \sin \vartheta = 0 \quad (7.13)$$

Solving equation 6.20 for v_{2f} , with $\theta = \vartheta = 45.0^\circ$, we get:

$$m_1 v_{1f} \sin 45.0^\circ = m_2 v_{2f} \sin 45.0^\circ$$

$$v_{2f} = \frac{m_1 v_{1f}}{m_2} \quad (7.14)$$

Substituting equation (6.14) into equation 6.12 for v_{2f} and making v_{1f} the subject of the formula, you get:

$$v_{1f} = \frac{v_{1i}}{2 \cos 45.0^\circ} \quad (7.15)$$

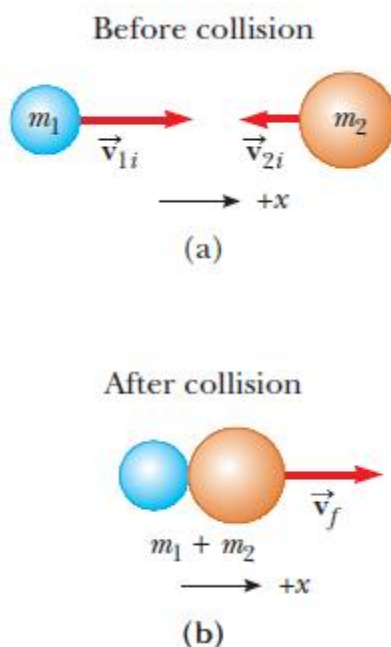
Activity 7.1

Billiard ball 1 is moving at a speed of $v_{1i} = 10.0 \text{ cm/s}$, when it has a glancing collision with an identical billiard ball that is at rest. After the collision, $\theta = \vartheta = 45.0^\circ$. The mass of the billiard ball is 0.170 kg .

- Find the speed of ball 1 and 2 after the collision.
- Is energy conserved in this collision?

7.4.2 Three types of Collisions

- Elastic collision** is a collision in which both momentum and kinetic energy are conserved
- Inelastic collision** is the one in which momentum is conserved but kinetic energy is not conserved
- Perfectly inelastic collision** is the one in which momentum is conserved but kinetic energy is not conserved and the colliding objects stick together and move with the same final velocity after collision.

7.4.2.1 Perfectly Inelastic Collision

Consider two objects of masses m_1 and m_2 moving with initial velocities v_{1i} and v_{2i} respectively before collision. They collide and move with common velocity v_{2i} as

shown in the figure above. You can find the common velocity using conservation of momentum alone:

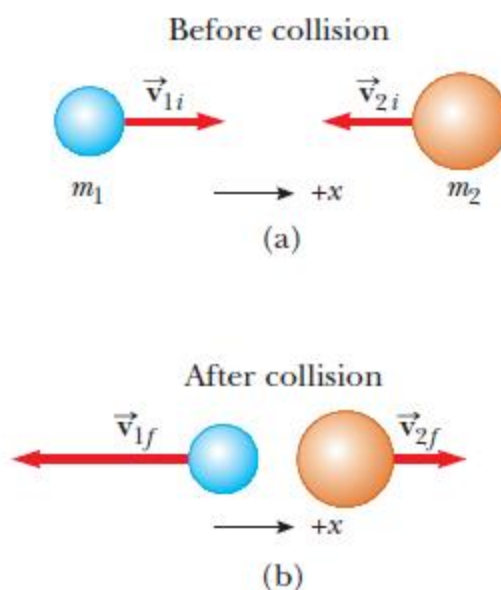
$$P_i = P_f$$

$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f \quad (7.16)$$

$$v_f = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2} \quad (7.17)$$

7.4.2.2 Elastic Collisions

Let's now consider the collision of two objects in which both momentum and kinetic energy are conserved as shown in the Figure below:



Conservation of momentum is given by:

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f} \quad (7.18)$$

Conservation of kinetic energy is given by:

$$\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2 \quad (7.19)$$

Cancelling $\frac{1}{2}$ and putting together those terms containing m_1 and those containing m_2 , equation 6.19 becomes:

$$m_1 (v_{1i}^2 - v_{1f}^2) = m_2 (v_{2f}^2 - v_{2i}^2) \quad (7.20)$$

Factorizing equation 6.20 we get:

$$m_1 (v_{1i} - v_{1f})(v_{1i} + v_{1f}) = m_2 (v_{2f} - v_{2i})(v_{2f} + v_{2i}) \quad (7.21)$$

You can rewrite equation 6.18 as:

$$m_1 (v_{1i} - v_{1f}) = m_2 (v_{2f} - v_{2i}) \quad (7.22)$$

Dividing equation 6.21 by Equation 6.22 produces:

$$v_{1i} + v_{1f} = v_{2f} + v_{2i}$$

Gathering initial and final terms together:

$$v_{1i} - v_{2i} = v_{2f} - v_{1f}$$

which can be rewritten as:

$$v_{1i} - v_{2i} = -(v_{1f} - v_{2f}) \quad (7.23)$$

Equation 7.23 together with Equation 7.18 can be used to solve problems dealing with perfectly elastic head-on collisions.

7.4.3 Summary

Linear momentum, P , of an object is defined as the product of its mass and its velocity,

$$\mathbf{P} = m\mathbf{v}$$

In terms of momentum, Newton's second law states that the net force is equal to the rate of change of momentum:

$$\mathbf{F}_{net} = \frac{\Delta \mathbf{P}}{\Delta t}$$

The Law of Conservation of Linear momentum states that in an isolated system of objects, the total momentum remains constant.

$$m_1 \mathbf{v}_1 + m_2 \mathbf{v}_2 = \text{constant}$$

An isolated system is one in which the net force acting on it is zero.

The Impulse of a force on an object is defined as $F\Delta t$, where F is the average force acting on an object during time interval Δt .

Impulse-Momentum theorem states that impulse is equal to the change of an object's linear momentum:

$$\text{Impulse} = F\Delta t = \Delta \mathbf{P}$$

Elastic collision is a collision in which both momentum and kinetic energy are conserved

Inelastic collision is the one in which momentum is conserved but kinetic energy is not conserved

Perfectly inelastic collision is the one in which momentum is conserved but kinetic energy is not conserved and the colliding objects stick together and move with the same final velocity after collision.

7.4.4 End of Unit 7 exercise

Unit 8

Uniform Circular Motion and Newton's law of Universal Gravitation

8.0 Introduction

The revolution of the Earth around the Sun, rotating fly wheel, electrons revolving around the nucleus, spinning top, the motion of a fan blade, revolution of the moon around the Earth etc. are some examples of circular motion. In all the above cases, the bodies or particles travel in a circular path. So, it is necessary to understand the motion of such bodies.

8.1 Objectives

By the end of this unit, you should be able to:

- Explain uniform circular motion
- Describe centripetal acceleration and centripetal force
- Discuss road banking
- Explain Newton's law of universal gravitation, gravitational force and weight

8.2 Key words

Angular displacement

Angular velocity

Angular acceleration

8.3 Uniform Circular motion

If a body moves in a circle or a circular arc with constant linear speed, it is said to undergo uniform circular motion. The magnitude of its velocity remains the same but changes direction continuously.

8.3.1 Angular displacement

Consider an object moving in a circular path from position A to position B in time Δt , it starts at an angle θ_i to an angle θ_f as in the diagram below:

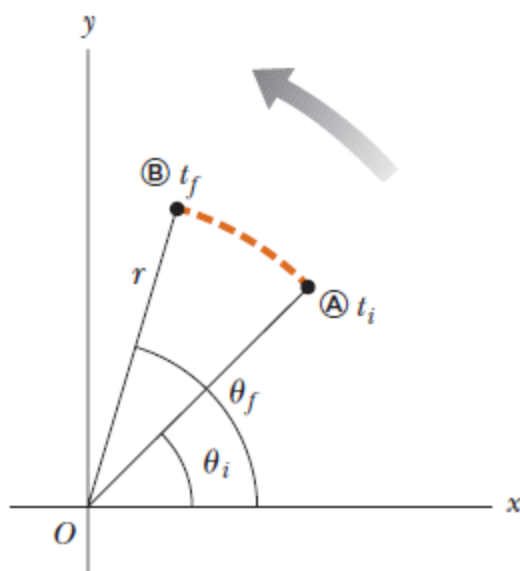


Figure 7.1

Then an object's angular displacement $\Delta\theta$ is the difference in its final and initial angles given by:

$$\Delta\theta = \theta_f - \theta_i \quad (8.01)$$

The SI unit of angular displacement is radian (rad)

From figure 8.1, an object covers a displacement along the circular arc equal to Δs with radius r , then angular displacement can also be expressed by:

$$\Delta\theta = \frac{\Delta s}{r} \quad (8.02)$$

The average angular speed ω_{av} of a rotating rigid object during the time interval Δt is the angular displacement divide by time interval:

$$\omega_{av} = \frac{\Delta\theta}{\Delta t} = \frac{\theta_f - \theta_i}{t_f - t_i} \quad (7.03)$$

The SI unit of angular speed is radian per second (rad/s)

Instantaneous angular speed ω of a rotating rigid object is the limit of average speed

$\frac{\Delta\theta}{\Delta t}$ as time interval Δt approaches zero:

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} \quad (8.04)$$

Its SI unit is radian per second squared (rad/s)

An average angular acceleration α_{av} during the time interval Δt is the change in its angular speed $\Delta\omega$ divide by Δt :

$$\alpha_{av} = \frac{\Delta\omega}{\Delta t} = \frac{\omega_f - \omega_i}{t_f - t_i} \quad (8.05)$$

Its SI unit is radian per second squared (rad/s²)

The instantaneous angular acceleration α is the limit of the average angular acceleration as time interval approaches to zero:

$$\alpha = \lim_{\Delta t \rightarrow 0} \frac{\Delta \omega}{\Delta t} \quad (8.06)$$

Its SI unit is radian per second squared (rad/s^2)

Activity 8.1

1. Define the following terms:
 - i. Angular velocity
 - ii. Angular acceleration
 - iii. Instantaneous angular velocity
 - iv. Instantaneous angular acceleration
2. Convert 270° into radian
3. A spot on centrifuge motor moves in a circular path of radius 0.2 m. when the spot has travelled a linear distance of 3 m, through what angle has the motor rotated? Write your answer in radians.

8.3.2 Rotational Kinematics

Rotational kinematics describes rotational motion in the same way linear kinematics was described in unit 2. Remember that we are considering acceleration to be constant both rotational and translation kinematics in this module. The equations of rotational motion can be derived in the same way as you derived the equations of motion in translational motion.

For example, consider a rotating object changing its angular speed from ω_i to ω_f during time interval $\Delta t = t$ i.e during time interval $t_i = 0$ to $t_f = t$, then:

$$\begin{aligned} \alpha &= \frac{\omega_f - \omega_i}{t_f - t_i} \\ \alpha &= \frac{\omega_f - \omega_i}{t} \\ \omega_f &= \omega_i + \alpha t \end{aligned} \quad (8.07)$$

You see that equation 8.07 is similar to equation 2.01 in unit 2 which is

$$v = u + at$$

You can therefore derive the other equations of rotational motion in the same way as you did in chapter 2. We summarize these equations in the Table 7.01

Table 7.01: Linear equations versus angular equations

Linear equations with $a = \text{constant}$	Angular acceleration with $a = \text{constant}$
$v = u + at$	$\omega_f = \omega_i + \alpha t$
$\Delta s = ut + \frac{1}{2}at^2$	$\Delta\theta = \omega_i t + \frac{1}{2}\alpha t^2$
$\Delta s = \left(\frac{u + v}{2}\right)t$	$\Delta\theta = \left(\frac{\omega_i + \omega_f}{2}\right)t$
$v^2 = u^2 + 2a\Delta s$	$\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$

The angular equations will be used in the same way equations of linear motion were used in solving problems in unit 2.

Example 8.1

A wheel rotates with a constant angular acceleration of 3.50 rad/s^2 . If the angular speed of the wheel is 2.00 rad/s at $t_i = 0$,

- Through what angle does the wheel rotate between $t_i = 0$ and $t_f = 2.00 \text{ s}$?
- What is the angular speed of the wheel at $t = 2.00 \text{ s}$?

Solution

- $\alpha = 3.50 \text{ rad/s}^2$, $\omega_i = 2.00 \text{ rad/s}$ and $t = 2.00 \text{ s}$

$$\Delta\theta = \omega_i t + \frac{1}{2}\alpha t^2$$

$$\Delta\theta = 2.00 \frac{\text{rad}}{\text{s}(2.00 \text{ s})} + \frac{1}{2} \left(\frac{3.50 \text{ rad}}{\text{s}^2} \right) (2.00)^2$$

$$\Delta\theta = 11.0 \text{ rad}$$

- $\omega_f = \omega_i + \alpha t$

$$\omega_f = 2.00 \text{ rad/s} + (3.5 \text{ rad/s}^2)(2.00 \text{ s})$$

$$\omega_f = 9.00 \text{ rad/s}$$

Example 8.2

An airplane propeller slows from an initial angular speed of 12.5 rev/s to final angular speed of 5.00 rev/s . During this process the propeller rotates through 21.0 revolutions. Find the angular acceleration of the propeller in radians per second squared, assuming its constant.

Solution

$$\Delta\theta = 21.0 \text{ rev} = (21.0)(2\pi \text{ rad}) = 42.0 \text{ rad}$$

$$\omega_i = 12.5 \frac{\text{rev}}{\text{s}} = (12.5) \left(2\pi \frac{\text{rad}}{\text{s}} \right) = 25.0 \text{ rad/s}$$

$$\omega_f = 5.00 \frac{\text{rev}}{\text{s}} = (5.00) \left(2\pi \frac{\text{rad}}{\text{s}} \right) = 10.0\pi \text{ rad/s}$$

$$(10.0\pi \text{ rad/s})^2 = (25.0\pi \text{ rad/s})^2 + 2\alpha(42.0 \text{ rad})$$

$$\alpha = -6.25\pi \text{ rad/s}^2$$

8.3.3 Relationship between angular and linear quantities

Angular variables are closely related to linear variables. Consider an object rotating about the z-axis through the point O. Assume the object rotates through an angle $\Delta\theta$, and hence point P moves through the arc length Δs , in time interval Δt . From the definition of radian measure:

$$\Delta\theta = \frac{\Delta s}{r}$$

Dividing both sides of this equation by Δt , we get:

$$\frac{\Delta\theta}{\Delta t} = \frac{\Delta s}{r\Delta t} \quad (8.08)$$

When Δt is very small, the angle $\Delta\theta$ is also small and $\frac{\Delta\theta}{\Delta t}$ is close to instantaneous angular velocity ω where as $\frac{\Delta s}{\Delta t}$ approaches to instantaneous linear velocity v , then equation 8.08 is equivalent to:

$$\omega = \frac{v}{r} \quad (8.09)$$

The magnitude of v is the linear speed $v = v_t$ called the tangential speed of a particle moving in circular path written as:

$$v_t = r\omega \quad (8.10)$$

If the object changes its angular speed by $\Delta\omega$ in the time interval Δt , this means that tangential speed changes by Δv_t . From equation 7.10, you have:

$$\Delta v_t = r\Delta\omega$$

Dividing both sides of this equation by Δt , you have:

$$\frac{\Delta v_t}{\Delta t} = r \frac{\Delta\omega}{\Delta t} \quad (8.11)$$

As the time interval approaches zero, $\frac{\Delta\omega}{\Delta t}$ approaches instantaneous angular acceleration α and $\frac{\Delta v_t}{\Delta t}$ approaches to instantaneous tangential acceleration α_t . Then equation 8.11 becomes:

$$a_t = r\alpha \quad (8.12)$$

Example 8.3

A compact disc rotates from rest up to an angular speed of 31.4 rad/s in time 0.892 s.

- What is the angular acceleration of the disc, assuming the angular acceleration is uniform?
- Through what angle does the disc turn while coming up to speed?
- If the radius of the disc is 4.45 cm, find the tangential of microbe riding on the rim of the disc when $t=0.892$ s.
- What is the magnitude of the tangential acceleration of the microbe at the given time?

Solution

$$(a) \omega_i = 0, \omega_f = 31.4 \frac{\text{rad}}{\text{s}} \text{ and } t = 0.892 \text{ s}$$

$$\omega_f = \omega_i + \alpha t$$

$$31.4 \frac{\text{rad}}{\text{s}} = 0 \text{ rad/s} + \alpha (0.892 \text{ s})$$

$$\alpha = 35.2 \text{ rad/s}^2$$

$$(b) \Delta\theta = \omega_i t + \frac{1}{2}\alpha t^2$$

$$\Delta\theta = 0(0.892) + \frac{1}{2}\left(35.2 \frac{\text{rad}}{\text{s}^2}\right)(0.892 \text{ s})^2$$

$$\Delta\theta = 14.0 \text{ rad}$$

$$(c) v_t = r\omega$$

$$v_t = (0.0445 \text{ m})(31.4 \text{ rad/s})$$

$$v_t = 1.40 \text{ m/s}$$

$$(d) a_t = r\alpha$$

$$a_t = 0.0445 \text{ m}(35.2 \text{ rad/s})$$

$$a_t = 1.57 \text{ m/s}^2$$

8.3.4 Centripetal acceleration

In uniform circular motion, acceleration of an object is always perpendicular to the path and always points towards the centre of the circular path. It is called centripetal acceleration. ‘Centripetal’ means centre seeking. The magnitude of centripetal acceleration is given by:

$$a_c = \frac{v^2}{r} \quad (8.13)$$

Where a_c is centripetal acceleration, v is the linear speed of the object, and r is the radius of the circular path. Centripetal acceleration is also called radial acceleration because its direction coincides with the radius of the circular path at every point.

Centripetal acceleration is also given by:

$$a_c = r\omega^2 \quad (8.14)$$

8.3.5 Centripetal force

The force that causes centripetal acceleration of an object undergoing circular motion is called centripetal force. It is this force that enables an object to move in a circular path.

For an object of mass m , moving in a circular path of radius r , with constant linear speed v , the magnitude of its centripetal force F_c , is given by Newton's second law of motion:

$$F_c = ma_c = m \frac{v^2}{r} \quad (8.15)$$

When an object moves in a circle but is speeding up or slowing down, a tangential component of acceleration $a_t = r\omega$, is also present. Because the tangential and centripetal components of acceleration are perpendicular to each other, we can find the magnitude of the **total acceleration** with the Pythagorean Theorem:

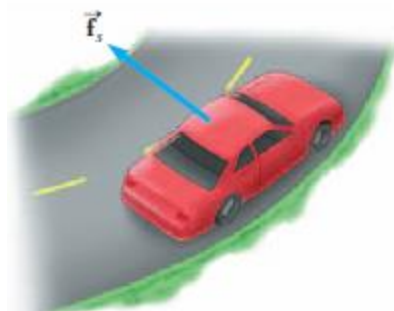
$$a_R = \sqrt{a_t^2 + a_c^2} \quad (8.16)$$

The direction of the resultant acceleration is found by :

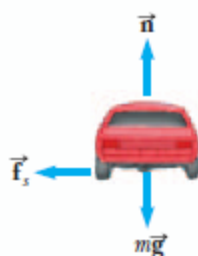
$$\Phi = \tan^{-1}(a_c/a_t) \quad (8.17)$$

Example 8.4

A car travels at a constant speed of 30.0 mi/h (13.4 m/s) on a level circular turn of radius 50.0 m, as shown in the bird's-eye view in Figure 7.13a. What minimum coefficient of static friction, μ_s , between the tires and roadway will allow the car to make the circular turn without sliding?

**Solution:**

The net force acting on the car in the radial direction is the force of static friction toward the centre of the circular path which causes the car to have centripetal acceleration.



$$n - mg = 0$$

$$n = mg$$

$$m \frac{v^2}{r} = f_{s \max} = \mu_s n = \mu_s mg$$

$$\mu_s = \frac{v^2}{rg} = \frac{(13.4 \text{ m/s})^2}{(50.0 \text{ m})(9.80 \text{ m/s}^2)} = 0.366$$

The value of μ_s for rubber on dry concrete is very close to 1, so the car can negotiate the curve with ease. If the road were wet or icy, however, the value for μ_s could be 0.2 or lower. Under such conditions, the radial force provided by static friction wouldn't be great enough to keep the car on the circular path, and it would slide off on a tangent, leaving the roadway.

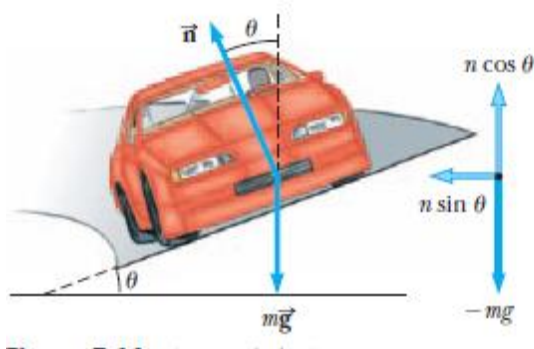
Banking of curved roads and tracks

When a car goes round a level curve, the force of friction between the tyres and the road provides the necessary centripetal force. If the frictional force, which acts as centripetal force and keeps the body moving along the circular road is not enough to provide the necessary centripetal force, the car will skid. In order to avoid skidding, while going round a curved path the outer edge of the road is raised above the level of the inner edge. This is known as banking of curved roads or tracks.

Example 8.5

The Daytona International Speedway in Daytona Beach, Florida, is famous for its races, especially the Daytona 500, held every spring. Both of its courses feature four-story, 31.0° banked curves, with maximum radius of 316 m. If a car negotiates the curve too slowly, it tends to slip down the incline of the turn, whereas if it's going too fast, it may begin to slide up the incline.

- Find the necessary centripetal acceleration on this banked curve so the car won't slip down or slide up the incline. (Neglect friction.)
- Calculate the speed of the race car.

**Solution**

- From the free-fall diagram:

$$n \cos \theta = mg$$

and

$$n = \frac{mg}{\cos \theta}$$

Centripetal force is given by

$$F_c = ma_c$$

This force is equal to $n \sin \theta = \frac{mg}{\cos \theta} \sin \theta = mg \tan \theta$

Equating the two equations, we get:

$$mg \tan \theta = ma_c$$

$$a_c = g \tan \theta = (9.8 \text{ m/s}^2)(\tan 31.0^\circ) = 5.89 \text{ m/s}^2$$

- We can get the speed of the race car by:

$$a_c = \frac{v^2}{r} \rightarrow v = \sqrt{ra_c} = \sqrt{316 \times 5.89} = 43.14 \text{ ms}^{-1}$$

Activity 8.1

1. A 1.50-kg bucket of water is tied by a rope and whirled in a circle with a radius of 1.00 m. At the top of the circular loop, the speed of the bucket is 4.00 m/s.

Determine the acceleration, the net force and the individual force values when the bucket is at the top of the circular loop

8.3.6 Newton's law of universal gravitation

Newton's law of universal gravitation states that: Every particle in the universe attracts every other particle with a force that is proportional to the product of their masses and inversely proportional to the square of the distance between them. This force acts along the line joining the two particles.

If two particles having masses m_1 and m_2 are separated by a distance r , a gravitational force F acts along a line joining them, with magnitude given by

$$F = G \frac{m_1 m_2}{r^2} \quad (9.18)$$

Where $G = 6.67 \times 10^{-11} \text{ kg}^{-1} \text{ m}^3 \text{ s}^{-2}$, is proportionality constant called the constant of universal gravitation. The gravitational force is always attractive.

Example:

A 50 kg Physics student and a 75 kg Physics teacher are sitting on a beach so that their centers are about 50 cm apart. Estimate the magnitude of gravitational force each exerts on the other.

Solution:

$$m_1 = 50 \text{ kg}, m_2 = 75 \text{ kg}, r = 50 \text{ cm} = 0.50 \text{ m}, \text{ and } G = 6.67 \times 10^{-11} \text{ kg}^{-1} \text{ m}^3 \text{ s}^{-2}$$

Then: $F = G \frac{m_1 m_2}{r^2} = \frac{6.67 \times 10^{-11} \text{ kg}^{-1} \text{ m}^3 \text{ s}^{-2} \times 50 \text{ kg} \times 75 \text{ kg}}{(0.50 \text{ m})^2} = 1.0 \times 10^{-6} \text{ N}$ This is unnoticeably small.

In everyday life experience, the gravitational force between two objects is not felt because the masses involved are too small to produce an appreciable value of attractive force.

NOTE

The law of universal gravitation describes a particular force, gravity, and how its magnitude varies with the distance and masses involved.

Newton's second law of motion relates the net force acting on an object to the mass and acceleration of that object.

8.3.7 Gravitational force and weight

Weight is the earth's gravitational force acting on a particular object. In the Equation of the Law of universal gravitation, if one of the object is the earth having mass, M_E , then gravitational force is given by:

$$F_g = mg = G \frac{M_E m}{r^2} \quad (9.19)$$

where g is the acceleration due to gravity.

From Equation 8.19 you can make g the subject of the formula to get:

$$g = G \frac{M_E}{r^2} \quad (9.20)$$

With $r = R_E + h$ where R_E is the radius of the earth and h is the height above sea level.

In equation 9.20, G and M_E are constants, therefore, g varies inversely as r^2 only. If r approaches infinity, g approaches zero. As a result the weight of the object also approaches to zero. This explains the reason why in the outer space object becomes weightless.

Activity 8.2

1. Explain the differences between mass and weight
2. Three 4-kg masses are each at a corner of an equilateral triangle located in space far from other masses.
 - a. If the sides of the triangle are 1.00 m long, find the magnitude of the net force exerted on each of the three masses.
 - b. How does the net force change if the sides of the triangle are halved in length?

8.3.8 Summary

Angles are conveniently measured in radians, where one radian is the angle subtended by an arc whose length is equal to radius, or

$$2\pi \text{ rad} = 360^\circ$$

or

$$1 \text{ rad} \approx 57.3^\circ$$

Angular velocity ω , is defined as the rate of change of angular position:

$$\omega = \frac{\Delta\theta}{\Delta t}$$

Angular acceleration, α , defined as the rate of change of angular velocity:

$$\alpha = \frac{\Delta\omega}{\Delta t}$$

The linear velocity v and acceleration a of a point fixed at a distance r from the axis of rotation are related to ω and α by:

$$v = r\omega, \quad a_{tan} = r\alpha, \quad a_c = \omega^2 r$$

Where a_{tan} and a_c are the tangential and centripetal components of the linear acceleration respectively.

The equations describing uniformly accelerated rotational motion have the same form as for uniformly accelerated linear motion:

$$\begin{aligned} \omega_f &= \omega_i + \alpha t; & \theta &= \omega_i t + \frac{1}{2}\alpha t^2; \\ \omega_f^2 &= \omega_i^2 + 2\alpha\theta; & \omega_{av} &= \frac{\omega_f + \omega_i}{2} \end{aligned}$$

8.3.8 End of unit 8 exercise

1. It is sometimes said that water is removed from clothes in spin dryer by centrifugal force throwing the water outward. Is this correct?
2. Will the acceleration of a car be the same when it travels around a sharp curve at 60 km/h as when it travels around a gentle curve at the same speed? Explain.
3. A bucket of water can be whirled in a vertical circle without water spilling out, even at the top of the circle when the bucket is upside down. Explain.
4. A 150 g ball at the end of a string is revolved uniformly in a horizontal circle of radius 0.600 m. the ball makes 2.00 revolutions in a second. What is its centripetal acceleration?
5. A jet plane travelling 1800 km/h pulls out of a dive by moving in an arc of radius 6.00 km. what is the plane's acceleration in g's?
6. A 25.0 kg child on a merry-go-go round is moving with a speed of 1.35 m/s when 1.20 m from the centre of the merry-go-round. Calculate:
 - (a) The centripetal acceleration of the child

- (b) The net horizontal force exerted on the child
7. A 1200 kg car rounds a curve of radius 70 m banked at an angle of 12.0° . if the car is traveling at 90.0 km/h, will a friction force be required? If so, how much and in what direction?
8. A racing car starts from rest in the pit area and accelerates at a uniform rate to a speed of 35 m/s in 11 s, moving on a circular track of radius 500 m. assuming constant tangential acceleration, find
- (a) The tangential acceleration
- (b) The centripetal acceleration when the speed is 30 m/s
- (c) The tangential and centripetal components of the net force exerted on the car when its speed is 30 m/s and the car's mass is 1000 kg.
9. How large must the coefficient of friction be between the tires and the road if a car is to round a level curve of radius 85 m at a speed of 95 km/h?
10. A 0.150 kg ball on the end of a 1.10 m long cord (negligible mass) is swung in a vertical circle.
- (a) Determine the minimum speed the ball must have at the top of its arc so that it continues moving in a circle.
- (b) Calculate the tension the cord at the bottom of the arc assuming the ball is moving at twice the speed of part (a).

Unit 9

Rotational Dynamics and Equilibrium

9.0 Introduction

In unit 7, you only concentrated on the description of rotational motion regardless of the cause of motion. In this unit your focus will be on rotational dynamics-the cause of rotational motion. This will help you to understand how a net force causes angular acceleration.

You will also deal with a quantity called torque which is similar to the net force in linear motion. Torque is experienced by you in everyday activities e.g when opening doors, walking. You will see how torques achieve static equilibrium of objects.

Centre of mass is another concept you will learn in this unit. You will proceed with this unit looking at rotational momentum and its conservation. Finally you will learn rotational work and energy.

9.1 Objectives

By the end of this unit, you should be able to;

- define torque
- state the two conditions for static equilibrium
- describe centre of mass, centre of gravity and centre of a system of particles
- describe conservation of angular momentum

9.2 Key words

- torque
- static equilibrium
- centre of mass
- centre of gravity
- moment of inertia

9.3 Conservation of angular momentum

9.3.1 Toque

Toque, τ , is defined the measure of the tendency of the force to give an object a turning effect about some axis of rotation.

Mathematically, toque can be defined as the product of the force component perpendicular to the lever arm and the lever arm. Lever arm is the perpendicular distance the axis of rotation to the line of action of the force. Toque is given by:

$$\tau = rF\sin\theta \quad (9.01)$$

Where r is the lever arm, F is the applied force and θ is the angle between r and F .

The SI unit of toque is Nm. This unit of toque is not the same as the unit of work. In work the displacement and the force are in the same line. In toque, however the lever arm and the force are perpendicular to each other. So the SI unit of toque is not a joule.

Equation 9.01 is a general equation for toque and it can be written in two ways:

$$1. \quad \tau = r(F\sin\theta) \quad (9.02)$$

Equation 9.02 defines toque as the product of lever arm, r , and the perpendicular component of the force, $F\sin\theta$.

The force component which is parallel to the lever arm gives zero toque because the angle, θ , is zero as shown in Figure 9.01.

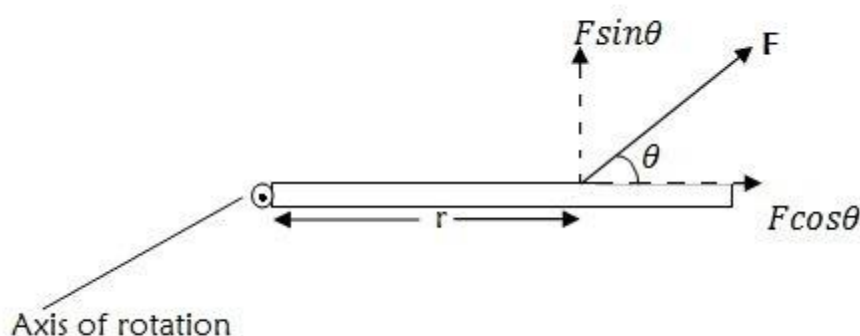


Figure 9.01

$$2. \quad \tau = (r\sin\theta)F \quad (9.03)$$

Equation 8.03 defines toque as the product of the perpendicular lever arm, $r\sin\theta$ to the applied force and the applied force, F as shown in Figure 8.02.

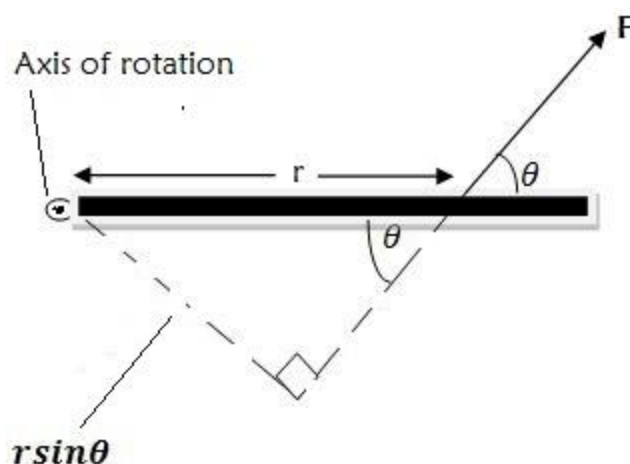


Figure 9.02

From the definition of torque, you see that torque varies directly as both force and lever arm. When the applied force, F , and lever arm are perpendicular (forming a right angle) to each other, then you get maximum torque given by:

$$\tau = rF \quad (9.04)$$

Recall that in linear motion, resultant force causes linear acceleration according to Newton's second law of motion. Likewise here in rotational motion it is the torque that causes angular acceleration of an object. Therefore, resultant torque acting on an object at rest will cause rotational motion of the object.

It is universally agreed that torque is positive in anticlockwise direction and negative in clockwise direction.

Example 9.01

The minimum torque required to remove a bottle top using an opener is 50.0 Nm at the distance from the fulcrum. What tangential force is required to remove the bottle top when:

- (a) Lever arm of the opener is 2 m ?
- (b) Lever arm of the opener is 5 m ?

Solution

- (a) $r = 2 \text{ m}, F = 50.0 \text{ N}$

$$F = \frac{\tau}{r} = \frac{50.0 \text{ Nm}}{2 \text{ m}} = 25.0 \text{ N}$$

- (b) $r = 5 \text{ m}, F = 50.0 \text{ N}$

$$F = \frac{\tau}{r} = \frac{50.0 \text{ Nm}}{5 \text{ m}} = 10.0 \text{ N}$$

Example 9.01 above shows that increasing the moment arm significantly reduces the force required to produce the same turning effect. This principle is very useful when it comes to unscrewing or screwing a nut using spanner. Holding the spanner closer the nut requires a lot of effort to unscrew or to screw the nut. Holding the spanner at further end requires less effort to unscrew or screw the nut.

9.3.2 Toque and angular acceleration

You recall that resultant toque or net toque causes angular acceleration, α . Our major concern now is to establish the relationship that exists between toque and angular acceleration.

You know that an object of mass, m , being acted upon by a net force, F , experiences linear acceleration whose magnitude is given by:

$$a = F/m \quad (9.05)$$

Equation 9.05 is a mathematical expression of Newton's second law of motion.

Equation 9.12 gives us the relationship between linear acceleration and angular acceleration as follows:

$$a = r\alpha \quad (9.06)$$

Substituting Equation 9.06 into Equation 8.05 for linear acceleration, a , you get:

$$\begin{aligned} r\alpha &= \frac{F}{m} \\ \alpha &= \frac{F}{mr} \end{aligned} \quad (9.07)$$

Multiplying both the denominator and the numerator to the right hand side of Equation 9.07 by r , you reach:

$$\alpha = \frac{rF}{mr^2} \quad (9.08)$$

Equation 9.07 tells you that for the applied force, F , which is perpendicular to the lever arm, r , angular acceleration is equal to the toque, $\tau = rF$, divide by moment of inertia, $I = mr^2$

You can rewrite Equation 8.08 as:

$$\alpha = \frac{\tau}{I}$$

From which you get:

$$\tau = I\alpha \quad (9.09)$$

Equation 9.09 is Newton's second law of rotational motion and it is similar to Newton's second law of motion described in kinematics.

Table 9.01 shows the linear quantities and their analogies of rotational quantities with respect to Newton's second law of motion.

Table 9.01

Linear Quantity	Angular Quantity
$F = ma$	$\tau = I\alpha$
m	I
a	α
F	τ

Example 9.2

Activity 9.2

9.3.3 Static Equilibrium

Static equilibrium is a condition in which an object is at rest and remains at a fixed position. The object at static equilibrium does not experience any form of motion in any direction apart from molecular motion.

9.3.3.1 Conditions for static equilibrium

1. The net force acting on the object must be equal to zero

$$\sum F_{net} = 0 \quad (9.10)$$

Writing Equation 6.10 in x and y components, you get:

$$F_x = 0 \text{ and } F_y = 0$$

Condition one is called **translational equilibrium**.

2. The net torque acting on the object must be zero

$$\sum \tau = 0 \quad (9.11)$$

Condition two is called **rotational equilibrium**

For an object to be in static equilibrium, it must satisfy both translational and rotational equilibrium

Example

A woman of mass $m = 55.0 \text{ kg}$ sits on the left end of see-saw-a plank of length $L=4.00 \text{ m}$, pivoted in the middle as shown in the figure below.

- (a) Where should a man of mass 75.0 kg sit if the system is to be balanced?
 (b) Find the normal force exerted by the pivot if the plank has a mass of $m_{pl} = 12.0 \text{ kg}$

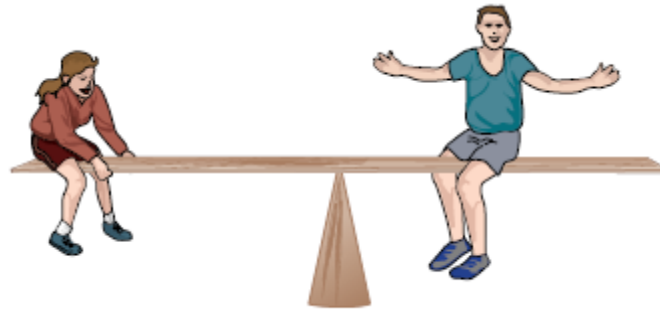


Figure 9.03

Solution

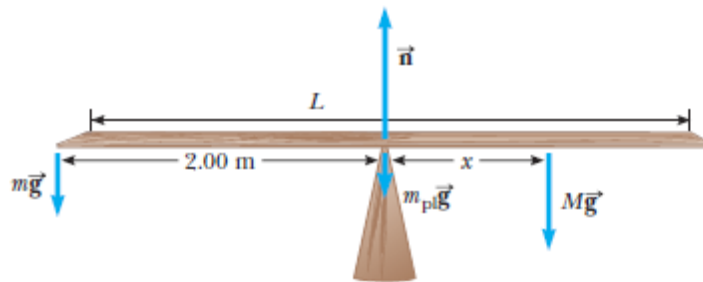


Figure 9.04

- (a) Let x be the distance from the pivot where the man is to sit to balance the system.

Applying the second condition for static equilibrium, $\sum \tau_{net} = 0$ you have:

$$\tau_{pivot} + \tau_{gravity} + \tau_{man} + \tau_{woman} = 0$$

The woman is at a distance of $L/2$ from the pivot. Therefore:

$$0 + 0 - Mgx + mgL/2 = 0$$

$$x = \frac{mgL/2}{Mg} = \frac{mL/2}{M} = \frac{(55.0 \text{ kg})(2.00 \text{ m})}{75.0 \text{ K}} = 1.47 \text{ m}$$

- (b) Apply the first condition of equilibrium to the plank, $\sum F_{net} = 0$:

$$n - mg - Mg - m_{pl}g = 0$$

$$n = (m + M + m_{pl})g = (55.0 \text{ kg} + 75.0 \text{ kg} + 12.0 \text{ kg})(9.8 \text{ m/s}^2)$$

$$n = 1.39 \times 10^3 \text{ N}$$

Activity 9.

1. Explain the two conditions required for an object to be in static equilibrium.

9.3.4 Centre of mass and center of gravity

Until now you have been mainly concerned with the motion of a single particle.

Centre of mass of an object is a single point on the object where the whole mass of the object is considered to concentrate.

Centre of gravity of an object is a single point on an object where the whole weight of the object is considered to act.

If an object has regular shape with uniform mass distribution (homogeneous, symmetric object), then the centre of mass and centre of gravity lie at their geometrical centers.

It is observed that if the object is supported under this centre of gravity point, it can be balanced, or it can be put in equilibrium by this single force. In the following two ways:

1. The whole weight of an object will be balanced by the upward applied force at the centre of gravity makes an object to attain translational equilibrium
2. Since the whole weight is considered to act at this pivot point, its moment arm is zero and the net torque is also zero resulting into rotational equilibrium.

Consider an object of arbitrary shape lying in the xy-plane as in Figure 8.05. the object has a large number of very small particles of weight $m_1g, m_2g, m_3g, \dots$ having coordinates $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots$. If the object is free to rotate around the origin, each particle contributes a torque about the origin that is equal to its weight multiplied by its lever arm. We want to locate a single point of application of a single force of magnitude $w = F_g = Mg$ (the total weight of the object), where the effect on the rotation of the object is the same as that of the individual particles. This point is called center of gravity. Equating the torque exerted by w at the center of gravity to the sum of the torques acting on the individual particles gives:

$$Mgx_{cg} = m_1gx_1 + m_2gx_2 + m_3gx_3 + \dots$$

$$x_{cg} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3 + \dots}{M} \quad (9.12)$$

M is the total mass of the object equal to $m_1 + m_2 + m_3 + \dots$ substituting this into (8.12) you get

$$x_{cg} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3 + \dots}{m_1 + m_2 + m_3 + \dots} = \frac{\sum m_i x_i}{\sum m_i} \quad (9.13a)$$

Where x_{cg} is the x-coordinate of the centre of gravity. Similarly y-coordinate of centre of gravity

$$y_{cg} = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3 + \dots}{m_1 + m_2 + m_3 + \dots} = \frac{\sum m_i y_i}{\sum m_i} \quad (9.13b)$$

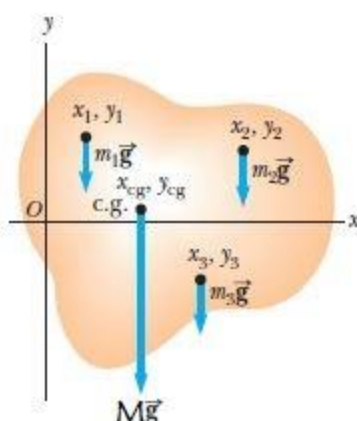


Figure 8.05

Equations 9.13a and 9.13b are similar for the equations for finding centre of mass.

Example

Three particles are located in a coordinate system as shown in Figure 8.06. find the centre of the gravity.

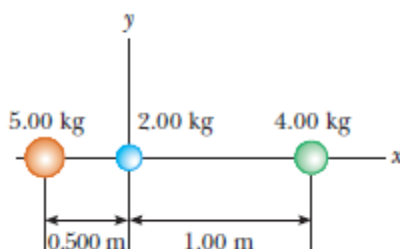


Figure 9.06

All particles lie on x-axis. The centre for gravity in y-coordinate is zero. Applying Equation 9.9a

$$x_{cg} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3 + \dots}{m_1 + m_2 + m_3 + \dots} = \frac{\sum m_i x_i}{\sum m_i}$$

$$x_{cg} = \frac{\sum m_i x_i}{\sum m_i} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3}$$

$$x_{cg} = \frac{(5.0 \text{ kg})(-0.500 \text{ m}) + (2.00 \text{ kg})(0) + (4.00 \text{ kg})(1.00 \text{ m})}{5.00 \text{ kg} + 2.00 \text{ kg} + 4.00 \text{ kg}} = 0.136 \text{ m}$$

9.3.5 Rotational kinetic Energy and moment of inertia

You know now that an object in motion possesses kinetic energy. Motion can either be translational or rotational or a combination of both. In translational motion, the kinetic energy, K , of an object is equal to $\frac{1}{2}mv^2$. In rotational motion this equation cannot work because the speed V of individual particles varies with the radius of every particle from the axis of rotation given by $v = r\omega$. You can see that v is not the same for every particle of the whole object. However, the angular speed, ω , of an object remains constant for all the particles in the object.

Lets now derive the equation that describe rotational kinetic energy.

You already know that kinetic energy for an object undergoing translational motion is given by:

$$KE_t = \frac{1}{2}mv^2 \quad (9.14)$$

But linear speed is given by $v = r\omega$, and substituting this into 8.14 for v you get:

$$KE_r = \frac{1}{2}m(r\omega)^2 = \frac{1}{2}mr^2\omega^2 \quad (9.15)$$

Equation 9.15 shows you that rotational kinetic energy depends on both the square of radius and the square of angular speed. While translational kinetic energy depends only on the square of linear speed. In other words, rotational kinetic energy depends on the distribution of mass of an object.

Now recall that the quantity $I = mr^2$ in equation 9.15 is called moment of inertia.

Then Equation 9.15 then reduces to

$$KE_r = \frac{1}{2}I\omega^2 \quad (9.16)$$

Equation 8.16 describes rotational kinetic energy (KE_r) as the product of moment of inertia, I , and the square of angular acceleration of the rotating object.

Lets now concentrate on how moment of inertial for arbitrary shaped object can be found. Consider a rotating object of arbitrary shaped having a large number of particles of masses $m_1, m_2, m_3, \dots, m_n$ located at distances of $r_1, r_2, r_3, \dots, r_n$ respectively from the axis of rotation as shown in Figure 9.07.

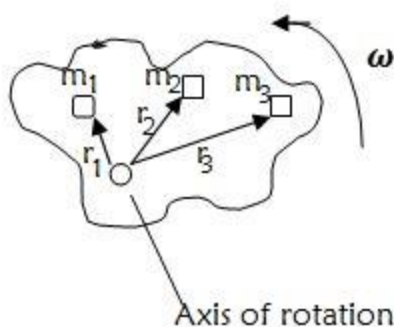


Figure 9.07

The total moment of inertia can be given by:

$$I = \sum m_i r_i^2 = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + \dots + m_n r_n^2 \quad (9.17)$$

The SI unit of moment of inertia is $kg.m^2$

Example

Figure 9.08 shows two objects having equal mass, m , connected either end of a rod rotating about its center. Find the moment of inertia of the system assuming that the mass of the rod is negligible.

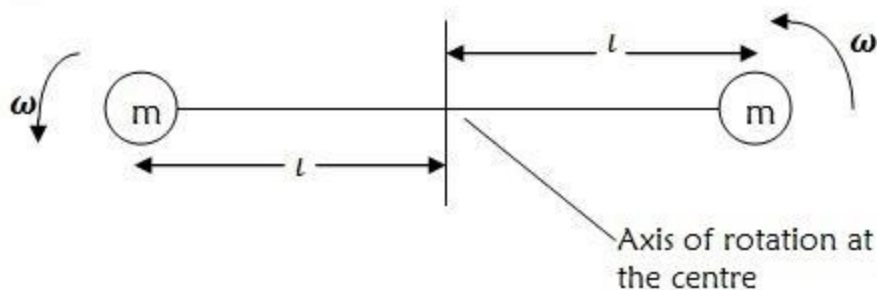


Figure 9.08

Solution

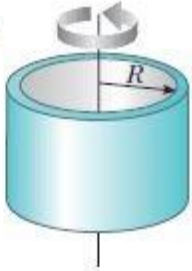
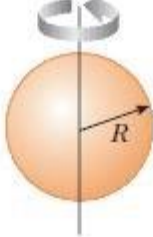
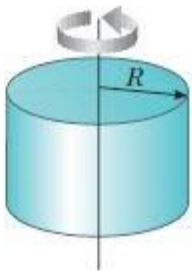
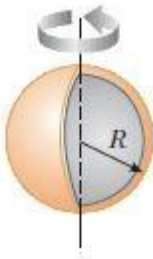
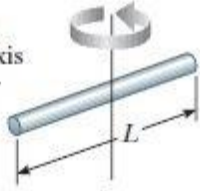
From the definition of moment of inertia:

$$I = \sum m_i r_i^2$$

$$I = ml^2 + ml^2 = 2ml^2$$

The table 9.02 below shows the moment of inertia for some uniform rigid objects of various shapes and having total mass of M .

Table 9.02: Moments of Inertia for Various Rigid objects of Uniform Composition

Hoop or thin cylindrical shell $I = MR^2$ 	Solid sphere $I = \frac{2}{5} MR^2$ 
Solid cylinder or disk $I = \frac{1}{2} MR^2$ 	Thin spherical shell $I = \frac{2}{3} MR^2$ 
Long thin rod with rotation axis through center $I = \frac{1}{12} ML^2$ 	Long thin rod with rotation axis through end $I = \frac{1}{3} ML^2$ 

Moment of inertia plays the same role in rotational motion that inertial mass plays in the translational motion. Thus, moment of inertia measures the resistance/reluctance of a body to start rotating when it is stationary or to stop rotating once it is in rotational motion.

9.3.6 Conservation of Energy

A system such as a bowling ball rolling down a ramp is described by three types of energy: **gravitational potential energy** PE_g , **translational kinetic energy** KE_t , and **rotational kinetic energy** KE_r . All these forms of energy, plus the potential energies of any other conservative forces, must be included in our equation for the conservation of mechanical energy of an isolated system:

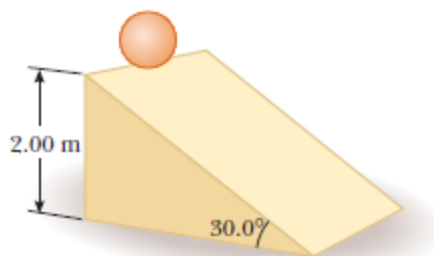
$$(KE_t + KE_r + PE)_i = (KE_t + KE_r + PE)_f \quad (9.18)$$

where i and f refer to initial and final values, respectively, and PE includes the potential energies of all conservative forces in a given problem. This relation is true *only* if we ignore dissipative forces such as friction. In that case, it's necessary to resort to a generalization of the work–energy theorem:

$$W_{nc} = \Delta KE_t + \Delta KE_r + \Delta PE \quad (9.19)$$

Example

A ball of mass M and radius R starts from rest at a height of 2.00 m and rolls down a 30.0° slope, as in Active Figure 8.25. What is the linear speed of the ball when it leaves the incline? Assume that the ball rolls without slipping.



Solution

Applying conservation of energy with $PE = PE_g$:

$$(KE_t + KE_r + PE_g)_i = (KE_t + KE_r + PE_g)_f$$

$$0 + 0 + Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}\left(\frac{2}{5}MR^2\right)\omega^2 + 0$$

The ball rolls without slipping, so $R\omega = v$

$$Mgh = \frac{1}{2}Mv^2 + \frac{1}{5}Mv^2 = \frac{7}{10}Mv^2$$

$$v = \sqrt{\frac{10gh}{7}} = \sqrt{\frac{10(9.8 \text{ m/s}^2)(2.00 \text{ m})}{7}} = 5.29 \text{ m/s}$$

9.3.7 Angular momentum

An object of mass, m , rotates in a circular path of radius r when acted on by a net force, F_{net} . The resulting net torque on the object increases its angular speed from the initial value ω_i to final value ω_f in a time interval Δt . Then, we can write:

$$\sum \tau = I\alpha = I \frac{\Delta\omega}{\Delta t} = I \left(\frac{\omega_f - \omega_i}{\Delta t} \right)$$

We can now define the product:

$$L = I\omega \quad (9.20)$$

as the angular momentum of the object. Then we can write:

$$\sum \tau = \frac{\text{change in angular momentum}}{\text{time interval}} = \frac{\Delta L}{\Delta t} \quad (9.21)$$

This is the rotational analog of Newton's second law and states that the net torque acting on an object is equal to the time rate of change of the object's angular momentum. This equation is similar to the impulse-momentum theorem.

When the net external torque acting on the system is zero, then $\frac{\Delta L}{\Delta t} = 0$, then

$$L_i = L_f \quad (9.22)$$

and so angular momentum is said to be conserved. This is called the conservation of angular momentum.

If the moment of inertia of an isolated rotating system changes, the system's angular speed will change. Therefore, conservation of angular momentum requires that

$$I_i \omega_i = I_f \omega_f \text{ if } \sum \tau = 0 \quad (9.23)$$

The Law of Conservation of angular momentum: it states that the total angular momentum of a rotating object remains constant if the net torque acting on it is zero.

Example

9.3.8 Summary

Torque:

Let $\mathbf{F}$ be force acting on an object, and let $\mathbf{r}$ be position vector from a chosen point O to the point of application of the force. Then the magnitude of the torque τ of the force $\mathbf{F}$ is given by:

$$\tau = rF \sin \theta$$

Where r is the length of the position vector, F is the magnitude of the force and θ is the angle between $\mathbf{F}$ and $\mathbf{r}$.

The torque at O depends on the distance to the point of application of force F and the force's magnitude and direction.

Torque and the two conditions for equilibrium:

1. The net external force must be zero: $\sum \mathbf{F} = 0$
2. The net external torque must be zero: $\sum \tau = 0$

The two conditions result into three equations when solving problems involving rotation in a plane. Two equations are from the first equation corresponding to x- and y-directions of force, and one equation is from the second condition, on torque. The three equations then are solved simultaneously.

The moment of inertia of a group of particles is:

$$I = \sum m v^2$$

For a rigid object which is free to rotate about a fixed axis, having a net external torque $\sum \tau$, acting on it, the object undergoes an angular acceleration α , so that the rotational equivalent of the second law of motion becomes:

$$\sum \tau = I\alpha$$

The relation $a = r\alpha$ is the key in relating the translational equations to the rotational equations.

If a rigid object rotates about a fixed axis with angular speed ω , its rotational kinetic energy is:

$$KE_r = \frac{1}{2}I\omega^2$$

I is the object's moment of inertia.

The conservation of energy for isolated rotating object is given by:

$$(KE_t + KE_r + PE)_i = (KE_t + KE_r + PE)_f$$

where i and f refer to initial and final values of respectively.

A ball rotating down inclined plane converts potential energy to translational and rotational energy.

When non-conservative forces are present, we can write the generalization of work-energy theorem as:

$$W_{nc} = \Delta KE_t + \Delta KE_r + \Delta PE$$

The angular momentum of a rotating object is given by:

$$L = I\omega$$

Angular momentum is related to torque as:

$$\sum \tau = \frac{\text{Change in angular momentum}}{\text{time interval}} = \frac{\Delta L}{\Delta t}$$

In an isolated rotating object where the net external torque acting on an object is zero, the total angular momentum of the system is constant i.e

$$L_i = L_f$$

and is said to be conserved. Conservation of angular momentum can also be written as:

$$I_i\omega_i = I_f\omega_f$$

9.3.9 End of unit 8 Exercise

1. What is the maximum exerted by a 55 kg person riding a bike if the rider puts all her weight on each pedal when climbing a hill? The pedals rotate in a circle of radius 17 cm.
2. A person exerts a force of 45 N on the end of a door 84 cm wide. What is the magnitude of the torque if the force is exerted (a) perpendicular to the door (b) at 60° angle to the face of the door?
3. A system consisting of two masses connected by a massless rod lies along the X-axis. A 0.4 kg mass is at a distance $x = 2$ m while a 0.6 kg mass is at $x = 7$ m. Find the x coordinate of the centre of mass.
4. Locate the centre of mass of a system of bodies of masses $m_1 = 1$ kg, $m_2 = 2$ kg and $m_3 = 3$ kg situated at the corners of an equilateral triangle of side 1 m.
5. A circular disc of mass m and radius r is set rolling on a table. If ω is its angular velocity, show that its total energy $E = \frac{3}{4}mr^2\omega^2$.
6. A thin metal ring of diameter 0.6 m and mass 1 kg starts from rest and rolls down on an inclined plane. Its linear velocity on reaching the foot of the plane is 5 ms^{-1} , calculate (i) the moment of inertia of the ring and (ii) the kinetic energy of rotation at that instant.
7. A solid cylinder of mass 200 kg rotates about its axis with angular speed 100 rads^{-1} . The radius of the cylinder is 0.25 m. What is the kinetic energy associated with the rotation of the cylinder? What is the magnitude of the angular momentum of the cylinder about its axis?
8. A person weighing 45 kg sits on one end of a seesaw while a boy of 15 kg sits on the other end. If they are separated by 4 m, how far from the boy is the centre of mass situated. Neglect weight of the seesaw.

PART 2

THERMAL PROPERTIES OF MATTER

Outline.

2.0 Introduction.

2.1 Objectives.

2.2 Key terms.

Unit 1: Thermodynamics.

Unit 2: Ideal gases.

Unit 3: Heat and the first law of thermodynamics.

1.0 Introduction

Thermal properties are the response of matter to applied heat or sources of different temperature. In everyday language we use the terms heat and temperature loosely as if they had the same meaning. In physics they have different meanings. Temperature is the property of matter which reflects the quantity of energy of motion of the component particles. On the other hand, Heat energy (or just heat) is a form of energy which transfers among particles in a substance (or system) by means of kinetic energy of those particle.

1.1 Objectives

By the end of this section, students must be able to:

- Understand the different scales used in measurement of temperature.
- Convert temperature from one scale to another.
- Explain the absolute temperature.
- Illustrate thermal expansion.
- Define the terms heat capacity and specific heat capacity.
- Deduce the formulae for heat capacity and specific heat capacity.
- Calculate specific heat capacities.
- Describe calorimeter.
- Identify latent heat capacities.
- Verify Newton's law of cooling.

1.2 Key terms

Temperature, heat, and internal energy, thermal contact, heat transfer, Thermal equilibrium, Heat capacity, specific heat capacity.

Unit 1

THERMODYNAMICS

1.0 Introduction

Thermodynamics is the branch of Physics devoted to study of heat and temperature and their relations to energy and work. It defines macroscopic variables such as internal energy, entropy and pressure that partly describe a body of matter of radiation. Most studies of thermodynamics are primarily concerned with two forms of energy – heat and work. Thermodynamics study includes quantitative analysis of machine and processes for transformation of energy and between work and heat. In classical thermodynamics a macroscopic viewpoint is taken regarding such matters. The term thermodynamics was first introduced by Lord Kelvin in 1849. The term comes from the Greek words *thermal* (heat) and *dynamics* (power).

1.1 Objectives

By the end of this unit, students must be able to:

- Distinguish between temperature scales.
- Describe the calibration of different thermometers.
- State the Zeroth Law of Thermodynamics.

1.2 Key terms;

Heat, temperature, pressure and internal energy.

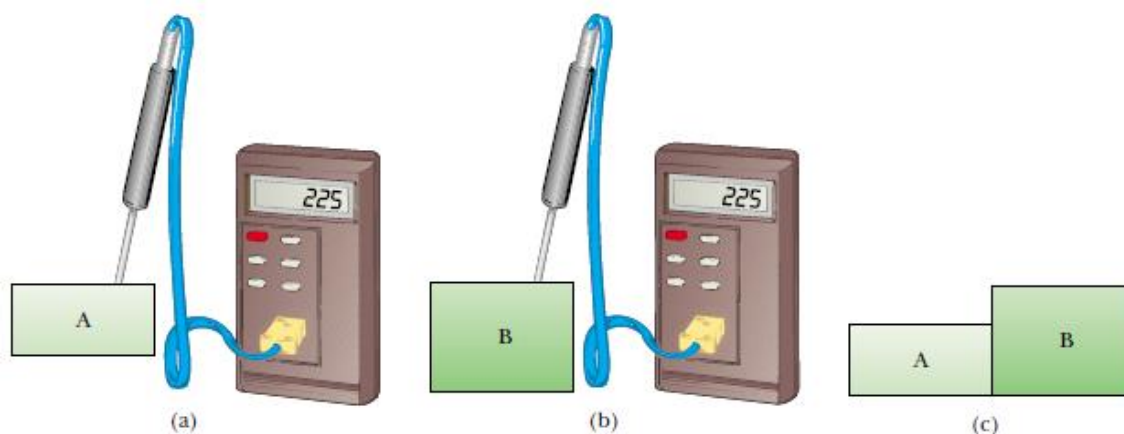
1.3 TEMPERATURE AND THE ZEROth LAW OF THERMODYNAMICS

1.3.1 Temperature

- We often associate the concept of temperature with how hot or cold an object feels when we touch it.
- To understand the concept of temperature, it is useful to define two often used phrases: *thermal contact* and *thermal equilibrium*.

- We are all familiar with the fact that two objects at different initial temperatures eventually reach some intermediate temperature when placed in contact with each other.
- For example, when an ice cube is dropped into a cup of hot coffee, it melts and the coffee's temperature decreases.
- *Thermal contact*; let us imagine that two objects are placed in an insulated container such that they interact with each other but not with the rest of the world. If the objects are at different temperatures, energy is exchanged between them, even if they are initially not in physical contact with each other.
- *Thermal equilibrium* is a situation in which two objects in thermal contact with each other cease to exchange energy by the process of heat.
- **Heat** is the transfer of energy from one object to another object as a result of a difference in temperature between the two.

i. Zeroth Law of Thermodynamics



The zeroth law of thermodynamics. (a) and (b) If the temperatures of A and B are measured to be the same by placing them in thermal contact with a thermometer (object C), no energy will be exchanged between them when they are placed in thermal contact with each other (c).

- *Zeroth law states that; if objects A and B are separately in thermal equilibrium with a third object C, then objects A and B are in thermal equilibrium with each other.*
- Two objects in thermal equilibrium with each other are at the same temperature.
- Conversely, if two objects have different temperatures, then they are not in thermal equilibrium with each other

1.4 THERMOMETERS AND THE TEMPERATURE SCALES

- *Thermometers* are devices that are used to define and measure temperatures.
- All thermometers are based on the principle that some physical property of a system changes as the system's temperature changes.
- Some physical properties that change with temperature are:
 1. The volume of a liquid.
 2. The length of a solid.
 3. The pressure of a gas at constant volume.
 4. The volume of a gas at constant pressure.
 5. The electric resistance of a conductor.
 6. The colour of an object. For a given substance and a given temperature range,
 - A temperature scale can be established on the basis of any one of these physical properties.
 - A common thermometer in everyday use consists of a mass of liquid—usually mercury or alcohol—that expands into a glass capillary tube when heated.
 - In this case the physical property that changes is the volume of a liquid.
 - Any temperature change in the range of the thermometer can be defined as being proportional to the change in length of the liquid column.
 - We can calibrate a thermometer by placing it in thermal contact with some natural systems that remain at constant temperature.
 - One such system is a mixture of water and ice in thermal equilibrium at atmospheric pressure.
 - On the ***Celsius temperature scale***, this mixture is defined to have a temperature of zero degrees Celsius, which is written as 0°C ; this temperature is called the *ice point* of water.
 - Another commonly used system is a mixture of water and steam in thermal equilibrium at atmospheric pressure; its temperature is 100°C , which is the *steam point* of water.
 - Once the liquid levels in the thermometer have been established at these two points, the length of the liquid column between the two points is divided into 100 equal segments to create the ***Celsius scale***.

- Thus, each segment denotes a change in temperature of one Celsius degree.

1.4.1.1 PROBLEMS WITH LIQUID IN GLASS THERMOMETERS.

1. No extreme accuracy when needed; because mercury and alcohol have different thermal expansion properties,
 2. The limited range of temperatures over which it can be used. E.g. A mercury thermometer cannot be used below the freezing point of mercury, which is -39°C , and an alcohol thermometer is not useful for measuring temperatures above 85°C .
- To overcome this problem, we need a universal thermometer whose readings are independent of the substance used in it. Hence the use of *Gas Thermometer*.

1.4.2 THE CONSTANT-VOLUME GAS THERMOMETER AND THE ABSOLUTE TEMPERATURE SCALE

- The Physical change exploited in the constant-volume gas thermometer is the *variation of pressure of a fixed volume of gas with temperature*.
- We can calibrate a gas thermometer like the one shown in the figure 2.4.2 below by immersing the flask in an ice-water bath, and mercury reservoir *B* raised or lowered until the top of the mercury in column *A* is at the zero point on the scale.

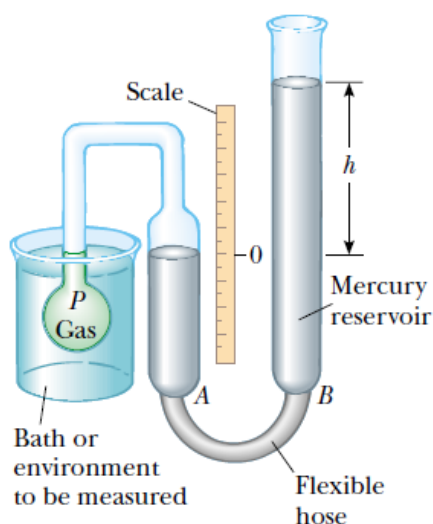


Figure 2.4.2

- The height h , the difference between the mercury levels in reservoir B and column A , indicates the pressure in the flask at 0°C .
- The flask is then immersed in water at the steam point, and reservoir B readjusted until the top of the mercury in column A is again at zero on the scale; this is to ensure that the gas's volume was the same as it was when the flask was in the ice bath. Hence "constant volume."
- This adjustment of reservoir B gives a value for the gas pressure at 100°C .
- These two pressure and temperature values can be plotted, as shown in Figure 2.4.3.
- The line connecting the two points serves as a calibration curve for unknown temperatures.

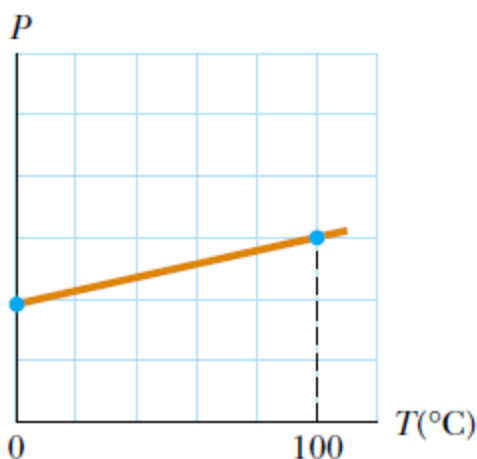


Figure 2.4.3

A graph of pressure versus temperature taken with a constant-volume gas thermometer. The two dots represent known reference temperatures (the ice and steam points of water).

- Now let us suppose that temperatures are measured with gas thermometers containing different gases at different initial pressures.
- Experiments show that the thermometer readings are nearly independent of the type of gas used, as long as the gas pressure is low and the temperature is well above the point at which the gas liquefies (Fig. 2.4.4).
- The agreement among thermometers using various gases improves as the pressure is reduced.

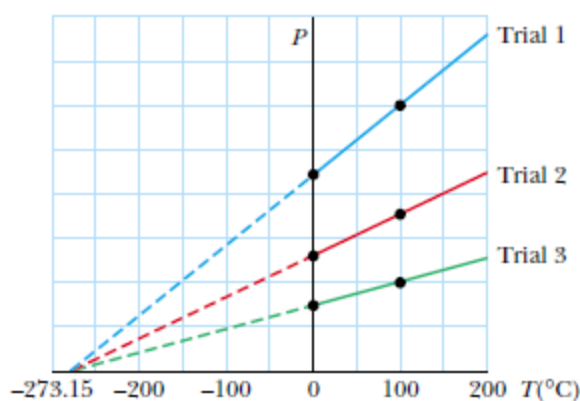


Figure 2.4.4

- If we extend the straight lines in Figure 2.4.4 toward negative temperatures, we find that **in every case, the pressure is zero when the temperature is -273.15°C.**
- This temperature is used as the basis for the **absolute temperature scale/kelvin scale**, which sets -273.15°C as its zero point.
- This temperature is often referred to as **absolute zero**.
- The size of a degree on the absolute temperature scale is chosen to be identical to the size of a degree on the Celsius scale.
- Because the size of a degree is the same on the two scales, a temperature difference of 5°C is equal to a temperature difference of 5 K.
- Thus, the conversion between these temperatures is

$$T_c = T - 273.15$$

Where T_c is the Celsius temperature and T is the absolute temperature.

- This new absolute temperature scale (also called the Kelvin scale) employs the SI unit of absolute temperature, the *kelvin*, which is defined to be **1/273.16 of the difference between absolute zero and the temperature of the triple point of water.**
- Triple point of water is the single combination of temperature and pressure at which liquid water, gaseous water, and ice (solid water) coexist in equilibrium.
- This triple point occurs at a temperature of 0.01°C and a pressure of 4.58 mm of mercury.

1.4.2.1. THE CELSIUS, FAHRENHEIT, AND KELVIN TEMPERATURE SCALES.

- A common temperature scale in everyday use in the United States is the **Fahrenheit scale.**
- This scale sets the temperature of the ice point at 32°F and the temperature of the steam point at 212°F.
- The relationship between the Celsius and Fahrenheit temperature scales is

$$T_F = \frac{9}{5}T_C + 32^\circ F$$

Example:

An object has the temperature of 50°F, what is its temperature in degrees Celsius and in kelvins?

Solution:

From the equation above we get,

$$\begin{aligned} T_C &= \frac{5}{9}(T_F - 32) \\ &= \frac{5}{9}(50 - 32) \\ &= 10^\circ \text{C} \end{aligned}$$

$$T_c = T - 273.15, \text{ hence } T = T_c + 273.15$$

$$= 10^\circ\text{C} + 273.15$$

$$= 283.15 \text{ K}$$

Activity 1.4

1. A pan of water is heated from 25°C to 80°C . What is the change in its temperature on the Kelvin scale and on the Fahrenheit scale? (phy 4 sci & engine pg 586)
2. A spray can containing a propellant gas at twice atmospheric pressure (202 kPa) and having a volume of 125 cm^3 is at 22°C . It is then tossed into an open fire. When the temperature of the gas in the can reaches 195°C , what is the pressure inside the can? Assume any change in the volume of the can is negligible.

1.5 THERMAL EXPANSION

- Thermal expansion is a consequence of the change in the average separation between the atoms in an object due to heat.
- Thermal expansion is as a result of increase in temperature, so when the temperature increases; the molecules gain more kinetic energy and they move faster than average speed and consequently things tend to expand.
- If thermal expansion is sufficiently small relative to an object's initial dimensions, the change in any dimension is, to a good approximation, proportional to the first power of the temperature change.
- Suppose that an object has an initial length L_i along some direction at some temperature and that the length increases by an amount ΔL for a change in temperature ΔT .
- Considering the fractional change in length per degree of temperature change, we define the **average coefficient of linear expansion** as:

$$\alpha \equiv \frac{\Delta L / L_i}{\Delta T}$$

- The equation above could be rewritten as:

$$\Delta L = \alpha L_i \Delta T \text{ (Linear expansion)}$$

This shows that “The change in length of an object is proportional to the change in temperature”

or as

$$L_f - L_i = \alpha L_i (T_f - T_i)$$

Where L_f is the final length, T_f and T_i are the final and the initial temperatures and the proportionality constant α is the average coefficient of linear expansion for a given material and has units of $^{\circ}\text{C}^{-1}$.

- Table 2.4 lists the average coefficient of linear and volume expansion for various materials

Table 2.4

Average Coefficients of Expansion for Some Materials Near Room Temperature			
Material	Average Coefficient of Linear Expansion [$(^{\circ}\text{C})^{-1}$]	Material	Average Coefficient of Volume Expansion [$(^{\circ}\text{C})^{-1}$]
Aluminum	24×10^{-6}	Ethyl alcohol	1.12×10^{-4}
Brass and bronze	19×10^{-6}	Benzene	1.24×10^{-4}
Copper	17×10^{-6}	Acetone	1.5×10^{-4}
Glass (ordinary)	9×10^{-6}	Glycerin	4.85×10^{-4}
Glass (Pyrex [®])	3.2×10^{-6}	Mercury	1.82×10^{-4}
Lead	29×10^{-6}	Turpentine	9.0×10^{-4}
Steel	11×10^{-6}	Gasoline	9.6×10^{-4}
Invar (Ni-Fe alloy)	0.9×10^{-6}	Air	3.67×10^{-3}
Concrete	12×10^{-6}	Helium	3.665×10^{-3}

- Note that for these materials α is positive, indicating an increase in length with increasing temperature.
- This is not always the case. Some substances-calcite (CaCO_3) is one example - expand along one dimension (positive α) and contract along another (negative α) as their temperatures are increased.
- Because the linear dimensions of an object change with temperature, it follows that surface area and volume change as well.
- The change in volume at constant pressure is proportional to the initial volume V_i and to the change in temperature according to the relationship

$$\Delta V = \beta V_i \Delta T$$

- Where β is the **average coefficient of volume expansion**.
- For a solid, the average coefficient of volume expansion is approximately three times the average linear expansion coefficient: $\beta = 3\alpha$

Proof

Consider a box of dimensions ℓ , w and h . Its volume at some temperature T_i is $V_i = \ell w h$.

If the temperature changes to $T_i + \Delta T$ its volume changes to $V_i + \Delta V$ where each dimension changes according to equation $[\Delta L = \alpha L_i \Delta T]$

Therefore,

$$\begin{aligned} V_i + \Delta V &= (\ell + \Delta \ell) (w + \Delta w) (h + \Delta h) \\ &= (\ell + \alpha \ell \Delta T) (w + \alpha w \Delta T) (h + \alpha h \Delta T) \\ &= \ell w h (1 + \alpha \Delta T)^3 \\ &= V_i [1 + 3\alpha \Delta T + 3(\alpha \Delta T)^2 + (\alpha \Delta T)^3] \end{aligned}$$

Now dividing both sides by V_i and then isolate the term $\Delta V/V_i$, we obtain:

$$\frac{\Delta V}{V_i} = 3\alpha \Delta T + 3(\alpha \Delta T)^2 + (\alpha \Delta T)^3$$

Because $\alpha \Delta T \ll 1$ for typical values of ΔT ($< \sim 100^\circ\text{C}$), we can neglect the terms $3(\alpha \Delta T)^2$ and $(\alpha \Delta T)^3$. Upon making this approximation, we see that

$$\frac{\Delta V}{V_i} = 3\alpha \Delta T$$

And making ΔV subject, the equation above becomes

$$\Delta V = 3\alpha V_i \Delta T.$$

Comparing with the equation $\Delta V = \beta V_i \Delta T$, $3\alpha = \beta$.

Example:

A segment of steel railroad track has a length of 30.000 m when the temperature is 0.0°C . What is its length when the temperature is 40.0°C ?

Solution

$$\Delta L = \alpha L_i \Delta T$$

From the table 2.4, α for steel is $11 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$

$$\begin{aligned}\therefore \Delta L &= [11 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}](30.000 \text{ m})(40.0 \text{ } ^\circ\text{C}) \\ &= 0.013 \text{ m}\end{aligned}$$

If the track is 30.000 m long at 0.0°C , its length at 40.0°C is 30.013 m

Activity 2.1

3. Show that the change in area of a rectangular plate is given by $\Delta A = 2\alpha A_i \Delta T$

1.5.1 BIMETALLIC STRIPS AND THERMOSTATS

- Bimetallic strip: When the temperatures of a brass rod and a steel rod of equal length are raised by the same amount from some common initial value, the brass rod expands more than the steel rod because brass has a larger coefficient of expansion than steel.
- Such strips can be found in the thermostats of certain home heating systems.
- The strip is made by securely bonding two different metals together. As the temperature of the strip increases, the two metals expand by different amounts and the strip bends. As figure 1.5.1 shows

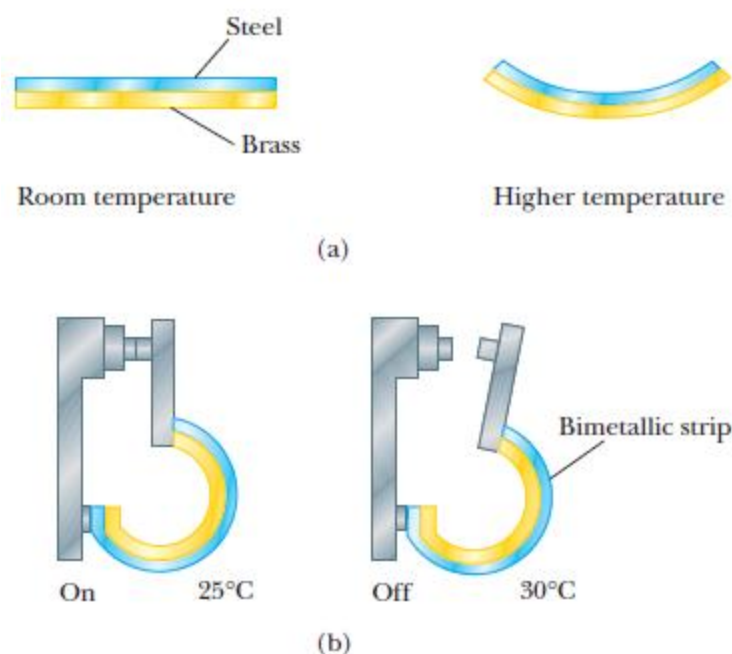


Figure 1.5.1

1.5.2 THE UNUSUAL BEHAVIOUR OF WATER

- Liquids generally increase in volume with increasing temperature and have average coefficients of volume expansion about ten times greater than those of solids.
- Cold water is an exception to this rule, as we can see from its density-versus-temperature curve, shown in Figure 1.5.2.

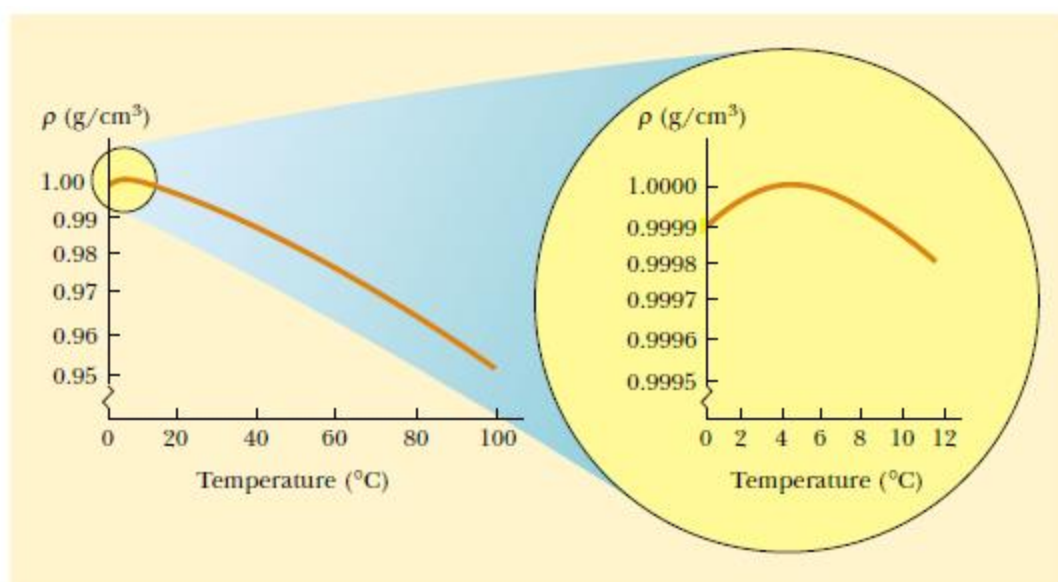


Figure 1.5.2

- As the temperature increases from 0°C to 4°C , water contracts and thus its density increases.
- Above 4°C , water expands with increasing temperature, and so its density decreases.
- Thus, the density of water reaches a maximum value of 1.000 g/cm^3 at 4°C .

UNIT SUMMARY

- Two objects are in **thermal equilibrium** with each other if they do not exchange energy when in thermal contact.
- The **zeroth law of thermodynamics** states that if objects A and B are separately in thermal equilibrium with a third object C, then objects A and B are in thermal equilibrium with each other.
- **Temperature** is the property that determines whether an object is in thermal equilibrium with other objects.
- **Two objects in thermal equilibrium with each other are at the same temperature.**
- The SI unit of absolute temperature is the **kelvin**, which is defined to be the fraction $1/273.16$ of the temperature of the triple point of water
- When the temperature of an object is changed by an amount ΔT , its length changes by an amount ΔL that is proportional to ΔT and to its initial length L_i :

$$\Delta L = \alpha L_i \Delta T$$

*where the constant α is the **average coefficient of linear expansion**. The **average coefficient of volume expansion** ' for a solid is approximately equal to 3α .*

- Thermometers measure temperature and are based on physical properties, such as the temperature-dependent expansion or contraction of a solid, liquid, or gas.
- The Kelvin temperature scale takes its zero point as **absolute zero** ($0\text{K} = -273.15\text{ }^\circ\text{C}$), the point at which, by extrapolation, the pressure of all gases falls to zero.

UNIT TEST

1. A piece of copper is dropped into a beaker of water. If the water's temperature rises, what happens to the temperature of the copper? Under what conditions are the water and copper in thermal equilibrium?
2. When the metal ring and metal sphere in Figure T1.2 are both at room temperature, the sphere can just be passed through the ring. After the sphere is heated, it cannot be passed through the ring. Explain. **What If?** What if the ring is heated and the sphere is left at room temperature? Does the sphere pass through the ring?



Figure T1.2

3. A copper telephone wire has essentially no sag between poles 35.0 m apart on a winter day when the temperature is -20.0°C . How much longer is the wire on a summer day when $T_c = 35.0^{\circ}\text{C}$?
4. A constant-volume gas thermometer is calibrated in dry ice (that is, carbon dioxide in the solid state, which has a temperature of -80.0°C) and in boiling ethyl alcohol (78.0°C). The two pressures are 0.900 atm and 1.635 atm.
 - a. What Celsius value of absolute zero does the calibration yield?
 - b. What is the pressure at
 - i. The freezing point of water?
 - ii. The boiling point of water?

5. A hollow aluminium cylinder 20.0 cm deep has an internal capacity of 2.000 L at 20.0°C. It is completely filled with turpentine and then slowly warmed to 80.0°C.
- How much turpentine overflows?
 - If the cylinder is then cooled back to 20.0°C, how far below the cylinder's rim does the turpentine's surface recede?

Unit 2

IDEAL GAS

2.0 Introduction

An ideal gas is defined as one in which all collisions between atoms or molecules are perfectly elastic and in which there are no intermolecular attractive forces. One can visualise it as a collection of perfectly hard spheres which collide but which otherwise do not interact with each other. In such a gas, internal energy is in the form of kinetic energy and any change in internal energy is accompanied by a change in temperature. An ideal gas can be characterized by three state variables; Absolute pressure (P), Volume (V) and absolute temperature (T).

2.1 Objectives

By the end of this unit, students must be able to:

- Explain relationship between the ideal gas constant and Boltzmann's constant.
- Describe the proportional relationships between pressure, volume, amount and temperature.
- Know how equations of state can be written in terms of the molar volume.

2.2 Keywords:

Ideal gas

2.3 *MACROSCOPIC DESCRIPTION OF AN IDEAL GAS*

- The volume expansion equation $\Delta V = \beta V_i \Delta T$ is based on the assumption that the material has an initial volume V_i before the temperature change occurs.
- This is the case for solids and liquids because they have a fixed volume at a given temperature.

- The case for gases is completely different because they don't have "standard" volume at a given temperature.
- As a result, we cannot express changes in volume ΔV in a process on a gas with Equation $\Delta V = \beta V_i \Delta T$
- For a gas, it is useful to know how the quantities volume V , pressure P , and temperature T are related for a sample of gas of mass m .
- In general, the equation that interrelates these quantities, called the *equation of state*, is very complicated, but is found experimentally to be relatively simple if the gas is maintained at a low pressure (or a low density)
- Most gases at room temperature and atmospheric pressure behave approximately as ideal gases.
- **An ideal gas is a collection of atoms or molecules that move randomly and exert no long-range forces on each other. Each particle of the ideal gas is individually point-like, occupying a negligible volume.**
- A gas usually consists of a very large number of particles, so it's convenient to express the amount of gas in a given volume in terms of the number of **moles**, n .

$$n = \frac{m}{\text{molar mass}} \quad \left(n = \frac{m}{M} \right)$$

Where n the numbers of moles, m is the mass of the substance and M is the mass of one mole

- One mole contains $N_A = 6.022 \times 10^{23}$ particles: Avogadro's number
- **One mole (mol) of any substance is that amount of the substance that contains as many particles (atoms, molecules, or other particles) as there are atoms in 12 g of the isotope carbon-12**
- Because there are 6.02×10^{23} particles in one mole of *any* element, the mass per atom for a given element is

$$m_{\text{atom}} = \frac{\text{molar mass}}{N_A}$$

- Now suppose an ideal gas is confined to a cylindrical container with a volume that can be changed by moving a piston, as in figure 2.1

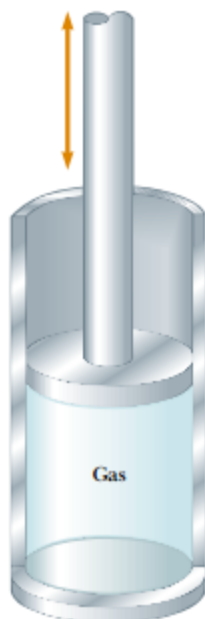


Figure 2.1

- Assume that the cylinder doesn't leak, so the number of moles remains constant. Experiments yield the following observations:
- First, when the gas is kept at a constant temperature, its pressure is inversely proportional to its volume (Boyle's law).
- Second, when the pressure of the gas is kept constant, the volume of the gas is directly proportional to the temperature (Charles's law.)
- Third, when the volume of the gas is held constant, the pressure is directly proportional to the temperature (Gay-Lussac's law).
- These observations can be summarized by the following equation of state, known as the **ideal gas law**:

$$PV = nRT$$

R is a constant and n is the number of moles of gas in the sample.

$$R = 8.314 \text{ J/mol.K}$$

(When the pressure is expressed in Pascal's and volume in cubic centimetres)

or

$$R = 0.08214 \text{ L.atm/mol.K}$$

(When the pressure is expressed in atmospheres and volume in litres)

The ideal gas law states that if the volume and temperature of a fixed amount of gas do not change, then the pressure also remains constant.

- The ideal gas law can be derived from the kinetic theory of gases and relies on the assumptions that:
 1. The gas consists of a large number of molecules, which are in random motion and obey Newton's laws of motion.
 2. The volume of the molecules is negligibly small compared to the volume occupied by the gas.
 3. No forces act on the molecules except during elastic collisions of negligible duration.
- It is common to call quantities such as P, V, and T the **thermodynamic variables** of an ideal gas.

Example:

An ideal gas occupies a volume of 100 cm³ at 20°C and 100 Pa. Find the number of moles of gas in the container.

Solution:

The quantities given are volume, pressure, and temperature: $V = 100 \text{ cm}^3 = 1.00 \times 10^{-4} \text{ m}^3$, $P = 100 \text{ Pa}$, $T = 20^\circ\text{C} = 293 \text{ K}$.

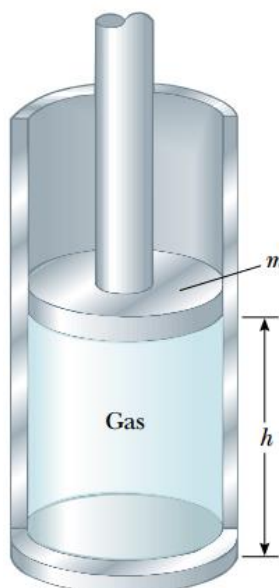
From the equation

$$PV = nRT \quad \text{we find that}$$

$$\begin{aligned} n &= \frac{PV}{RT} \\ &= \frac{(100 \text{ Pa})(1.00 \times 10^{-4} \text{ m}^3)}{(8.314 \text{ J/mol.K})(293 \text{ K})} \\ &= 4.11 \times 10^{-6} \text{ mol} \end{aligned}$$

Activity

1. A vertical cylinder of cross-sectional area $A=0.001 \text{ m}^2$ is fitted with a tight-fitting, frictionless piston of mass $m = 20.0 \text{ kg}$ (see the Figure below)



If $n = 0.200$ moles of an ideal gas are in the cylinder at a temperature of $T = 350 \text{ K}$, what is the height h at which the piston is in equilibrium under its own weight?

- The number of molecules contained in one mole of any gas is Avogadro's number, $N_A = 6.02 \times 10^{23}$ particles/mol, so.

$$n = \frac{N}{N_A}$$

where n is the number of moles and N is the number of molecules in the gas.

With Equation above, we can rewrite the ideal gas law in terms of the total number of molecules as:

$$PV = nRT = \frac{N}{N_A} RT$$

Or

$$PV = Nk_B T$$

Where $k_B = \frac{R}{N_A} = 1.38 \times 10^{-23} \text{ J/K}$. Boltzmann's constant.

UNIT SUMMARY

An **ideal gas** is one for which PV/nT is constant. An ideal gas is described by the **equation of state**,

$$PV = nRT$$

where n equals the number of moles of the gas, V is its volume, R is the universal gas constant (8.314 J/mol.K), and T is the absolute temperature. A real gas behaves approximately as an ideal gas if it has a low density.

Avogadro's number is $N_A = 6.02 \times 10^{23}$ particles/mol.

A mole of anything, by definition, consists of an Avogadro's number of particles.

The number is defined so that one mole of carbon-12 atoms has a mass of exactly 12g.

The mass of one mole of a pure substance in grams is the same, numerically, as that substance's atomic (or molecular) mass.

The quantity k_B is **Boltzmann's constant** ($1.38 \times 10^{-23} \text{ J/K}$).

UNIT TEST

1. An auditorium has dimensions 10.0 m x 20.0 m x 30.0 m. How many molecules of air fill the auditorium at 20.0°C and a pressure of 101 kPa?
2. The mass of a hot-air balloon and its cargo (not including the air inside) is 200 kg. The air outside is at 10.0°C and 101 kPa. The volume of the balloon is 400 m³. To what temperature must the air in the balloon be heated before the balloon will lift off? (Air density at 10.0°C is 1.25 kg/m³).
3. An ideal gas at 20.0°C and a pressure of 1.50×10^5 Pa is in a container having a volume of 1.00 L.
 - a. Determine the number of moles of gas in the container.
 - b. The gas pushes against a piston, expanding to twice its original volume, while the pressure falls to atmospheric pressure. Find the final temperature.
4. Suppose the temperature of 4.50 L of ideal gas drops from 375 K to 275 K.
 - a. If the volume remains constant and the initial pressure is atmospheric pressure, find the final pressure.
 - b. Find the number of moles of gas.

Unit 3

HEAT AND FIRST LAW OF THERMODYNAMICS

3.0 introduction

This unit will focus on the concept of internal energy, the processes by which energy is transferred, the first law of thermodynamics, and some of the important applications of the first law. The first law of thermodynamics is a statement of conservation of energy. It describes systems in which the only energy change is that of internal energy and the transfers of energy are by heat and work.

3.1 Objectives

By the end of this unit, students should be able to:

- Define the terms *heat*, *internal energy*, *latent heat* and *specific latent heat*.
- To be able to state the First Law of thermodynamics

3.2. Keywords;

Heat, internal energy, Heat capacity, calorimetry, latent heat

3.3 HEAT AND INTERNAL ENERGY

- There is a major difference between internal energy and heat.
- Heat involves a *transfer* of internal energy from one location to another.
- **Internal energy** (E_{int}) is the energy associated with the microscopic components of a system—the atoms and molecules of the system. The internal energy includes kinetic and potential energy associated with the random translational, rotational, and vibrational motion of the particles that make up the system, and any potential energy bonding the particles together.
- **Heat** is the transfer of energy between a system and its environment due to a temperature difference between them.

3.3.1 UNITS OF HEAT

- Heat was defined in terms of the temperature changes it produced in an object, and a separate unit of energy, the **calorie**, was used for heat.
- The calorie (cal) is defined as **the energy necessary to raise the temperature of 1 g of water from 14.5° to 15.5°C.**
- The **British thermal unit (Btu)**, was defined as **the energy required to raise the temperature of 1 lb of water from 63°F to 64°F.**
- Heat (like work) is a measure of the transfer of energy, its SI unit should be the joule. The calorie is now defined to be exactly 4.186 J

$$1 \text{ cal} \equiv 4.186 \text{ J}$$

This definition makes no reference to raising the temperature of water.

3.4 SPECIFIC HEAT

- The historical definition of the calorie is the amount of energy necessary to raise the temperature of one gram of a specific substance (water) by one degree. That amount is 4.186 J.
- Raising the temperature of one kilogram of water by 1° requires 4 186 J of energy.
- The amount of energy required to raise the temperature of one kilogram of an arbitrary substance by 1° varies with the substance.
- For example, the energy required to raise the temperature of one kilogram of copper by 1.0°C is 387 J.
- Every substance requires a unique amount of energy per unit mass to change the temperature of that substance by 1.0°C.

The **heat capacity** C of a particular sample of a substance is defined as the amount of energy needed to raise the temperature of that sample by 1°C

- From this definition, we see that if energy Q produces a change ΔT in the temperature of a sample, then;

$$Q = C\Delta T$$

The **specific heat** c of a substance is the heat capacity per unit mass

- If a quantity of energy Q is transferred to a substance of mass m , changing its temperature by $\Delta T = T_f - T_i$, the **specific heat** c of the substance is defined by

$$c \equiv \frac{Q}{m\Delta T}$$

- SI unit: Joule per kilogram-degree Celsius (J/kg °C)

Table 2.3.4. Specific Heats of some Materials at 25°C and Atmospheric Pressure

Substance	Specific Heat c	
	J/kg · °C	cal/g · °C
Elemental Solids		
Aluminum	900	0.215
Beryllium	1 830	0.436
Cadmium	230	0.055
Copper	387	0.092 4
Germanium	322	0.077
Gold	129	0.030 8
Iron	448	0.107
Lead	128	0.030 5
Silicon	703	0.168
Silver	234	0.056
Other Solids		
Brass	380	0.092
Glass	837	0.200
Ice (− 5°C)	2 090	0.50
Marble	860	0.21
Wood	1 700	0.41
Liquids		
Alcohol (ethyl)	2 400	0.58
Mercury	140	0.033
Water (15°C)	4 186	1.00
Gas		
Steam (100°C)	2 010	0.48

Example

Find the energy required to raise the temperature of 0.500 kg of water by 3.00°C.

Solution:

From $c \equiv \frac{Q}{m\Delta T}$ we get

$$\begin{aligned} Q &= cm\Delta T \\ &= (4186 \text{ J/kg}\cdot^\circ\text{C})(0.500 \text{ kg})(3.00^\circ\text{C}) \\ &= 6.28 \times 10^3 \text{ J} \end{aligned}$$

Activity 3.4 a

1. The temperature of a silver bar rises by 10.0°C when it absorbs 1.23 kJ of energy by heat. The mass of the bar is 525 g . Determine the specific heat of silver.
2. A 1.50-kg iron horseshoe initially at 600°C is dropped into a bucket containing 20.0 kg of water at 25.0°C . What is the final temperature? (Ignore the heat capacity of the container, and assume that a negligible amount of water boils away).
3. A steel strut near a ship's furnace is 2.00 m long, with a mass of 1.57 kg and cross-sectional area of $1.00 \times 10^{-4} \text{ m}^2$. During operation of the furnace, the strut absorbs thermal energy in a net amount of $2.50 \times 10^5 \text{ J}$.
 - i. Find the change in temperature of the strut
 - ii. Find the increase in length of the strut.

3.5 CALORIMETRY

- One technique for measuring the specific heat of a solid or liquid is to raise the temperature of the substance to some value, place it into a vessel containing cold water of known mass and temperature, and measure the temperature of the combination after equilibrium is reached.
- Define the system as the substance and the water.
- If the vessel is assumed to be a good insulator, so that energy doesn't leave the system, then we can assume the system is isolated.
- Vessels having this property are called **calorimeters**, and analysis performed using such vessels is called **calorimetry**.

Conservation of energy allows us to write the equation;

$$Q_{\text{cold}} = -Q_{\text{hot}}$$

- Q_{cold} is positive because energy is flowing into cooler objects, and Q_{hot} is negative because energy is leaving the hot object.

- The negative sign on the right-hand side of Equation ensures that the right-hand side is a positive number, consistent with the left-hand side.
- The equation is valid only when the system it describes is isolated.

Example:

A 125-g block of an unknown substance with a temperature of 90.0°C is placed in a Styrofoam cup containing 0.326 kg of water at 20.0°C. The system reaches an equilibrium temperature of 22.4°C. What is the specific heat, c_x of the unknown substance if the heat capacity of the cup is neglected?

Solution:

Let T be the final temperature, and let T_w and T_x be the initial temperatures of the water and block, respectively

Using equations: $Q = cm\Delta T$ and $Q_{cold} = -Q_{hot}$

The water gains thermal energy Q_{cold} , while the block loses thermal energy Q_{hot}

$$Q_{cold} = -Q_{hot}$$

$$\therefore c_w m_w (T - T_w) = -c_x m_x (T - T_x)$$

Solving for c_x and substitute numerical values:

$$\begin{aligned} c_x &= \frac{c_w m_w (T - T_w)}{m_x (T_x - T)} \\ &= \frac{(0.326 \text{ kg})(4190 \text{ J/kg} \cdot ^\circ\text{C}(22.4^\circ\text{C} - 20.0))}{(0.125 \text{ kg})(90.0^\circ\text{C} - 22.4^\circ\text{C})} \\ c_x &= 388 \text{ J/kg} \cdot ^\circ\text{C} \end{aligned}$$

Activity 3.5

1. A 255-g block of gold at 85.0°C is immersed in 155 g of water at 25.0°C. Find the equilibrium temperature, assuming the system is isolated and the heat capacity of the cup can be neglected.

2. A 0.050 0-kg ingot of metal is heated to 200.0°C and then dropped into a beaker containing 0.400 kg of water initially at 20.0°C. If the final equilibrium temperature of the mixed system is 22.4°C, find the specific heat of the metal.

3.6 LATENT HEAT

- A substance often undergoes a temperature change when energy is transferred between it and its surroundings.
- There are situations, however, in which the transfer of energy does not result in a change in temperature.
- This is the case whenever the physical characteristics of the substance change from one form to another; such a change is commonly referred to as a **phase change**.
- Two common phase changes are from solid to liquid (melting) and from liquid to gas (boiling).
- Another is a change in the crystalline structure of a solid.
- All such phase changes involve a change in internal energy but no change in temperature.
- The increase in internal energy in boiling, for example, is represented by the breaking of bonds between molecules in the liquid state.

The energy Q needed to change the phase of a given pure substance is

$$Q = \pm mL$$

where L , called the **latent heat** of the substance, depends on the nature of the phase change as well as on the substance. The unit of latent heat is the joule per kilogram (J/kg)

- **Latent heat of fusion** L_f is the term used when the phase change is from solid to liquid (*to fuse* means “to combine by melting”).
- **Latent heat of vaporization** L_v is the term used when the phase change is from liquid to gas (the liquid “vaporizes”)

Table 2.3.5: Latent Heats of Fusion and Vaporization

Latent Heats of Fusion and Vaporization

Substance	Melting Point (°C)	Latent Heat of Fusion		Boiling Point (°C)	Latent Heat of Vaporization	
		(J/kg)	(cal/g)		(J/kg)	(cal/g)
Helium	-269.65	5.23×10^3	(1.25)	-268.93	2.09×10^4	(4.99)
Nitrogen	-209.97	2.55×10^4	(6.09)	-195.81	2.01×10^5	(48.0)
Oxygen	-218.79	1.38×10^4	(3.30)	-182.97	2.13×10^5	(50.9)
Ethyl alcohol	-114	1.04×10^5	(24.9)	78	8.54×10^5	(204)
Water	0.00	3.33×10^5	(79.7)	100.00	2.26×10^6	(540)
Sulfur	119	3.81×10^4	(9.10)	444.60	3.26×10^5	(77.9)
Lead	327.3	2.45×10^4	(5.85)	1 750	8.70×10^5	(208)
Aluminum	660	3.97×10^5	(94.8)	2 450	1.14×10^7	(2 720)
Silver	960.80	8.82×10^4	(21.1)	2 193	2.33×10^6	(558)
Gold	1 063.00	6.44×10^4	(15.4)	2 660	1.58×10^6	(377)
Copper	1 083	1.34×10^5	(32.0)	1 187	5.06×10^6	(1 210)

- Consider the energy required to convert a 1.00-g cube of ice at - 30.0°C to steam at 120.0°C.
- Figure 2.3.6 indicates the experimental results obtained when energy is gradually added to the ice. Let us examine each portion of the red curve.

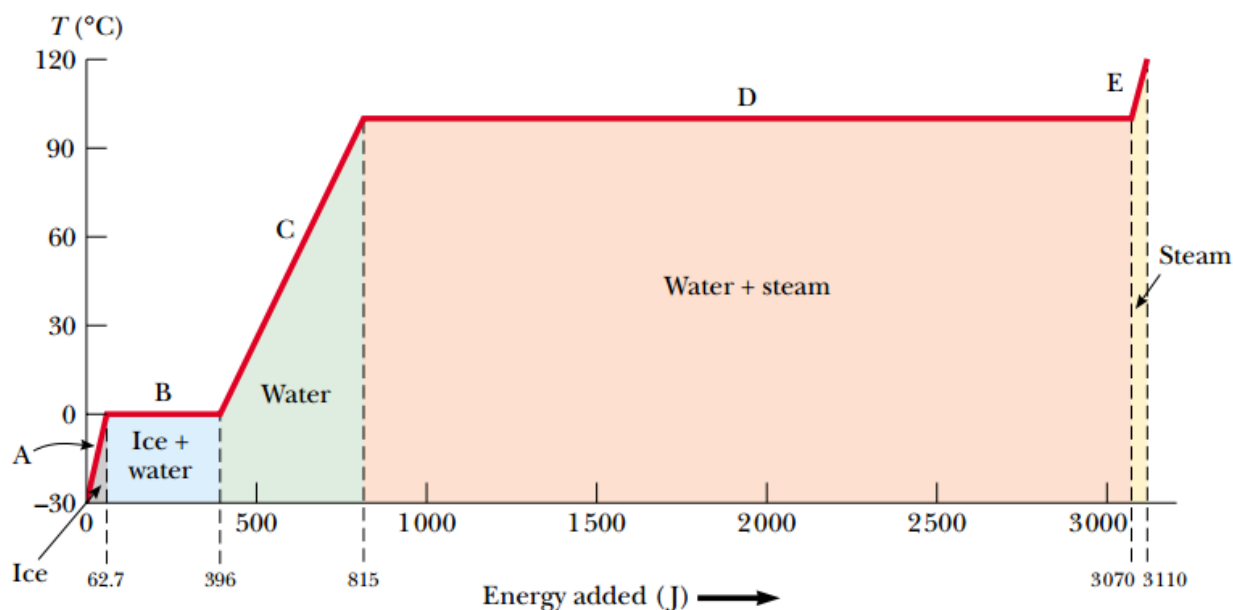


Figure 2.3.6: A plot of temperature versus energy added when 1.00 g of ice initially at - 30.0°C is converted to steam at 120.0°C.

- Part A.** On this portion of the curve, the temperature of the ice changes from -30.0°C to 0.0°C. Because the specific heat of ice is 2 090 J/kg·°C, we can calculate the amount of energy added by using;

$$Q = m_i c_i \Delta T = (1.00 \times 10^{-3} \text{ kg})(2090 \text{ J/kg} \cdot ^\circ \text{C})(30.0^\circ \text{C}) = 62.7 \text{ J}$$

- **Part B.** When the temperature of the ice reaches 0.0°C , the ice–water mixture remains at this temperature—even though energy is being added—until all the ice melts. The energy required to melt 1.00 g of ice at 0.0°C is, from;

$$Q = m_i L_f = (1.00 \times 10^{-3} \text{ kg})(3.33 \times 10^5 \text{ J/kg}) = 333 \text{ J}$$

Thus, we have moved to the 396 J ($= 62.7 \text{ J} + 333 \text{ J}$) mark on the energy axis in Figure 2.3.6

- **Part C.** Between 0.0°C and 100.0°C , nothing surprising happens. No phase change occurs, and so all energy added to the water is used to increase its temperature. The amount of energy necessary to increase the temperature from 0.0°C to 100.0°C is

$$Q = m_w c_w \Delta T = (1.00 \times 10^{-3} \text{ kg})(4.19 \times 10^3 \text{ J/kg} \cdot ^\circ \text{C})(100.0^\circ \text{C}) = 419 \text{ J}$$

- **Part D.** At 100.0°C , another phase change occurs as the water changes from water at 100.0°C to steam at 100.0°C . Similar to the ice–water mixture in part B, the water–steam mixture remains at 100.0°C —even though energy is being added—until all of the liquid has been converted to steam. The energy required to convert 1.00 g of water to steam at 100.0°C is

$$Q = m_w L_v = (1.00 \times 10^{-3} \text{ kg})(2.26 \times 10^6 \text{ J/kg}) = 2.26 \times 10^3 \text{ J}$$

- **Part E.** On this portion of the curve, as in parts A and C, no phase change occurs; thus, all energy added is used to increase the temperature of the steam. The energy that must be added to raise the temperature of the steam from 100.0°C to 120.0°C is

$$Q = m_s c_s \Delta T = (1.00 \times 10^{-3} \text{ kg})(2.01 \times 10^3 \text{ J/kg} \cdot ^\circ \text{C})(20.0^\circ \text{C}) = 40.2 \text{ J}$$

The total amount of energy that must be added to change 1 g of ice at -30.0°C to steam at 120.0°C is the sum of the results from all five parts of the curve, which is $3.11 \times 10^3 \text{ J}$. Conversely, to cool 1 g of steam at 120.0°C to ice at -30.0°C , we must remove $3.11 \times 10^3 \text{ J}$ of energy.

3.6.1. PROBLEM-SOLVING HINTS

CALORIMETRY PROBLEMS

If you have difficulty in solving calorimetry problems, be sure to consider the following points:

- Units of measure must be consistent. For instance, if you are using specific heats measured in $\text{J/kg} \cdot ^{\circ}\text{C}$, be sure that masses are in kilograms and temperatures are in Celsius degrees.
- Transfers of energy are given by the equation $Q = cm\Delta T$ only for those processes in which no phase changes occur. Use the equations $Q = \pm mL_f$ and $Q = \pm mL_v$ only when phase changes are taking place; be sure to select the proper sign for these equations depending on the direction of energy transfer.
- Often, errors in sign are made when the equation $Q_{\text{cold}} = -Q_{\text{hot}}$ is used. Make sure that you use the negative sign in the equation, and remember that ΔT is always the final temperature minus the initial temperature.

Activity 3.6

1. What mass of steam initially at 130°C is needed to warm 200 g of water in a 100-g glass container from 20.0°C to 50.0°C ?
2. The temperature of a silver bar rises by 10.0°C when it absorbs 1.23 kJ of energy by heat. The mass of the bar is 525 g. Determine the specific heat of silver.

3.7. THE FIRST LAW OF THERMODYNAMICS

- The **first law of thermodynamics** is another energy conservation law that relates changes in internal energy
- Suppose that a system undergoes a change from an initial state to a final state. During this change, energy transfer by heat Q to the system occurs, and work W is done on the system.
- Suppose that the system is a gas in which the pressure and volume change from P_i and V_i to P_f and V_f .
- If the quantity $Q + W$ is measured for various paths connecting the initial and final equilibrium states, we find that it is the same for all paths connecting the two states.
- We conclude that the quantity $Q + W$ is determined completely by the initial and final states of the system, and we call this quantity the **change in the internal energy** of the system.
- Although Q and W both depend on the path, **the quantity $Q + W$ is independent of the path.**
- If we use the symbol E_{int} to represent the internal energy, then the change in internal energy ΔE_{int} can be expressed as:

$$\Delta E_{int} = Q + W$$

The first law of thermodynamics

The first law of Thermodynamics states that energy can be converted from one form to another with the interaction of heat, work and internal energy, but it cannot be created nor destroyed under any circumstances

3.7.1. WORK IN THERMODYNAMIC PROCESSES

- Energy can be transferred to a system by heat and by work done on the system.
- In most cases of interest treated here, the system is a volume of gas, which is important in understanding engines.

- All such systems of gas will be assumed to be in thermodynamic equilibrium, so that every part of the gas is at the same temperature and pressure.
- Consider a gas contained by a cylinder fitted with a movable piston as in figure 3.7.1a and in equilibrium.
- The gas occupies a volume V and exerts a uniform pressure P on the cylinder walls and the piston.
- The gas is compressed slowly enough so the system remains essentially in thermodynamic equilibrium at all times.
- As the piston is pushed downward by an external force F through a distance y , the work done on the gas is;

$$W = -F\Delta y = -PA\Delta y$$

Where the magnitude F of the external force equal to PA , possible because the pressure is the same everywhere in the system (by the assumption of equilibrium).

The **work W done on a gas** at constant pressure is given by

$$W = -P\Delta V$$

where P is the pressure throughout the gas and ΔV is the change in volume of the gas during the process.

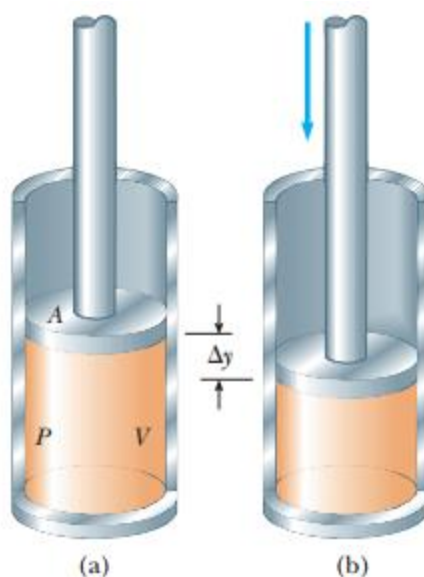


Figure 3.7.1

Example:

An ideal gas absorbs $5.00 \times 10^3 \text{ J}$ of energy while doing $2.00 \times 10^3 \text{ J}$ of work on the environment during a constant pressure process.

- (a) Compute the change in the internal energy of the gas.
- (b) If the internal energy now drops by $4.50 \times 10^3 \text{ J}$ and $2.00 \times 10^3 \text{ J}$ is expelled from the system, find the change in volume, assuming a constant pressure process at $1.01 \times 10^5 \text{ Pa}$.

Solution:

- (a) Substitute values into the first law, noting that the work done on the gas is negative

$$\begin{aligned}\Delta E_{int} &= Q + W \\ &= 5.00 \times 10^3 \text{ J} - 2.00 \times 10^3 \text{ J} \\ &= 3.00 \times 10^3 \text{ J}.\end{aligned}$$

- (b) Substitute the equation for work done at constant pressure into the first law: then Solve for the change in volume, ΔV ,

$$\begin{aligned}\Delta E_{int} &= Q + W = Q - P\Delta V \\ -4.5 \times 10^3 \text{ J} &= -2.00 \times 10^3 \text{ J} - (1.01 \times 10^5 \text{ Pa})\Delta V \\ \Delta V &= 2.48 \times 10^{-2} \text{ m}^3\end{aligned}$$

Activity 3.7

1. A thermodynamic system undergoes a process in which its internal energy decreases by 500 J. At the same time, 220 J of work is done on the system. Find the energy transferred to or from it by heat.
2. An ideal gas initially at 300 K undergoes an isobaric expansion at 2.50 kPa. If the volume increases from 1.00 m³ to 3.00 m³ and 12.5 kJ is transferred to the gas by heat, what are:
 - a. The change in its internal energy
 - b. Its final temperature?

3.8. MOLAR SPECIFIC HEAT OF AN IDEAL GAS

- Molar specific heat capacity is the amount of *heat energy* required to raise the *temperature of 1 mole* of a substance by 1°C.
- The energy required to raise the temperature of n moles of gas from T_i to T_f depends on the path taken between the initial and final states.
- If you supply heat to a gas that is *allowed to expand at constant pressure*, some of the heat that you supply goes to doing external work, and only a part of it goes towards raising the temperature of the gas.
- On the other hand, *if you keep the volume of the gas constant*, all of the heat you supply goes towards raising the temperature.
- Consequently, more heat is required to raise the temperature of the gas by one degree if the gas is allowed to expand at constant pressure than if the gas is held at constant volume and not allowed to expand.
- Thus the heat capacity of a gas (or any substance) is greater if the heat is supplied at constant pressure than if it is supplied at constant volume.
- Thus distinguishing between the *heat capacity at constant volume* C_v and the *heat capacity at constant pressure* C_p , and, as we have seen $C_p > C_v$.

3.8.1 CONSTANT VOLUME SPECIFIC HEAT

- The molar specific heat at constant volume is defined by:

$$Q = nC_v\Delta T$$

- Using the First Law of Thermodynamics, this can be put in the form:

$$\Delta E_{int} + P\Delta V = nC_v\Delta T \text{ where } W = P\Delta V$$

but since $\Delta V = 0$, the expression becomes $\Delta E_{int} = nC_v\Delta T$

- The expression for C_v then becomes, $C_v = \frac{1}{n} \frac{\Delta E_{int}}{\Delta T}$
- If the molar specific heat is constant, we can express the internal energy of a gas as;

$$E_{int} = nC_vT$$

- For a monatomic ideal gas, $E_{int} = \frac{3}{2}nRT$, so

$$C_v = \frac{3}{2}R = 12.5 \text{ J/mol} \cdot \text{K}$$

3.8.2 CONSTANT PRESSURE SPECIFIC HEAT

- The molar specific heat at constant pressure is defined by:

$$Q = nC_p\Delta T \quad (1)$$

- Using the First Law of Thermodynamics, for a constant pressure process this can be put in the form:

$$\Delta E_{int} + P\Delta V = nC_p\Delta T \quad (2)$$

- From the ideal gas law, ($PV = nRT$) under constant pressure condition, it can be seen that;

$$P\Delta V = nR\Delta T \quad (3)$$

- Substituting $P\Delta V$ in equation (2) we get,

$$\Delta E_{int} + nR\Delta T = nC_p\Delta T \quad (4)$$

$$\therefore nC_p = \frac{\Delta E_{int}}{\Delta T} + nR \quad (5)$$

- Dividing equation (5) by n on both sides we get,

$$C_p = \frac{1}{n} \frac{\Delta E_{int}}{\Delta T} + R \quad (6)$$

- Since the constant volume specific heat is $C_v = \frac{1}{n} \frac{\Delta E_{int}}{\Delta T}$

- Substituting $C_v = \frac{1}{n} \frac{\Delta E_{int}}{\Delta T}$ in equation (6) we get

$$C_p = C_v + R \quad (7)$$

- For an ideal monatomic gas

$$C_p = C_v + R = \frac{5}{2}R = 20.8 \text{ J/mol} \cdot \text{K}$$

- From the equation (7) we see that $C_p - C_v = R$, this expression applies to *any* ideal gas.
- It predicts that the molar specific heat of an ideal gas at constant pressure is greater than the molar specific heat at constant volume by an amount R , the universal gas constant (which has the value 8.31 J/mol K).
- The ratio of these heat capacities is a dimensionless quantity

$$\gamma = \frac{C_p}{C_v} = \frac{(5/2)R}{(3/2)R} = \frac{5}{3} = 1.67$$

Ratio of molar specific heats
for a monatomic ideal gas

Table 3.8: Molar Specific Heats of Various Gases

Molar Specific Heats of Various Gases				
Gas	Molar Specific Heat (J/mol · K)^a			
	C_P	C_V	$C_P - C_V$	$\gamma = C_P/C_V$
Monatomic Gases				
He	20.8	12.5	8.33	1.67
Ar	20.8	12.5	8.33	1.67
Ne	20.8	12.7	8.12	1.64
Kr	20.8	12.3	8.49	1.69
Diatomic Gases				
H ₂	28.8	20.4	8.33	1.41
N ₂	29.1	20.8	8.33	1.40
O ₂	29.4	21.1	8.33	1.40
CO	29.3	21.0	8.33	1.40
Cl ₂	34.7	25.7	8.96	1.35
Polyatomic Gases				
CO ₂	37.0	28.5	8.50	1.30
SO ₂	40.4	31.4	9.00	1.29
H ₂ O	35.4	27.0	8.37	1.30
CH ₄	35.5	27.1	8.41	1.31

^aAll values except that for water were obtained at 300 K.

Example:

A cylinder contains 3.00 mol of helium gas at a temperature of 300 K.

- If the gas is heated at constant volume, how much energy must be transferred by heat to the gas for its temperature to increase to 500 K?
- How much energy must be transferred by heat to the gas at constant pressure to raise the temperature to 500 K?

Solution

- For the constant-volume process, we have

$$Q_1 = nC_V\Delta T$$

Because $C_V = 12.5 \text{ J/mol} \cdot \text{K}$ for helium and $\Delta T = 200 \text{ K}$, we obtain;

$$\begin{aligned} Q_1 &= (3.00 \text{ mol})(12.5 \text{ J/mol} \cdot \text{K})(200 \text{ K}) \\ &= 7.50 \times 10^3 \text{ J} \end{aligned}$$

b. For constant-Pressure process

$$\begin{aligned}
 Q_2 &= nC_p\Delta T \\
 &= (3.00 \text{ mol}) \left(20.8 \frac{\text{J}}{\text{mol}} \cdot \text{K} \right) (200 \text{ K}) \\
 &= 12.50 \times 10^3 \text{ J}
 \end{aligned}$$

Activity 3.8

1. A system of monatomic ideal gas at $2.00 \times 10^5 \text{ Pa}$ and an initial temperature of 293 K slowly expands at constant pressure from a volume of 1.00 L to 2.50 L .
 - a. Find the work done on the environment.
 - b. Find the change in internal energy of the gas.
 - c. Use the first law of thermodynamics to obtain the thermal energy absorbed by the gas during the process.
 - d. Use the molar heat capacity at constant pressure to find the thermal energy absorbed.
2. Ideal monatomic gas at an initial temperature of 475 K is compressed from 3.00 L to 2.00 L while its pressure remains constant at $1.00 \times 10^5 \text{ Pa}$.

Find

- a. The work done on the gas,
- b. The change in internal energy,
- c. The energy lost by heat, Q

3.9 NEWTON'S LAW OF COOLING

- Newton's law of cooling states, "For a body cooling in a draught (i.e., by forced convection), the rate of heat loss is proportional to the difference in temperatures between the body and its surroundings." Since the temperature change ΔT is proportional to the heat change.

$$\left(\begin{array}{c} \text{Rate of heat loss} \\ \text{to surrounding} \end{array} \right) \propto \left(\begin{array}{c} \text{Excess} \\ \text{temperature} \end{array} \right)$$

We can write

$$\frac{dT}{dt} = -k(T - T_s)$$

Where $\frac{dT}{dt}$ = The rate of loss of heat to surrounding,

T_s = The temperature of the surroundings,

T = The temperature of the body

k = Proportionality constant whose value depend on both the nature and the area of the body's surface.

3.9.1 THE FIVE-FOURTHS POWER LAW

- This applies when a body is cooling under conditions of natural convection. It can be stated as:

$$\left(\begin{array}{c} \text{Rate of heat loss} \\ \text{to surrounding} \end{array} \right) = -k(T - T_s)^{5/4}$$

Unit summary

- **Internal energy** is all of a system's energy that is associated with the system's microscopic components. Internal energy includes kinetic energy of random translation, rotation, and vibration of molecules, potential energy within molecules, and potential energy between molecules.
- **Heat** is the transfer of energy across the boundary of a system resulting from a temperature difference between the system and its surroundings. We use the symbol Q for the amount of energy transferred by this process.
- The **heat capacity** C of any sample is the amount of energy needed to raise the temperature of the sample by 1°C . The energy Q required to change the temperature of a mass m of a substance by an amount ΔT is

$$Q = mc\Delta T$$

where c is the **specific heat** of the substance

- The energy required to change the phase of a pure substance of mass m is:

$$Q = \pm mL$$

where L is the **latent heat** of the substance and depends on the nature of the phase change and the properties of the substance. The positive sign is used if energy is entering the system, and the negative sign is used if energy is leaving.

- The **first law of thermodynamics** states that when a system undergoes a change from one state to another, the change in its internal energy is

$$\Delta E_{int} = Q - W$$

Unit Test.

1. Using the first law of thermodynamics, explain why the total energy of an isolated system is always constant.
2. A 3.00-g lead bullet at 30.0°C is fired at a speed of 240 m/s into a large block of ice at 0°C , in which it becomes embedded. What quantity of ice melts?
3. In an insulated vessel, 250 g of ice at 0°C is added to 600 g of water at 18.0°C .
 - a. What is the final temperature of the system?
 - b. How much ice remains when the system reaches equilibrium?
4. An ideal gas is enclosed in a cylinder with a movable piston on top of it. The piston has a mass of 8 000 g and an area of 5.00 cm^2 and is free to slide up and down, keeping the pressure of the gas constant. How much work is done on the gas as the temperature of 0.200 mol of the gas is raised from 20.0°C to 300°C ?
5. A combination of 0.250 kg of water at 20.0°C , 0.400 kg of Aluminium at 26.0°C , and 0.100 kg of copper at 100°C is mixed in an insulated container and allowed to come to thermal equilibrium. Ignore any energy transfer to or from the container and determine the final temperature of the mixture.

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